3. The following shall apply when straight conical sections are used:
   a. Design shall be per the ASME Code, Division 2, Sections 4.3.11 and 4.4.13.
   b. Joint efficiency of 1.0 shall not be permitted for conical transitions without a knuckle due to the difficulty in examining the joints.
   c. Checks shall be made for design conditions and hydrotest condition.
4. Stiffening rings, if used, shall be far enough away from circumferential butt joints to allow easy nondestructive examination (NDE) (ultrasonic testing [UT] and magnetic particle testing [MT]) of the adjacent butt joints to occur during vessel manufacturing and while in service.
5. Ellipsoidal heads, torispherical heads, toriconical heads, and toriconical transition sections shall have straight flanges at least 1½ inches (40 mm) long.

5.4 Nozzles, Manways, Bosses, and Other Openings

1. Nozzles for nominal pipe size (NPS) size 1½ (diameter nominal [DN] 40) and larger shall be in accordance with [PVM] GF-C14311.
   a. Seamless long weld neck and integrally reinforced nozzle forgings shall be preferred for nozzles and manways. However, pad reinforced openings may be used as an alternate.
   b. Smaller nozzles shall not be used except as required to meet other standards or specifications such as in Section 5.4, item 9.
2. Nozzles shall be integrally reinforced (i.e., reinforcing pads shall not be permitted) in vessel services if one or more of the following situations apply:
   a. Design temperature exceeding 650°F (345°C).
   b. Cyclic service.
   c. Severe process environments having stress corrosion cracking potential.
   d. Vessels in mild or benign services but having defined process upsets that may carry over stress corrosion cracking agents.
3. Nozzle reinforcement design per the current ASME Code shall be acceptable, provided that the design conforms to all other requirements listed in this specification.
4. Flanged nozzles NPS 4 (DN 100) and smaller shall be long weld neck flanges or fabricated from seamless pipe and weld neck flanges.
   a. Minimum pipe thickness for NPS 2 (DN 50) and NPS 3 (DN 80) shall be Schedule 160.
   b. Minimum pipe thickness for NPS 4 (DN 100) shall be Schedule 120.
   c. These pipe thicknesses shall be inclusive of CA and exclusive of cladding or weld overlay.
5. ‘Set-on’ nozzle designs shall be acceptable only for small bore services and only for Division 1 vessels. Proposed designs shall be accepted by Owner.
6. Manways shall be in accordance with [PVM] GF-C14311 and [PVM] GF-C87280.
   a. For vessels with a nominal shell inside diameter (ID) less than 48 inches (1.2 m), manways shall be a minimum NPS 20 (DN 500) with a minimum uncorroded ID of 17.9 inches (454 mm).
   b. For vessels with a nominal shell ID 48 inches (1.2 m) and larger, manways shall be a minimum NPS 24 (DN 600) with a minimum uncorroded ID of 21.5 inches (546 mm).

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7. Threaded connections may be used only if required by another code or standard such as Tubular Exchanger Manufacturers Association (TEMA).
   a. Threads shall be chased after installation or post weld heat treatment (PWHT).
   b. Threaded connections shall not be acceptable for use on internally coated or lined vessels.
8. Threaded connections greater than NPS 1 (DN 25) shall not be used.
9. Threaded connections NPS 1 (DN 25) and less, when shown on the vessel drawing, shall be full coupling, ASME Class 6000, 'set through' the vessel shell with a full penetration weld, per Section 5.2, item 2. This application shall be limited to shell thickness less than 3/4 inch (19 mm) and design temperature less than 450°F (230°C).
10. Openings should not penetrate, and reinforcing pads (when permitted) should not cover welded joints. When unavoidable, the weld seam shall be ground flush, 100 percent radiographed to a minimum of 2 inches (50 mm) beyond each side of the opening or repad per Section 9.4, item 1 prior to pad installation.
11. Tongue-and-groove flange facing shall not be used.
12. For vessels supported by a skirt, the flange of any nozzle in the bottom head shall be located outside the skirt.
    a. The scope of the ASME Code, Section VIII shall be limited to the vertical nozzle with the elbow, horizontal piping, and flange placed in the scope of ASME B31.3.
    b. The code specification break between the ASME Code, Section VIII, and ASME B31.3 shall be clearly documented on the drawings and manufacturer's data report.
13. Process nozzle connections shall be designed for local stresses due to external loads from the attached piping. The external loads shall be consistent with the loads provided by PIM-DU-5155, unless otherwise stated by Purchaser.
14. Local stresses on the vessel/nozzle connection imposed by external piping or attachments may be evaluated. The use of WRC Bulletin 537 or WRC Bulletin 297 shall be acceptable to meet this requirement, when applicable.

    Note: *WRC Bulletin 537 is an updated version of WRC Bulletin 107, which has been withdrawn.*

    a. Extrapolation of geometric parameter curves in WRC 537 and WRC 297 shall not be permitted.
    b. If WRC 537 and WRC 297 are not applicable or not desired, more rigorous analysis methods such as FEA shall be used with results evaluated per the ASME Code, Division 2, Part 5.
15. Integrally reinforced welded branch fittings (such as weldolets, thredolets, and sockolets) shall not be acceptable for vessel connections.
16. Nozzles for vessels identified as being in cyclic or hydrogen service shall be designed to accommodate ultrasonic shear wave (UTSW) inspection and surface examination by MT or liquid penetrant testing (PT).

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17. The following connections shall not have internal projections but shall be finished flush with the inside surface of the vessel:
    a. Vents and drains.
    b. Manways.
    c. Nozzles for two-phase flow.
    d. Top vapor outlets.
    e. Bottom liquid outlets except when a vortex breaker compliant with [PVM] GA-C99913 is used.

5.5 Attachments

1. External attachments (such as saddles, pads, insulation support rings, stiffening rings, lifting lugs, platform and ladder clips, and nameplate brackets) shall be continuously welded to the vessel unless otherwise agreed to by Owner.
2. Pads and saddles shall be circular or shall have a corner radius of at least five times the pad thickness or 2 inches (50 mm), whichever is smaller.
3. Vents shall be provided in all pads and similar attachments that create a void space.
    a. The vent shall be drilled and tapped national pipe thread (NPT) 1/4 (DN 8) prior to installation on the vessel.
    b. The vent shall be located in the lowest practical point of the pad or attachment.
    c. Split or two-piece reinforcement pads shall comply with the ASME Code, Division 1, paragraph UG-37(h).
4. Lifting lugs, tailing lugs, and lifting trunnions shall be analyzed for loads transmitted to the shell, heads, or support skirt during lifting.
    a. Trunnions shall be welded through the pad into the shell.
    b. Lifting lugs shall be seal welded to the vessel or reinforcement pad on four sides to prevent corrosion under the lugs. A gap in the weld on the bottom side for drainage shall be acceptable provided that the remaining weld length is of adequate strength for the design.
5. Fillet welds shall be at least 3/16 inch (5 mm), shall be welded with multiple passes, and shall be subject to the same PWHT as the vessel.
    a. Fillet welds shall have a smooth flat or slightly concave profile at the specified design throat thickness.
    b. Undercut shall not be permitted.
6. When external or internal attachments cover welds in the vessel, the covered area of the weld shall be ground flush and 100-percent inspected by radiography or UTSW 2 inches (50 mm) beyond each side of the attachments.
7. For internal attachments, efforts shall be made to minimize covering welds in the vessel.
8. Internal attachments shall be continuously welded to the vessel, except that a 1-inch (25-mm) skip in the bottom of an internal pad or other attachment detail that creates a void space shall be made for a vent.

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9. A grounding lug shall be supplied in accordance with PIP VEFV1103.
10. Insulation and fireproofing supports shall conform to PIP VEFV1122 and PIP VEFV1123.

    Note: *PIP drawings VEFV1103, VEFV1122, and VEFV1123 can be found in the PIP document, VEFV1100.*

11. Railings, stairways, walkways, platforms, and ladders shall conform to the requirements of SID-SU-5106.
12. All materials welded directly to the pressure-retaining envelope shall be material of the same P number and group number in the applicable table in the ASME Code, Section II, Part D.
13. Galvanized clips or attachments shall be welded to the vessel only after galvanizing has been removed to a minimum of 1/2 inch (12 mm) beyond the extent of welding.

5.6 Vessel Supports

1. Localized shell stresses at support-to-shell locations shall be considered, as applicable, for wind load, earthquake, and all other loadings described in the ASME Code, Division 1, paragraph UG-22 or, for Division 2 vessels, in the ASME User's Design Specification.
2. Support steel shall be no thinner than the smaller of 1/4 inch (6 mm) thick including CA (new) or the nominal thickness of the vessel component to which it is attached.
3. Support welding shall use continuous seal-welded construction, except as allowed by applicable standard drawing details.
4. Design of support skirts for vertical vessels shall conform to [PVM] GD-C78876.
    a. Skirt openings shall be provided with rings or collars, sized to provide 100 percent area reinforcement of the opening.
    b. Provision shall be made for venting of the skirt. Location of the vent shall take into account any required insulation of the bottom head and internal fireproofing of the skirt.
5. For vertical vessels on legs, design of support legs shall conform to [PVM] GB-C99945.
6. Generally, leg supports should not be used under the following conditions:
    a. Operating temperature exceeds 450ºF (230ºC).
    b. Service is cyclic.
    c. Vessel height-to-diameter ratio exceeds 5, where height is measured from the base of the support to the top tangent line of the vessel.
    d. Vessel ID is greater than 60 inches (1.5 m).
7. Design of support saddles for horizontal vessels shall conform to [PVM] GF-C99694.
    a. Calculation of localized shell stress may be determined using the L.P. Zick method described in the article, "Stresses in Large Horizontal Cylindrical Pressure Vessels on Two Saddle Supports." This method is valid only for vessels with two saddles.
    b. Distance between head tangent line and the saddle centerline shall not be greater than 25 percent of the tangent-to-tangent length of the vessel.

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8. Horizontal vessels subject to partial or FV and requiring external stiffening rings shall also include rings located over and integral with each saddle. The saddle rings shall be designed and sized to accommodate the circumferential bending from the saddle reaction and the inward radial load due to external pressure.
9. Vertical vessels directly supporting heavy equipment, such as thermosyphon reboilers, shall be checked to determine if they require circumferential ring reinforcement to accommodate the radial thrust and weight from each reboiler on the shell. If the vessel is subject to any vacuum condition, the resulting vacuum radial load shall also be accommodated in the ring design.

5.7 Internals

1. Dimensional changes or alterations to details shall not be made without prior written acceptance by Purchaser.
2. Fabricator shall be responsible for clarification of any questionable points when preparing detailed fabrication drawings.
3. Fabricator shall be solely responsible for the effect of the internals upon the pressure-containing components of the vessel.
4. Unless specified otherwise, for low alloy internals and supports welded to the shell or not easily removable, the CA per surface shall be 75 percent of the single side CA for the shell as stated in the data sheet.
5. Unless specified otherwise, for bolted and easily removable low alloy internals, the CA per surface shall be 50 percent of the single side CA for the shell as stated in the data sheet.
6. Unless otherwise stated, Supplier shall provide all internals in accordance with the following requirements:
    a. Internals furnished by others shall be installed by the fabricator.
    b. Baffles, partitions, and other parts shall be designed to accommodate differential thermal expansion between vessel shell and internal parts.
    c. All components subject to differential pressure shall be designed to accommodate such loads.
    d. Baffles, partitions, and other parts shall be designed to avoid trapping water or flammable liquids that cannot be drained or purged.
    e. Internal ladders shall comply with [PVM] GB-C99772.
    f. Internal non-pressure-containing parts shall be designed using appropriate stress for material and temperature as specified in the ASME Code, Section II, Part D.

Note: *Designs for internals are shown on Purchaser drawings in a manner intended to permit the fabricator to prepare detailed shop fabrication drawings that comply with Purchaser's design with regard to process requirements and CA (not MACA).*

5.8 Lifting Requirements

1. A minimum impact factor of 1.5 shall be applied to the lift weight for designing lifting devices.
2. Lift weight shall include complete fabricated weight of the vessel plus all components to be included in the lift (e.g., trays, ladders/platforms, insulation, additional piping with insulation, lifting slings, and shackles).

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3. Lifting shall be done when wind speed is below 25 mph (40 km/h or 11m/s). This wind load shall be included in the consideration of the lift.
4. Stresses shall be checked for the lifting condition.
   a. Calculated combined stress shall not exceed 80 percent of yield strength at ambient temperature.
   b. Compressive stress shall not exceed 1.2 times the B factor for the Code.
5. Load calculations shall be performed and sent to Purchaser for review and acceptance.
6. Design shall consider a maximum sling angle of 30 degrees from vertical along the weak axis of the lifting lug unless a spreader bar is used. If a spreader bar is used, then this requirement shall be called out on the vessel drawing along with any angle limitations.

5.9 Wind Requirements for Flexible Columns or Tall Vessels

1. Vertical vessels or columns that have a fundamental natural frequency less than 2 Hz shall be checked for wind-induced vibration.
2. Critical wind velocity of the vessel shall be greater than 3 times the maximum mean hourly wind speed at the top of the vessel determined in accordance with ASCE 7-16.

Note: *For details and corrective actions to prevent excessive vibrations, refer to ASME STS-1.*

5.10 Details for Clad or Lined Pressure Vessels

1. For pressure vessels that are to be clad or weld overlayed, fabrication details shall comply with PVM-SU-1322.
2. Strip lining on shells and loose liners on nozzles shall not be used.
3. For pressure vessels that are to be internally lined with a non-metallic lining or coating, fabrication details shall comply with PIP VESV1003.
4. Lining systems shall comply with COM-SU-5191.

6.0 Materials

1. When specified by Purchaser, materials shall be furnished to the specification and grade shown on the vessel drawing, data sheet, or list sheets.
2. Base metal specifications for the pressure boundary and all directly external welded attachments shall conform with the following conventional 2¼ Cr-1 Mo ASME material specifications:
   a. **Plate:** SA-387 Gr. 22, Class 2.
   b. **Forgings:** SA-182 Gr. F22, Class 3; SA-234 Gr. WP22, Class 1 or 3; or SA-336 Gr. F22, Class 3.
   c. **Pipe:** SA-335, Gr. P22; SA-369, Gr. FP22.
3. Fabricator may propose alternate materials in a separate quotation, but the alternate materials shall comply with all requirements of this specification.

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6.1 Pressure Containing Parts

1. All plate and forging grades of material shall be vacuum degassed and produced with a fine grain practice.

   a. This shall be stated in the material test reports.

   b. Chemical requirements by heat analysis shall be as follows:

$$J - factor = (Si + Mn) \times (P + Sn) \times 10^4 \leq 100$$

Equation 1

   Where:

   Si, Mn, P, and Sn are in wt %.

   Additionally, limit chemistry as follows:

   – P to 0.007 wt % maximum.

   – S to 0.007 wt % maximum.

   – Cu to 0.20 wt % maximum.

   – Ni to 0.30 wt % maximum.

2. All materials shall be normalized and tempered (N&T) or quenched and tempered (Q&T) to meet the required mechanical properties.

3. All plate grades of material 2 inches (50 mm) thick and greater shall be UT examined in accordance with the requirements of the ASME Code, Division 2, paragraph 3.3.3.

4. All forgings shall be forged as close as practical to the finished shape and size to develop metal flow in a direction most favorable for resisting the stresses encountered in service.

   a. All forgings 2 inches (50 mm) thick and thicker shall be UT examined in accordance with the requirements of the ASME Code, Division 2, paragraph 3.3.4.

   b. All surfaces of forgings, regardless of thickness, shall be wet fluorescent magnetic particle testing (WFMT) examined in accordance with the requirements of the ASME Code, Division 2, paragraph 3.3.5 after finish machining.

5. Castings shall be prohibited.

6. Tensile and Charpy V-Notch impact tests required for certification of all plate, forging, and pipe grades of material according to the ASME Code, Division 2, paragraph 3.11 shall represent the vessel in the completed condition.

7. For Code Division 1 and Division 2 vessels, compliance with the ASME Code, Division 2, paragraph 3.11 shall be verified by subjecting specimens for tensile and Charpy V-notch impact tests to the total thermal history that is required for fabrication of the vessel, plus two additional PWHT thermal cycles. Thermal history shall include all heat treatments, hot forming, heat treatment after hot forming, PWHT, and at least two additional PWHT cycles to allow for a shop and field repair.

   Note: *Compliance with these tensile and impact requirements can be met by including additional testing of the base metals in the welding procedure qualification (Section 7.3) or production test plates (Section 7.8) requirements of this specification, with the acceptance of Owner.*

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8. For cold-formed heads and other components for both Division 1 and Division 2 vessels, post-forming stress relief heat treatment shall be required.
    a. Complete reheat treatment (N&T or Q&T) in accordance with the ASME materials specification shall be required if the forming strain (extreme fiber elongation) exceeds 9 percent. Forming strains shall be calculated per the formulas in Table 6.1 of the ASME Code, Division 2.
    b. Fabricator shall submit calculations clearly showing calculated fiber elongation and required heat treatments. When fiber elongation calculations require stress relief or complete reheat treatment, it shall be clearly shown on the vessel drawings.
    c. Cold sizing shall be permitted after stress relief heat treatment, if necessary to correct for distortion, providing maximum strain does not exceed 3 percent (as calculated in item 8a). Supplementary calculations shall be required for any component that requires cold sizing to confirm that stress relief or complete reheat treatment is not necessary.
9. For hot-formed components, complete reheat treatment (N&T or Q&T) in accordance with the SA materials specifications and the plate mill heat treatment operation shall be required. Certified documentation from Supplier shall state if hot formed, and if associated heat treatment was performed.
10. Components that require reheat treatment by fabricator shall not need to be obtained from the materials Supplier in the final heat treatment condition.
11. Fabricator shall prepare a flow chart showing the sequence of all forming, welding, heat treating, and inspection operations for review by Owner prior to the start of production.
12. Plate, forging, and pipe grades of material shall meet the Charpy V-notch impact test requirements of the ASME Code, Division 2, paragraphs 3.11 through 3.11.8.4 at the MDMT shown on the vessel drawing or data sheet, with the following additional requirements:
    a. The ASME Code, Division 2, Figures 3.7 and 3.8 shall be used to determine Charpy V-notch impact test exemptions for both Division 1 and Division 2 vessels. In plates and forgings where thickness is greater than 2 inches (50 mm), impact testing shall be required.
    b. When Charpy V-notch impact tests are required, the minimum Charpy V-notch impact energy required shall be 40 ft-lbf (54 J) average at 0°F (–18°C) for three specimens and 35 ft-lbf (47 J) minimum at 0°F (–18°C) for any single specimen for both Division 1 and Division 2 vessels. The percent ductile fracture and lateral expansion (in mils or mm) shall also be reported.
13. Charpy V-notch impact testing shall be required for MDMTs below −20°F (−30°C).
14. Test coupons shall be taken from 1/2 T of the material's thickness.
15. Charpy V-notch impact specimens shall be oriented transverse to the direction of major working with the notch perpendicular to the surface.
16. Temper embrittlement screening of plate and forging materials per Section 14.0 shall be required if the shell thickness is greater than 3 inches (75 mm) or if temper embrittlement screening is required on the vessel drawing or data sheet.
17. Materials for reinforcing pads (where allowed) for openings and attachments shall be the same (specification, grade, class, etc.) as that of the shell or head where the pad is to be attached.

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6.2 Non-Pressure-Containing Components

1. Skirts and other vessel supports for a length of at least 12 inches (300 mm) from the point of attachment to the vessel shall be fabricated from the same material specification and grade as the vessel plates.
   a. Skirts and other vessel supports beyond 12 inches (300 mm) from the point of attachment to the vessel may be fabricated from any material listed in the ASME Code, Section II, Part D that meets the requirements of SA-516.
   b. All plate, forging, and pipe grades of material shall meet the Charpy V-notch impact test requirements for the MDMT for the vessel support shown on the vessel drawing.
2. Skirts and other vessel supports shall be suitable, per paragraph UCS-66 of the ASME Code, Division 1, or Section 3.11 of Division 2, for the MDMT for the support.
3. Except for the nameplate, all materials welded to any part of the vessel shall be listed in the ASME Code, Section II, Part D materials.
4. All material shall be suitably identified as to the material specification and grade.

6.3 Bolting

1. For vessels with an MDMT warmer than or equal to –20°F (–30°C), external bolting used to assemble pressure-containing components shall be studs and nuts conforming to SA-193, Gr B7, and SA-194, Gr 2 or 2H for maximum design temperatures not exceeding 750°F (400°C), and SA-193, Gr B16 and SA-194, Gr 7 for maximum design temperatures of 850°F (455°C).
2. Austenitic grades of material with strength increased by strain hardening shall not be used.
3. Bolts of 1 3/8-inch (35-mm) diameter and larger shall be provided with extra length equal to one bolt diameter and a second nut shall be furnished for protection of the extra bolt length.
4. For studs in blind holes, either the end of the stud or the bottom of the hole shall be relieved to prevent galling.

   Note: *The approach described in ASME PCC-2, Article 3.3, Appendix I is recommended.*

7.0 Welding

7.1 General

1. All welding, qualification of welding procedures, and qualification of welders and welding operators shall be done in accordance with the ASME Code, Section IX, except as restricted herein.
2. Vessel fabricator shall be responsible for the following, with referenced documents made available for use by the welders or welding operators and for reference by the Code-authorized inspector and Purchaser:
   a. Selecting and preparing the welding procedure specification (WPS) for each type of joint, material, and thickness.
   b. Preparing and providing a procedure qualification record (PQR) documenting proof of the weldability of the variables described in the WPS.

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c. Preparing and providing a welder performance qualification (WPQ) documenting proof of the welder's ability to deposit sound weld metal within the range of WPS parameters being used.

d. Submitting welding procedures to Purchaser for acceptance prior to beginning fabrication of the vessel.

3. Structural welding involving rolled shapes and plates of carbon steel shall, as a minimum, conform to the requirements of the general welding specifications of AWS D1.1/D1.1M. Structural welding of 2¼ Cr-1 Mo material shall use welding procedures qualified to the ASME Code, Section IX, and this specification.

7.2 Welding Procedure Specifications

1. The welding procedure specification (WPS) shall clearly identify the essential variables, materials and specifications, consumables by name and specification, as well as the standard ASME Code requirements.

2. The application for each WPS shall be shown on a separate weld map or welding and testing plan that shall clearly indicate the WPS for each joint and for attachments. The map or plan shall be submitted to the Purchaser prior to the start of welding.

7.3 Welding Procedure Qualification

1. The weld procedure shall be qualified for all welding to be used for fabrication of the vessel, including repair welding.

2. When Charpy V-notch impact tests are required, the heat input used as determined by welding current and travel speed shall be considered essential variables.

   a. Stringer bead welding technique shall be preferred; however, weave beads shall be allowed for out-of-position welding if the maximum width of weave does not exceed three times the electrode core wire diameter.

   b. The heat input range specified on the WPS shall not exceed the highest qualified value on the supporting PQR.

3. Welding procedures shall be qualified with the maximum PWHT as applicable for each vessel or vessel part.

4. The PWHT total time at (soak) temperature shall be three times the ASME Code minimum required for the thickness in order to qualify the weld procedure for repair welding and possible further heat treatments.

5. For all hydrogen-containing services, each welding procedure shall have three Vickers hardness traverses of the weld joint on a weld sample in the minimum PWHT condition as shown in Figure 1.

   a. Hardness traverses shall be made 0.04 inch to 0.08 inch (1 mm to 2 mm) below both surfaces and at 1/2 t, where t is the material thickness. Traverses shall include weld metal, heat-affected zones (HAZs), and base metals.

   b. Each traverse shall include 10 hardness readings for a total of 30 hardness readings per weld sample.

   c. The hardness shall not exceed 235 HV10.

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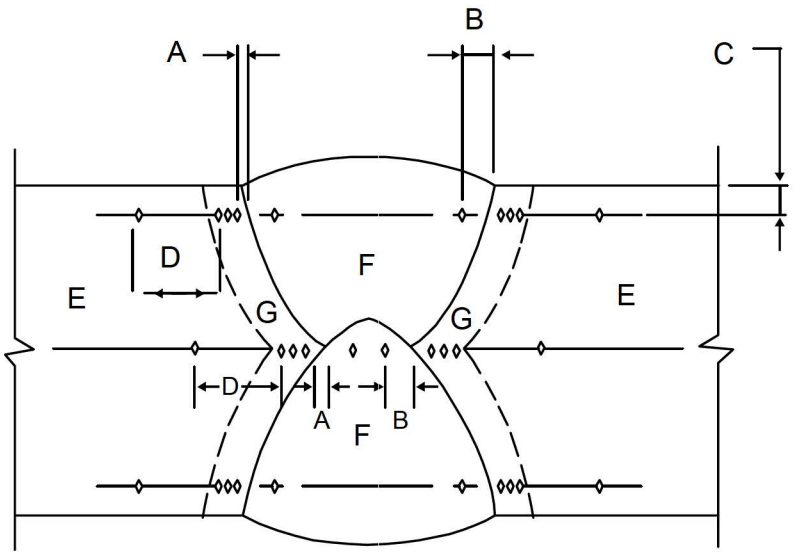
d. Hardness test results shall be part of the PQR.

Figure 1: Location of Vickers Hardness Indentations



Key:

A	Approx. 8 mil (0.2 mm)	E	Base Metal
B	0.04 in.–0.12 in. (1 mm–3 mm) (TYP)	F	Weld Metal
C	0.04 in.–0.08 in. (1 mm–2 mm) (TYP)	G	Heat-Affected Zone (HAZ)
D	0.12 in.–0.24 in. (3 mm–6 mm) (TYP)		

7.4 Acceptable Methods

1. Acceptable methods of welding shall be shielded metal arc welding (SMAW), submerged arc welding (SAW), gas metal arc welding (GMAW), GTAW, and flux-core arc welding gas-shielded (FCAW-G) processes.
2. Other welding processes may be used with Owner's written acceptance.
3. The interrupted-arc (short circuit transfer) GMAW process may be used only for the following applications:
   a. Root passes on circumferential, longitudinal, or nozzle-to-shell welds if back-gouged and back-welded.
   b. Root passes on circumferential piping welds for fabricated nozzles or internal piping.
   c. Root passes on non-pressure-containing vessel internals.
4. Joints welded from two sides shall be ground, back-chipped, or back-gouged to sound metal before continuing welding on the backside of the joint.
5. Before welding or hot cutting, weld surfaces and HAZs shall be clean and free of lead, sulfur, dirt, grease, and any other deleterious material.
6. Fit-up of any Type 1 single welded butt joint shall be inspected by Owner's inspector prior to welding.

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7.5 Selection of Consumables

1. Selection of consumables shall be the responsibility of the fabricator.
   a. Consumables shall be selected to meet the same strength and toughness requirements as the base metal.
   b. Low-hydrogen consumables shall be used unless others are accepted in writing by Owner.
2. Material test reports for the specific consumables to be used shall be submitted to Owner for acceptance prior to beginning fabrication.
3. Welding procedures (WPSs and PQRs) shall be submitted to Purchaser for acceptance prior to beginning vessel fabrication.
4. Dissimilar metal welds shall be prohibited for all Category A, B, C, and D welds unless accepted in writing by Owner.
5. Low-alloy weld metals shall have Cr and Mo contents in the same range as the plate and forging grades of material. The following chemical composition restrictions shall apply to the heat analysis of the consumable:

$$\bar{X} = (10P + 5Sb + 4Sn + As)/100 \leq 15\ ppm$$

Equation 2

Where:

P, Sb, Sn, and As are in ppm.

Additionally, limit chemistry as follows:

– C to 0.15 wt % maximum

– Cu to 0.20 wt % maximum

– Ni to 0.30 wt % maximum

6. Each heat of low-alloy consumables shall undergo positive material identification (PMI) before use (with weld buttons made and PMI performed per CPM-PU-5074 and Section 9.5 of this specification).
7. All consumables shall be of a low-hydrogen type that is certified by the consumable manufacturer, or tested by fabricator, to result in less than 8 ml of diffusible hydrogen per 100 gm (H-8 designation) of deposited weld metal.
   a. Diffusible hydrogen content shall be measured by the mercury displacement or gas chromatograph methods per AWS A4.3.
   b. Vessel fabricator shall provide written evidence of compliance with this paragraph with welding procedure submittal.
   c. A specification sheet for the proposed brand of filler metal from the consumable supplier shall be sufficient.
8. Consumables shall be stored, handled, and used in accordance with the electrode manufacturer's instructions.
   a. Submerged arc flux shall be kept and stored in low humidity (60 percent maximum), warm (minimum 70°F [20°C]), controlled storage areas.
   b. Flux shall be dried per flux manufacturer requirements prior to being used on the shop floor.

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c. Flux may be recovered and recycled only twice before it shall be discarded.

9. For vessels requiring impact testing, the same brand name and manufacturer source consumables used for welding procedure qualification shall be used for production welding. Any change or variation shall require re-qualification.

10. For vessels with design temperatures greater than 650°F (345°C), temper embrittlement screening of the weld consumables as specified in Section 14.0 shall be required (regardless of the thickness of the shell).

    a. This testing shall be conducted prior to start of fabrication.

    b. Each heat and lot of welding consumables, as well as filler metal heat/flux batch combination, shall be subject to the step-cooling tests.

11. For vessels designed for severe low-temperature applications (less than –20°F [–5°C]) or where high-impact strength is required (greater than 40 ft-lb [54 Joules]), the following requirements shall be met:

    a. Batches of consumables shall be proven suitable and certified to be in accordance with the specification before their use on production welding.

    b. Batch testing may be performed by the consumable supplier.

    c. Batch testing of SAW consumables shall be done by combining the heat number for the electrode and lot number for the flux and performing all the required mechanical tests per the appropriate AWS designation.

7.6 Tensile Strength

1. Tensile properties of the deposited weld metal shall meet those of the base metal specified in Section 6.1, item 6.

2. Tensile strength of the weld metal shall be limited to 100 ksi (690 MPa) maximum.

3. This shall be verified by a hardness test of the weld metal in the welding procedure qualification and shown on the PQR and a hardness test of each Category A and B weld seam.

7.7 Hardness of Weld Deposit and Adjacent Weld Metal

1. After final PWHT (see Section 8.3), hardness determinations shall be made for each pressure-retaining weld using a Telebrineller portable hardness tester (or other Purchaser-accepted testing technique).

2. Each hardness test result shall be the average of three impressions at each test location.

    a. Test locations shall include the weld metal and base metals adjacent to the fusion line on both sides.

    b. Hardness values of all three impressions shall be reported.

3. Hardness values shall not exceed 235 HBW (Brinell hardness using tungsten carbide indenter) or equivalent.

4. Hardness tests shall be performed on each 10-foot (3-m) length of weld, or fraction thereof, with a minimum of one test per weld.

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7.8 Vessel Production Test Plates

1. When Charpy V-notch impact testing is required by Section 6.1, item 12, one test plate shall be made for each vessel and for each welding procedure used for Category A, B, C, and D joints. Vessel production test plates shall be in accordance with paragraph 3.11.2.10 of the ASME Code, Division 2.
2. Production (vessel) test plates shall be subjected to the same total thermal history expected for the finished vessel, including hot forming, preheat, and PWHT.
   a. Production test plates shall be taken from hot-formed components or shall require a separate thermal cycle to mimic hot forming thermal history.
   b. Production test plates shall be heat treated with the vessel unless accepted otherwise by Owner.
3. Charpy V-notch impact tests shall be made for the base metals, weld metals, and HAZs when required by Section 6.1, item 12, and they shall comply with all requirements of Section 6.1, item 12.

7.9 Welding After Post Weld Heat Treatment or Tests

All welding, including attachment welds, shall be completed prior to final post weld heat treatment (PWHT) except welding of internal attachments to the corrosion resistant overlay or cladding.

8.0 Heat Treatment

8.1 Preheat

1. For welding and high-temperature cutting operations, minimum required preheats shall be 300°F (150°C) for all thicknesses.
2. Preheat shall be maintained continuously until PWHT, an intermediate stress relief (ISR), or a dehydrogenation heat treatment (DHT) is performed in accordance with Section 8.2.
3. Preheat shall extend for a minimum distance equal to the material thickness but not less than 3 inches (75 mm) on each side of the weld.

8.2 Intermediate Stress Relief or Dehydrogenation Heat Treatment

8.2.1 Intermediate Stress Relief

1. For 2¼ Cr-1 Mo steel, the preheat for high restraint weld joints shall be maintained until PWHT or until an intermediate stress relief (ISR) can be performed at temperatures greater than 1100°F (600°C) for 2 hours minimum before cooling to ambient.
   a. ISR shall be required for all shell butt welds, nozzle-to-shell attachment welds, areas of weld build-up, larger structural attachments, and other welds having high restraint.
   b. All other welds may have DHT performed.
   c. ISR shall be performed in a furnace or local ISR treatment may be used, provided circumferential heating band that includes the full circumference of the shell is used.
2. When ISR is required, it shall also be used for all weld repairs.
3. Manufacturer shall be responsible for maintaining temperature gradients that are not harmful to the vessel during heating and cooling.

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8.2.2 Dehydrogenation Heat Treatment

1. Dehydrogenation heat treatment (DHT) shall be performed at 575–800°F (300–425°C) for a duration of 1 hour minimum.
2. Manufacturer may request written authorization from Purchaser to use DHT in lieu of the ISR required for high restraint areas.
3. Owner shall require qualification of the manufacturer's fabrication shop with regard to consumables procurement, storage, and shop floor handling prior to accepting the use of DHT in lieu of ISR.

8.3 Post Weld Heat Treatment (Stress Relief)

1. Final post weld heat treatment (PWHT) shall be performed at 1275°F ± 25°F (690°C ± 14°C) and shall be held for 1 hour per inch (25 mm) of thickness (2 hour minimum).
2. PWHT shall be sufficient to meet the strength and hardness limits in Section 7.6 and Section 7.7, as demonstrated by PQR tests.
3. The process of heating for PWHT shall be at fabricator's option, except that direct flame heating and exothermic methods shall not be permitted.
4. PWHT shall meet the minimum temperature in item 1 above and the minimum time required by the ASME Code, Division 1, Table UCS-56 or Division 2, Table 6.8, as applicable.
5. The use of ASME Code, Division 1, Table UCS-56.1 or Division 2, Table 6.16 for applying PWHT at a lower temperature for an increased length of time shall not be acceptable.
6. A detailed procedure for all PWHT, ISR, and DHT shall be required for Purchaser review and acceptance, and shall include a vessel sketch of the following:
   a. Locations and types of thermocouples.
   b. Method of thermocouple attachment.
   c. Insulation details for local PWHT.
   d. Methods used to support the vessel or part thereof.
   e. Bracing to be used to prevent distortion.
   f. Heating control system details.
7. A sufficient number of thermocouples shall be attached to the vessel, and recorders shall be used to verify to Purchaser's inspector that the required temperature and holding time have been obtained.
8. Details of any methods used to prevent scaling on machined surfaces shall be included in the procedure.
9. During PWHT, vessels shall be blocked and supported as necessary to avoid any deformation or damage.
10. Lead- or sulfur-containing substances such as oil, grease, thread lubricants, and paint shall be removed from vessel surfaces prior to PWHT.

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11. Any internal parts installed in the vessel prior to PWHT shall be assessed to ensure that their corrosion and material properties are not affected. The following shall be done after PWHT:
    a. Parts shall be realigned.
    b. Warpage shall be corrected.
    c. Bolts shall be retightened to comply with vessel drawing.
12. PWHT of the completed vessel as whole shall be preferred. The temperature difference between any two thermocouples during heating and cooling cycles shall not exceed 150°F (70°C).
13. Manufacturer shall be responsible for maintaining temperature gradients that are not harmful to the vessel during heating and cooling.
14. When accepted by Purchaser, local heat treatment shall be permitted for welds as long as the local PWHT is a circumferential band around the entire perimeter of the pressure vessel and there are no geometric anomalies such as nozzles in the transition temperature zone.

9.0 Examination

Quality assurance and quality control throughout procurement and fabrication shall be performed in accordance with QAM-SU-6053 and QAM-SU-6163.

9.1 Dimensional Tolerances

1. Tolerances for vessel dimensions, nozzle and manway locations and orientations, and other design details shall be in accordance with IOGP S-619, Annex E.
2. Vessel fabricator shall calculate maximum permissible *deviation e* from a perfect circle for all shell sections and skirt (if one is used), per the ASME Code, Division 1, UG-80(b) or Division 2, paragraph 4.4.4. These allowable deviations shall be clearly shown on vessel drawings.

9.2 Minimum Examination Requirements Common to All Weld Joints

1. Weld bevels with a shell component thickness of 2 inches (50 mm) or greater shall receive MT prior to welding.
    a. All indications of linear flaws shall be removed by grinding or chipping, and the area shall be re-examined to ensure satisfactory removal of the flaw.
    b. Repairs shall be made with a welding procedure qualified in accordance with Section 7.3.
2. Category A, B, C, and D butt-welded root areas shall receive MT of the weld preparation following back-gouging and prior to back welding. This examination shall be randomly witnessed by a Purchaser inspector for Category C and D joints.
3. When specified in the data sheet, Category A, B, C, and D weld joints in shell components with a thickness of 3 inches (75 mm) or greater shall receive full coverage UT after final heat treatment.

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9.3 Minimum Examination Requirements for Specific Joint Categories

1. Vessels shall have 100 percent inspection of Category A and B welds.
   a. NDE shall be performed per ASME Code, Division 2, Table 7.2 for the specific weld configuration and joint category at a 1.0 joint efficiency.
   b. NDE extent shall be 100 percent if a lower value is allowed per ASME Code, Division 2, Table 7.2.
   c. Completed welds shall receive either UT or radiographic testing (RT) before PWHT.
   d. Final UT inspection shall be after final PWHT.
   e. MT shall be performed on completed welds before PWHT and repeated after PWHT.
2. Category C and D welds shall receive the following inspections:
   a. NDE shall be performed per ASME Code, Division 2, Table 7.2 for the specific weld configuration and joint category at a 1.0 joint efficiency.
   b. The NDE extent shall be 100 percent even if a lower value is allowed per ASME Code, Division 2, Table 7.2.
   c. Unless otherwise specified, MT in item 1e above shall be randomly witnessed by an Owner's inspector or representative.
   d. MT shall be performed on completed welds before PWHT and repeated after PWHT.
   e. UT shall be witnessed as per Section 9.8, item 3.

9.4 Examination Requirements for Structural Attachments and Miscellaneous Joints

1. Skirt-to-vessel welds shall receive UT unless another type of examination is accepted by Owner.
   a. Where possible, skirt-to-vessel welds shall be designed to permit UT.
   b. If impossible or impractical to design so that UT can be applied, skirt-to-vessel welds shall receive MT after the root pass is completed and every third layer thereafter.
2. Internal and external attachment welds, including those for lifting attachments when supplied, shall receive MT after grinding, if grinding is required. MT shall be performed after heat treatment.
3. Surfaces from which temporary attachments have been removed shall receive MT after grinding.
4. Weld pad buildups for attachment of internal or external components and supports shall receive UT.

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9.5 Visual Examination

1. Purchaser's inspector may be present during fabrication of the vessel and shall be granted free access, but the fabricator shall provide shop inspection to verify quality of workmanship and that the work is being performed in accordance with this specification and the ASME Code.
2. Any evidence of poor workmanship or damage due to mishandling shall be repaired at the fabricator's expense before the vessel is accepted.

9.6 Magnetic Particle Examination

1. Magnetic particle examination shall comply with ASME Code, Division 2, paragraph 7.5.6.1.
2. Acceptance criteria shall comply with ASME Code, Division 2, paragraph 7.5.6.2.
3. Prod burns shall be removed by grinding.
4. WFMT shall be used for all accessible internal welds on vessels 40-inch (1000-mm) ID and larger except as noted in item 6 below.
5. MT for external welds shall be of a wet visible type.
6. Dry powder MT techniques may be used in the following situations:
   a. On welds that receive a preheat in excess of 125°F (52°C).
   b. For in-process welding checks, such as after back-gouging and weld prep.
7. MT procedures shall be subject to Owner review and acceptance.
8. Entire surfaces of forgings, including welding edges, shall be examined by MT after finish machining but before welding.
9. All surfaces on which temporary attachments have been removed shall receive an MT examination after grinding.
10. All accessible surfaces of category A, B, C, and D welds shall receive an MT examination after hydrostatic testing.

9.7 Radiography

1. Radiography (RT) shall be in accordance with the ASME Code, Division 2, paragraph 7.5.3.
2. Acceptance criteria shall be per the ASME Code, Division 2, paragraph 7.5.3.2.
3. Acceptance RT shall be required after final PWHT.
4. Radiographs shall be retained by fabricator for at least 1 year after the date of shipment and shall be made available for review upon request by Owner.

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9.8 Ultrasonic Examination

1. UT shall be made in accordance with the ASME Code, Division 2, paragraph 7.5.4.1, and the acceptance criteria of paragraph 7.5.4.2 shall be applied unless otherwise required in this specification.
2. UT on Category A and B joints will be considered "in lieu of RT" per the ASME Code, Division 1, UW-51(a)(4) or Section 7.5.5 of the ASME Code for Division 2 and shall use a recordable UT method.
3. UT of welds shall be witnessed by a Purchaser's inspector.
4. UT methodology shall require prior acceptance by Owner of detailed UT procedures including weld and nozzle configuration, mode of the UT wave, reflection angle, scanning position, and calibration blocks.
5. The selected UT method shall be one that creates a permanent record such as phase array ultrasonic testing (PAUT), time of flight diffraction (ToFD), or other equivalent method accepted by the Purchaser.
6. UT shall be performed after PWHT, unless otherwise specified.

9.9 Positive Materials Identification

1. Fabricator or fabricator's approved sub-contractor shall perform the positive material identification (PMI) inspection in accordance with CPM-PU-5074.
2. Purchaser's inspector shall be notified and may elect to witness some or all of the PMI inspection.

10.0 Hydrostatic Testing

10.1 Preparation

1. Fabricator shall furnish facilities and test materials, including blinds, bolts, nuts, and gaskets.
2. Vessel shall be cleaned internally and externally of dirt, debris, welding slag, and weld spatter before the hydrostatic test is performed.
3. No paint or primer shall be applied to the vessel prior to hydrostatic testing.
4. Purchaser shall be notified 3 working days in advance so that the hydrostatic test can be witnessed.
5. Reinforcing pads shall be pneumatically tested per the ASME Code, Section V, Article 10, Appendix I, with 15 psig (1 barg) dry air or nitrogen and a bubble-forming solution.

10.2 Required Stress Levels

1. Hydrostatic test pressure for Division 1 vessels shall be based on the MAP (new and cold) as described in the ASME Code, UG-99(c). This test pressure shall be reduced as necessary to comply with the limits of the ASME Code, Division 2, paragraph 4.1.6.2.
2. Hydrostatic test pressure for Division 2 vessels shall be determined in accordance with the ASME Code, Division 2, paragraph 8.2.(e).

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10.3 Other Requirements

1. Test water shall not contain more than 50 ppm chlorides.
2. During hydrostatic testing, the vessel metal temperature shall be at least 30°F (17°C) above the MDMT, or 60°F (15°C), whichever is warmer.
3. Any suitable sheet gaskets may be used on ASME raised-face flanges for the hydrostatic test, with the following exceptions:
   a. Elastomeric gaskets shall not be used.
   b. ASME raised-face openings with hinged covers shall require service-type gaskets.
4. All other gasketed joints shall require service-type gaskets.
5. Hydrotest pressure shall be held for 1 hour minimum unless stated otherwise by Purchaser.
6. After hydrostatic testing, the vessel shall be promptly and carefully drained and completely dried, especially around trays and internals.
7. The vessel shall be internally inspected to ensure that no residual moisture remains.

11.0 Painting

11.1 General

1. Exterior surfaces that require painting shall be prepared in accordance with the referenced coating system as specified on the vessel data sheet.
2. Application of coating systems shall comply with COM-SU-4743.
3. Coating systems shall comply with COM-SU-5191.

11.2 Priming

When priming is specified on the vessel drawing, priming shall be done after hydrostatic testing and in accordance with the coating system as specified in the vessel data sheet.

11.3 Fireproofing

1. Fireproofing shall comply with FPM-DU-5091 for onshore applications or FPM-DU-1508 for offshore topside applications.
2. Intumescent (epoxy) fireproofing coatings shall be strongly preferred over cementitious fireproofing.

12.0 Nameplate

1. Each vessel shall be identified by a bracket-mounted, austenitic stainless steel nameplate.
2. Bracket shall be continuously welded to the vessel prior to stress relief, and the nameplate continuously welded to the bracket after stress relief.
3. Bracket shall be so mounted that the nameplate is not inadvertently covered by insulation.
4. In addition to the stamping required by the ASME Code, the nameplate shall show the purchase order number and item number, as well as the MAEP (at design temperature and steamout temperature if applicable) per Section 5.1.3, item 7.

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13.0 Preparation for Shipment

13.1 Flanges

1. Flange faces shall be protected against corrosion and damage by applying stiff grease and by bolting on wooden or steel covers with a minimum of four bolts.
2. A plastic sheet shall be inserted between a wooden cover and the greased flange surface.
3. Nozzles fitted with blind flanges (including manways and handholes) shall be bolted with specified studs, nuts, and gaskets in place, ready for service. Nuts shall be torqued as recommended by the gasket manufacturer.

13.2 Threaded Connections

1. Female threaded connections shall be fitted with solid round steel bar stock plugs of the same material as the connection.
2. Plugs shall have a length of shank equal to at least twice the diameter and shall not be free-machining steels.
3. Plugs shall be marked by stamping to designate the material used.
4. Plugs shall be threaded with pipe threads to match the connection.

13.3 Vent Holes

Vent holes and skips in welds provided for venting in reinforcing pads, saddles, etc., shall be plugged with stiff grease.

13.4 Thread Lubricant

Bolting shall be lubricated with a recognized brand of high-temperature molybdenum disulfide (MoS_2)-based thread lubricant.

13.5 Security

Fabricator shall load and block the vessel in a manner to avoid damage during shipment.

13.6 Atmosphere

1. Fabricator shall use dry air and desiccant for maintaining a non-corrosive environment in the vessel during shipment.
2. Fabricator shall use stainless steel wire to hang stainless steel tags on the vessel manways or handholes to indicate that they contain desiccant.
3. The use of nitrogen or other means of creating an oxygen-deficient atmosphere shall not be used without prior Owner acceptance.

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13.7 Labeling

1. Text shall be stenciled or neatly written in characters at least 4 inches (100 mm) high using coating requirements described in Section 11.0. A contrasting color of paint shall be used for the text.
2. Vessel tag number, purchase order number, and fabricator's name shall be written on the skirt or near a saddle.
3. If vessel is PWHT, vessel shall be labeled "Vessel Is PWHT – Do Not Weld."

14.0 Supplemental Requirement: Temper Embrittlement Screening for Chromium-Molybdenum Steels

Note: *The requirements in this section are used only when specified on the vessel drawing or data sheet.*

1. For this specification, temper embrittlement screening of the base metals and the welding consumables shall only be required on vessels with a maximum design metal temperature equal to or greater than 650°F (345°C) and a shell thickness greater than 3 inches (75 mm).
2. Temper embrittlement screening shall be required on the welding consumables only when the maximum design metal temperature is equal to or greater than 650°F (345°C) and the shell thickness is less than or equal to 3 inches (75 mm).

14.1 General

1. Sampling for step-cooled impact tests shall be as described in Section 6.1, item 12.
2. Step-cooling testing shall be required for base metal materials to determine susceptibility to temper embrittlement.
3. Prior to start of fabrication, step-cooling tests shall be performed on the weld consumables to determine susceptibility to temper embrittlement. Each heat and lot of welding consumables, as well as each filler metal/flux batch combination, shall be subject to step-cooling tests.
4. For each step-cooling testing, the following shall apply:
   a. Test specimens shall be heat treated to the minimum PWHT condition.
   b. Average impact values at 0°F (−18°C) of three Charpy V-notch test specimens shall not be less than 40 ft-lb (55 J), with no single value below 35 ft-lb (47 J).
   c. The percent ductile fracture and lateral expansion in mils shall also be reported.
   d. Testing shall comply with the ASME Code, Division 2, Section 3.11.7.

14.2 Step-Cooling Tests

1. Step-cooling tests of the base metals shall be required for the following:
   a. The shell plate with the highest J-factor for each vessel.
   b. The nozzle forging with the highest J-factor for each vessel.
   c. The head plate/forging with the highest J-factor for each vessel.

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2. Step-cooling tests shall be performed on the weld metal and HAZ for each welding procedure as specified in Section 14.1, item 3. Previously qualified WPS with step-cooling tests for the same materials (same ASME Code material specification and nominal chemistry) may be accepted by Purchaser.

3. Test procedure and acceptance criteria shall be as follows:

   a. Two sets of Charpy V-notch test specimens, with a minimum of 18 specimens per set, shall be prepared and subjected to the following heat treatments:

      1) **Set 1** – Minimum PWHT only to establish a transition temperature curve before step cooling.

      2) **Set 2** – Minimum PWHT plus the step-cooling heat treatment specified in item b below to establish a transition temperature curve after step cooling.

   b. Step-cooling heat treatment shall be as follows:

      1) Heat to 600°F (316°C); heating rate not critical.

      2) Heat at 100°F (56°C)/hour maximum to 1100°F (593°C).

      3) Hold at 1100°F (593°C) for 1 hour.

      4) Cool at 10°F (6°C)/hour maximum to 1000°F (538°C).

      5) Hold at 1000°F (538°C) for 15 hours.

      6) Cool at 10°F (6°C)/hour maximum to 975°F (524°C).

      7) Hold at 975°F (524°C) for 24 hours.

      8) Cool at 10°F (6°C)/hour maximum to 925°F (496°C).

      9) Hold at 925°F (496°C) for 60 hours.

      10) Cool at 5°F (3°C)/hour maximum to 875°F (468°C).

      11) Hold at 875°F (468°C) for 100 hours.

      12) Cool at 50°F (28°C)/hour maximum to 600°F (316°C).

      13) Cool to ambient temperature in still air.

   c. After the Charpy V-notch test specimen sets are heat treated, each set of specimens shall be impact tested at a minimum of six selected test temperatures to establish a transition temperature curve.

      1) One of the tests shall be performed at 0°F (−18°C).

      2) Three specimens shall be tested at each test temperature.

      3) The number of test specimens used shall be sufficient to establish a Charpy V-notch impact toughness transition temperature curve.

      4) The curve shall clearly define the transition as well as the lower and upper shelves.

   d. The 40 ft-lb (55 J) transition temperatures shall be determined from the transition temperature curves established from the two sets of Charpy V-notch specimens. The impact properties shall meet the following requirement:

$$CvTr40 + 2.5\,\Delta CvTr40 \leq 50°F\ (CvTr55J + 2.5\,\Delta CvTr55J \leq 10°C)$$

Equation 3

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Where:

CvTr40 = 40 ft-lb (55J) transition temperature of material subjected to the minimum PWHT only.

ΔCvTr40 = The shift of the 40 ft-lb (55J) transition temperature between the two sets of specimens (minimum PWHT vs. minimum PWHT + step cool).

14.3 Record Keeping and Reporting

1. Supplier shall maintain for each vessel an up-to-date record showing the identity of each heat of wire, lot of electrodes, or lot of flux used in each weld.
2. A copy of this record shall be provided to Owner upon completion of the vessel.
3. Test results shall be reported to Owner before the material represented by the specimens is used in the vessel.

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CATALYTIC DEWAXING UNIT
MANALI, CHENNAI, INDIA

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GN-1. ABBREVIATIONS

The following abbreviations are used throughout this document:

CI	=	cast iron
CS	=	carbon steel
CS (Per GN-9)	=	carbon steel with modifications specified in General Note 9 to improve resistance to hydrogen-induced cracking
SS	=	stainless steel
12 Cr	=	family of martensitic and ferritic stainless steels (Types 410, 410S and the cast equivalents)
18-8	=	family of austenitic stainless steels (Types 304, 316, 321, 347, etc.)
c.a.	=	corrosion allowance
pp	=	partial pressure
ppmw	=	parts per million by weight
ppbw	=	parts per billion by weight
PWHT	=	postweld heat treatment
THS	=	thermally stabilized
ga.	=	Birmingham Wire Gage; minimum wall thickness equivalents are summarized below:

10 ga. = 0.134 in. (3.40 mm)	16 ga. = 0.065 in. (1.65 mm)
12 ga. = 0.109 in. (2.77 mm)	18 ga. = 0.049 in. (1.25 mm)
14 ga. = 0.083 in. (2.11 mm)	20 ga. = 0.035 in. (0.89 mm)

GN-2. HIGH TEMPERATURE HYDROGEN ATTACK

Where applicable, materials have been selected for resistance to high temperature hydrogen attack. Material recommendations are based on curves for hydrogen resistance, API Publication 941, latest edition. CLG practice is to limit maximum operating conditions to 3.5 kg/cm^2 (50 psia) hydrogen partial pressure and 28°C (50°F) below the applicable hydrogen resistance curve. We do not recommend C-1/2 Mo steel for new construction in high temperature hydrogen service.

GN-3. HIGH TEMPERATURE H_2-H_2S CORROSION

Where applicable, materials have been selected for resistance to high temperature H_2-H_2S corrosion and to minimize corrosion products in the system. CLG uses in-house proprietary corrosion data to determine the proper materials of construction and to predict corrosion rates.

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GN-4. HIGH TEMPERATURE H_2S CORROSION

Where applicable, materials have been selected for resistance to high temperature H_2S corrosion (when no H_2 is present) and to minimize corrosion products in the system. CLG uses in-house proprietary corrosion data and industry data such as the Modified McConomy curves to determine the proper materials of construction and to predict corrosion rates.

GN-5. ORGANIC ACID CORROSION

Materials for hot, raw feed piping and equipment have been selected for resistance to organic acid corrosion and to minimize corrosion products in the system. (Organic acids decompose in the reactors, so this is not a concern in downstream equipment.) The organic acid concentration in raw feed is indicated by the neutralization number (neut. no.), as determined by ASTM Test Methods D664 or D974. CLG uses in-house proprietary corrosion data and published data, such as that of CLG's R. L. Piehl, to determine the proper materials of construction and to predict corrosion rates.

GN-6. INTERGRANULAR CORROSION AND POLYTHIONIC SCC

CLG and other refiners have had adverse experience with regular carbon grades and low carbon grades of 18-8 SS in high temperature H_2-H_2S services above 700°F (371°C). Intergranular corrosion and stress corrosion cracking have occurred due to the action of sulfur-derived (polythionic) acids in hydrodesulfurizer plants. Accordingly, we require that chemically stabilized grades (Types 321 and 347) be used, both for their superior sensitization resistance and higher design stresses.

Laboratory investigations reported by Samans at NACE and CLG's R. L. Piehl at API meetings indicated superior sensitization resistance and performance of Type 347 SS over Type 321 SS. Consequently, we use Type 347 SS for services subject to severe sensitization, whether during fabrication or during operation. However, for tube bundles and piping, we believe that the exposure time to sensitization temperatures during fabrication is short enough to justify the use of Type 321 SS. Type 321 SS will normally be less expensive and less likely to develop welding and heat treating problems. Type 347 SS may be used as an alternative material if the wall thickness is less than 19 mm (0.75 in.). Type 347 SS weld metal is used for welding both Type 321 and 347 SS base metal.

For service temperatures above 454°C (850°F), Types 321 and 347 SS can sensitize. CLG recommends furnace tubes [and other SS equipment that will operate above 454°C (850°F)] be thermally stabilized to prevent sensitization. Thermal stabilization consists of heating the entire piece of equipment, after all welding is complete, at 899°C (1650°F) for 4 hours.

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GN-7. CHLORIDE STRESS CORROSION CRACKING AND CHLORIDE CORROSION

CHLORIDE STRESS CORROSION CRACKING

Stainless steels are subject to chloride stress corrosion cracking (Cl SCC) in service if proper precautions are not exercised. The following precautions are recommended to mitigate this problem. (**See GN-22 for protection of stainless steel from the atmosphere**.)

Postweld Heat Treatment

To provide resistance to Cl SCC, all 18-8 SS cold bends should be heat treated at 843-899°C (1550-1650°F), 1 hour per 25 mm (1 inch) of thickness, 1 hour minimum, followed by air cooling.

Drain Lines and Low Points

Industrial experience has shown that drain lines and low points in austenitic stainless steel lines are particularly prone to Cl SCC. Although PWHT lowers the incidence of Cl SCC of drain lines, it does not eliminate it. Therefore, CLG requires that low points be avoided and drain lines in stainless steel piping be made of Alloy 625 or Alloy 825. All components of the drain assembly, from the weld between the drain line and the stainless steel main process piping, through the first block valve off of the main process line, should be Alloy 625 or Alloy 825.

This requirement includes utility, vent, and drain lines on all high pressure vessels clad/weld overlayed with 3XX SS. Although we typically avoid these types of connections in high pressure vessels, where present, these lines are potential deadlegs and Cl SCC is a concern. The piping for these connections should be Alloy 625 or Alloy 825. The requirement for Alloy 625 or Alloy 825 is also appropriate for utility, vent, and drain lines in low pressure 3XX SS clad equipment which could be exposed to wet chlorides.

Soda Ash Washing

Chevron USA Hydroprocessing Units have elected to stop using soda ash during unit shutdowns. This decision was based on the minimal benefit provided by soda ash washing in preventing polythionic stress corrosion cracking and intergranular corrosion due to chlorides. These damage mechanisms affect sensitized stainless steel materials and our designs utilize materials that are resistant to sensitization under the design conditions. Therefore, there is a low risk of these damage mechanisms occurring. On the other hand, Chevron has found a relatively high risk of chloride and/or caustic cracking of equipment during startup when the soda ash solution has not been completely drained from the equipment. Sodium carbonate may evolve into sodium hydroxide under normal operational conditions experienced during the unit start-up.

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Those Licensees who currently use soda ash washing should perform their own evaluation to determine if the benefits of soda ash washing outweigh the costs and risks associated with soda ash washing. It would be recommended that any such evaluation include a survey of the Unit's entire operational history to assess the possibility of material sensitization. The potential for material sensitization on equipment exposed to temperatures above normal design values due to operational issues should be considered.

CHLORIDE CORROSION

General

Although any chloride can cause corrosion, in hydroprocessing plants we are most concerned with ammonium chloride. Feedstocks usually contain nitrogen, which can react with the chloride present. Normal practice is to maintain feed and hydrogen chloride levels to less than 1 ppm. Higher chloride levels increase the risk of ammonium chloride salt fouling and corrosion.

Chlorides present in the feed/hydrogen streams convert to hydrochloric acid in the reactor and exit in the reactor effluent stream. As the reactor effluent stream cools, ammonia (formed from nitrogen in feed) and HCl react to form ammonium chloride (NH_4Cl).

Ammonium chloride salts are hygroscopic (they attract water and dissolve very readily). **Ammonium chloride is highly corrosive under wet conditions. It can cause severe corrosion and pitting of carbon steel, chrome-moly steels, austenitic stainless steels, and even some high alloys like the Incoloys.**

Specific Hydroprocessing Exchangers

Reactor Effluent Cooling System Heat Exchangers

In hydroprocessing plants, ammonium chloride deposits can accumulate on heat exchanger tubes in the reactor effluent system as the process fluid cools. Typically, salt deposits are dry and non-corrosive. They are objectionable primarily because they are foulants that interfere with heat transfer and increase pressure drop through equipment. Exchangers that may experience salt fouling are designed for intermittent water wash in the upstream piping. While water is being injected, corrosive conditions exist in the exchanger. However, the duration and frequency of water wash in most plants is low enough that metal loss is minimal. To minimize damage, we recommend limiting water washing to once a month maximum, and when washing, to fully remove all salts. See the Process Manual for more detailed limits on intermittent water washing of these exchangers.

For plants with high chloride feeds, salts may deposit rapidly and more frequent water washing may be necessary. Serious corrosion can occur under these conditions. If conditions are so severe that no reasonably priced alloy will be resistant to corrosion, the plant flow scheme may need to be modified.

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REAC's

Ammonium chloride corrosion may occur in the REAC and its inlet piping downstream of the wash water injection point, from improper design or operation. Causes are inadequate amounts of injection water or poor wash water distribution. The reactor effluent stream contains ammonia, H_2S, and some chloride. As the effluent cools below the dew point, the first drops of water to condense absorb the chloride and become acidic and corrosive. The intent of the continuous water washing is to quench through the dew point instantaneously from no water to lots of water, so that both ammonium chloride and ammonium bisulfide salts are immediately dissolved and diluted. **When water washing, it is imperative to add enough water so that at least 25% remains unvaporized at the injection point.** See the Process Manual for more water washing guidance.

Separator Liquid Heat Exchangers

Chloride corrosion is possible in the exchanger or exchangers that heat separator liquid on its way to distillation. Separator water can contain ammonium chloride, and water may be entrained in the hydrocarbon coming out of the separators. As the hydrocarbon heats, the water can evaporate and salts can concentrate. This may lead to highly corrosive wet salts on heat exchanger tubes, so appropriate alloys must be chosen[1].

Other Equipment

Chloride corrosion is also possible in the top of the first distillation column, as chloride-containing water can be carried over. For plants with chlorides above 1 ppmw in the feed, Alloy 625 WOL/cladding is recommended for the top portion of the first distillation column (typically the product stripper or fractionator, if no product stripper is present).

GN-8. WET H_2S CRACKING/SULFIDE STRESS CRACKING

Wet H_2S cracking is a blanket term that covers all cracking mechanisms that require H_2S and water to be active. In all forms of cracking, the culprit is hydrogen, produced from a corrosion mechanism, entering the steel. The result can be cracking, blistering, or a combination of both. Generally, 50 ppm H_2S is the accepted minimum threshold limit for wet H_2S cracking, but cracking can sometimes occur at lower concentrations.

Sulfide stress cracking (SSC) is one form of wet H_2S cracking, and is addressed in this GN-8. Other forms of wet H_2S cracking are addressed in GN-9.

SSC can occur rapidly when high strength steels (generally materials with yield strengths above 6300 kg/cm^2(a) (90 ksi) or with hardnesses above HRC 22 (237 Brinell) are exposed to wet sour

[1] This issue is discussed in Corrosion 2000 Paper 00687.

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environments. SSC is entirely preventable by controlling the strength and hardness of materials exposed to wet H_2S environments.

Equipment and Piping

Materials with yield strengths above 6300 kg/cm^2(a) (90 ksi) or with hardnesses above HRC 22 (237 Brinell) should be avoided for wet H_2S services. For CS, CLG also recommends following NACE Standard SP0472, which includes further limiting the hardness of carbon steel welds and heat-affected zones (HAZs) to 200 Brinell. See NOTE E on the Notes for Materials of Constuction Drawings. Note that CLG requires adherence to NACE SP0472 for most CS in refinery services to avoid any harmful materials mix-ups and to allow for possible exposure to wet H_2S environments in the future.

If higher strength materials are absolutely required, materials such as some precipitation hardening SS and hardenable nickel alloys may be used. NACE Standard MR0103 may be used as a guide for requirements for using these materials, which are often necessary in rotating equipment and valves.

See GN-21 for limitations of dissimilar welds in wet H_2S service.

GN-9. WET H_2S CRACKING/HYDROGEN BLISTERING AND HYDROGEN-INDUCED CRACKING

While GN-8 addresses SSC, this GN-9 addresses the other forms of wet H_2S cracking, including hydrogen blistering, hydrogen-induced cracking (HIC), and stress oriented hydrogen-induced cracking (SOHIC). SOHIC is simply linking-up of hydrogen blisters at highly stressed areas like weld HAZs. SOHIC is sometimes called stepwise cracking. These forms of wet H_2S cracking are not entirely preventable by material controls, and periodic inspection of equipment in more severe wet H_2S cracking environments is often warranted.

Equipment

Carbon steel materials used for pressure vessel construction are subject to hydrogen blistering, HIC, and SOHIC in sour service environments. Hydrogen blistering, HIC, and SOHIC may attack standard grades of CS with HAZ hardness below HRC 22 (the lower limit for sulfide stress cracking).

CLG has based our materials recommendations on our extensive experience and inspection of equipment in wet H_2S service. We have defined three severity levels of service, and corresponding materials recommendations as follows:

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- **For equipment in the most severe sour services where hydrogen blistering, HIC and SOHIC are a concern, we recommend carbon steel with 18-8 SS cladding or weld overlay per GN-15, as shown on the Materials Selection Diagram.**

- For other sour service equipment, where hydrogen blistering, HIC and SOHIC are a concern, but are somewhat less likely based on our experience, we specify

- **"per GN-9"** on the Materials Selection Diagram. **For these services, follow PVM-SU-4750, including Supplement "C" requirements. Supplement "C" includes heat treatment of all welds and cold-formed parts, the use of normalized steel, chemistry control, NACE MR0103, and special weld procedure qualification and NDE requirements. Initial and follow-up wet H2S inspection is also recommended.**

- Finally, for some equipment items, where wet H_2S cracking is possible, but not as likely, we specify **heat treatment of all welds and cold-formed parts for resistance to sulfide stress cracking (SSC) and to enhance resistance to stress oriented hydrogen-induced cracking (SOHIC). NACE MR0103 compliance, and initial and follow-up wet H2S inspection is also recommended.** These equipment items are indicated on the Materials Selection Diagram (MSD) as requiring PWHT. **For these items, follow PVM-SU-4750, including the requirements of Supplement "B". (**Note that the MSD will also indicate "PWHT" for equipment items in amine and caustic service. These require PWHT per Supplement A of PVM-SU-4750, but Supplement B requirements are not necessary.)

When specified on the Materials Selection Diagram, the vessel shall be internally coated in accordance with CLG Specification COM-SU-5014 after PWHT and WFMT are performed as specified above.

Piping

The occurrence of hydrogen blistering, HIC, and SOHIC in seamless carbon steel piping has been rare. Therefore, to date, CLG has not specified any special metallurgical requirements for piping materials to avoid these mechanisms. We do require piping base materials, welds and HAZs to be resistant to SSC, as given by the requirements in GN-8 and in NOTE E on the Notes for Materials Selection Drawings.

GN-10. AMMONIUM BISULFIDE CORROSION

Ammonia and hydrogen sulfide are present in varying amounts in the reactor effluent, depending primarily on the sulfur and nitrogen content of the feed. The NH_3 and H_2S react and form ammonium bisulfide (NH_4HS) salts, typically in the reactor effluent air cooler (REAC). Dry NH_4HS salts are not corrosive, but can create significant fouling, so typically are removed through continuous water washing of the REAC. Aqueous NH_4HS is corrosive. Corrosion rates

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are influenced by NH_4HS concentration, velocity (shear stress), H_2S partial pressure, and injection water quality. The influence of each of these factors is discussed briefly below.

Note that formation of ammonium chloride (NH_4Cl) salts can also cause fouling, typically in the effluent exchanger upstream of the REAC. Often intermittent wash water injection is necessary in this location. Like NH_4HS, dry NH_4Cl is not corrosive, but wet salts can be very corrosive.

FACTORS AFFECTING NH_4HS CORROSION RATES

NH_4HS Concentration & Use of Alloys

Other factors being equal, increasing NH_4HS concentration will result in increased corrosion. Depending on the NH_4HS concentration and H_2S partial pressure, use of alloys may be needed to control the corrosion rate to reasonable levels.

While corrosion rates of carbon steel are generally low at NH_4HS concentrations below 3 wt%, the use of Alloy 825 is usually required for systems operating in excess of this amount. In general, Alloy 825 systems should be limited to 12 wt% max., but this limit is affected by other factors, as described below. Note that Alloy 825 requires extra care in fabrication and installation. See CLG specification CPM-SU-5013 for fabrication requirements for Alloy 825.

H_2S Partial Pressure

Increasing H_2S partial pressure will result in increased corrosion rates. This influence has only recently been recognized and quantified.

For each hydroprocessing plant, process conditions are categorized into one of several severity regions, based on the maximum NH_4HS concentration and H_2S partial pressure, given on the MSD. Materials of construction and velocity limits are defined for each of the severity categories.

Velocity (Shear Stress)

Aqueous NH_4HS corrosion in REAC systems is an erosion-corrosion mechanism, with the corrosion rate strongly influenced by the shear stress between the fluid and the metal surface. In these systems, shear stress is a function of velocity. In the REAC system, CLG has occasionally experienced damage in the form of localized erosion-corrosion in areas of high velocity or turbulence (such as reducers at control valves, elbows, and tees). Another problem has been pitting in piping dead legs and tube bottoms when bulk fluid velocities in multi-phase flow lines are low. Under low flow conditions, solids accumulate and under deposit pitting occurs.

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Strict velocity ranges are necessary to control the erosion-corrosion rates and solids accumulation. The maximum velocity limits will depend on the alloy used, the service severity, and the location in the system, as discussed below. A minimum velocity limit of 10 fps (3 m/s) (for all service severities) is necessary only for the REAC tubes and REAC inlet/outlet piping to ensure even flow distribution into the REAC and to prevent solids accumulation and resultant under deposit corrosion.

For carbon steel REAC tubes and REAC inlet & outlet piping, CLG recommends that bulk fluid velocities be designed for 3 to 6 m/s (10 to 20 ft/s). It is important that the velocities in all portions of the piping be checked to ensure the maximum limit is not exceeded. Velocities of 3 to 6 m/s (10 to 20 ft/s) are acceptable; however, a design velocity of 4.5 m/s (15 ft/s) provides future flexibility to allow increases in recycle gas rate. In some very low severity systems, the velocity limits for carbon steel can be relaxed to 3-9 m/s (10-40 ft/s). See the MSD for specific recommendations.

For Alloy 825 (UNS N08825) REAC tubes and inlet/outlet piping systems, velocities of 3-12 m/s (10 to 40 ft/s) are acceptable, with a design velocity of 11m/s (35 ft/s) recommended.

Again, note that these above limits are for REAC tubes and inlet/outlet piping. These velocity limits are not necessary for the vapor stream from the CHPS or for the sour oil stream from the CHPS. They are also not intended for intermittent water wash locations.

For the sour water line from the CHPS, maximum velocities may be restricted to lower levels than those stated above, depending on the system severity and alloy used. Any such limits are given on the MSD. There are no minimum velocities required for 100% liquid lines, including the sour water lines, but deadlegs should be avoided in all streams.

Wash Water Injection Rates and Water Quality

Wash water injection quality is critical to reliable operation. Contaminants such as free cyanides, oxygen, iron and particulates can increase corrosion rates. Injection water quality must be strictly enforced to avoid corrosion.

Stripped sour water is the preferred source of wash water. The vacuum column overhead water should not be used since it likely contains oxygen; the atmospheric column overhead water will have too low a pH. The following limits for injection water quality must be strictly adhered to:

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Item	Water Quality
O_2	<15 ppb
pH	8-10
Fe (Soluble)	<1 ppm
Cl-	<100 ppm in injection water
Cyanides	Nil Free CN-
Total Dissolved Solids (TDS)	<250 ppm Desirable
Total Suspended Solids (TSS) & TDS >1.5 micron	<1000 ppmw

Of these limits, oxygen content is the most important, as its presence can greatly accelerate corrosion. Note that most refineries can readily maintain oxygen content of the injection water in the 5-15 ppbw range. However, some locations must add an oxygen scavenger into their injection water to meet this O_2 limit.

The quantity of wash water is also important. The continuous water injection rate must be enough to provide 100% vapor-phase water saturation plus at least 25% excess liquid-phase water at each injection point. The water injection rate will also determine the amount of ammonium bisulfide in the sour water at the outlet of the REAC, and so, the system corrosion severity.

When intermittent water wash is used, it is required that all the water used for continuous wash be diverted to the intermittent wash location. CLG has strict guidelines on intermittent water wash; see Process Manual for details. For intermittent wash water locations, positive shutoff (double block and bleed) of the water is critical when injection is not being used, to avoid leakage and potentially severe localized corrosion.

REAC SYSTEM PIPING CONFIGURATION

General

Beginning three diameters upstream of the wash water injection location, the piping shall match the same alloy requirements and velocity limits as for the REAC tubes and header boxes. For all materials, inlet and outlet piping of reactor effluent coolers should be completely balanced with at least 10 pipe diameters of length between each bend and branch to insure equal flow velocities and distribution of condensate to all exchanger banks. Long radius (5 diameter) elbows or bends are recommended (1.5 diameter or above are required). Piping dead legs should be avoided. See CLG drawing RC-960360 for details.

Spray nozzles shall be Alloy C-276 or Alloy 825. Full cone spray type nozzles positioned in the same direction as process flow shall be used. In addition to earlier injection rate requirements, the minimum water injection rate necessary to produce a full spray pattern must be maintained,

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as defined by the nozzle vendor specifications. This helps ensure water reaches all tubes and no corrosive, wet salts are left behind.

Special care should be taken to avoid/eliminate all dead legs from three diameters upstream of the point of wash water injection, past the lines leaving the separator. This includes the vapor, oil, and sour water lines. Such dead legs are a serious trap for build-up of highly corrosive contaminants in the system.

Single Point vs. Multi-Point Injection

In the past, CLG had recommended multi-point wash water injection systems for some plants. In theory, multi-point injection systems provide more uniform water distribution than single point injection systems. However, in practice, these systems are more difficult to install and to maintain. Therefore, we have now taken a more conservative approach and standardized on using single point injection for all plants. See P&IDs for details.

GN-11. HIGH PRESSURE HYDROGEN SERVICE

Hydrogen service is defined as any fluid that contains more than21.1 kg/cm^2(a) [300 psia] hydrogen partial pressure. CLG's requirements for hydrogen service are as follows:

CS Piping

1. Seamless pipe is required for pipe diameters up to and including 405 mm (16 in.). For pipe diameters above 405 mm (16 in.), ASTM A672 Grade C60, C65 or C70, Class 32 may be used.

2. All welds shall be 100% radiographed.

3. The MDMT shall be no greater than +0°C (32°F). Figure 323.2.2.A and Table 323.3.5 of ASME B31.3 shall be used to determine impact testing requirements.

4. All welds on piping above 19 mm (3/4 in.) thick shall be postweld heat treated. (Note this requirement applies to ALL services, not just high pressure hydrogen; see also GN-17).

Cr-Mo Steel Piping

1. Seamless pipe is required for pipe diameters up to and including 405 mm (16 in.). For pipe diameters above 405 mm (16 in.), ASTM A691 Class 32 or Class 42 may be used.

2. All welds shall be 100% radiographed.

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3. All welds, including socket and seal welds, shall be PWHT'd.

4. The MDMT shall be no greater than +0°C (32°F). Curve B of Figure 323.2.2.A and Table 323.3.5 of ASME B31.3 shall be used to determine impact testing requirements. (Note that this Figure is intended for carbon steel, but we want Cr-Mo materials to meet these impact testing requirements.)

18-8 SS Piping

1. Welded pipe and fittings are acceptable, but shall be solution annealed.

2. All welds shall be 100% radiographed.

3. Weld deposits shall be checked for ferrite content. A ferrite number (FN) of not less than 3% and not more than 10% is required. FN shall be determined by use of a Ferritescope or another equivalent device. All measurement of the ferrite number shall be performed on material in the "as welded" condition, i.e., prior to any heat treatment.

Alloy 825 Piping

1. The pipe shall conform to all requirements of CLG Specification CPM-SU-5013, "Alloy 825 for Heat Exchangers and Piping in Hydroprocessing Reactor Effluent Service."

2. All welds shall be 100% radiographed.

GN-12. STAINLESS STEEL AND HIGH ALLOY MATERIALS

1. Welded 18-8 SS parts must be the low carbon grades (304L or 316L) or the preferred stabilized grades (321 or 347). When 304L is specified, 321 or 347 may be used if more economical. Regular carbon grades (304 or 316) can be used only for non-welded parts, which will not be exposed to temperatures above 371°C (700°F). Stabilized grades (321 or 347) must be used for continuous service above 371°C (700°F) (see GN-6).

2. When 317 or 317L is specified, 316, 316Ti, 316Nb, or 316L materials with 2.5 wt% Mo minimum are acceptable alternatives. Note that standard 316 materials can be used for demisters, coalescers, or filter internals.

3. Welded 12 Cr SS parts must be Type 410S, limited to a maximum macrohardness of 225 Brinell. Non-welded 12 Cr SS parts can be Type 410.

4. For 18-8 SS heat exchanger tubing, CLG uses welded SA249 tubes provided they have been ultrasonic or eddy current tested and hydrotested.

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5. For 18-8 SS furnace tubes, CLG uses only seamless tubes.

6. For 18-8 SS piping, CLG considers welded pipe acceptable. For ANSI 150 psi flange rated piping and lower, a minimum of spot radiography is required of the longitudinal welds. For ANSI 300 psi flange rated piping and higher, a minimum of 100% radiography is required for the longitudinal welds. (We require 100% radiography of all welds in piping in the high pressure reactor loop as specified in GN-11.)

7. Ferrite content of all 18-8 SS welds should be limited to 3 to 10%. The ferrite content shall be measured with a Ferritescope or another equivalent device. All measurement of the ferrite number shall be performed on material in the "as welded" condition, i.e., prior to any heat treatment.

8. Where 18-8 SS is specified and differential thermal expansion is such that unacceptably high stresses may develop, e.g., in some high pressure closure designs, either Alloy 825 or Alloy 625 may be used provided the design temperature does not exceed 454°C (850°F). Alloy 600 can be used up to 343°C (650°F) at H_2S partial pressures up to 7.03 kg/cm^2(a) [100 psia] and up to 399°C (750°F) for H_2S partial pressures below 7.0 x 10^{-3} kg/cm^2(a) [0.1 psia].

GN-13. VALVE MATERIALS

1. The following guidelines apply to most valves. See separate specification for emergency depressuring valves.

2. In general, valve bodies should be the same material as the piping material. An example of a typical exception is for 321 piping where cast valve bodies must be 347 SS.

3. "Valve trim" refers to stems, discs, and seats.

4. 18-8 SS valve trim that will be welded (tack welding included) must be the low carbon grades (304L or 316L) or preferably stabilized grades (321 or 347). Regular SS grades (304 or 316) can only be used for non-welded parts, which will not be exposed to temperatures above 371°C (700°F). Stabilized grades (321 or 347) must be used for continuous service above 371°C (700°F) (see GN-6).

5. 18-8 SS valve trim may be substituted for 12 Cr (410 or 410S) SS trim in any service, but 12 Cr (410 or 410S or 405) may not be substituted for 18-8 SS. In addition, other ferritic or martensitic stainless steels 400 series SS are not acceptable when 18-8 SS is specified.

6. Stellite valve trim is an acceptable and preferred substitute for 18-8 SS trim in any service.

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7. When 18-8 SS valve trim is specified, there are some acceptable alternative materials. Valve manufacturers typically provide higher strength materials like precipitation hardening SS, cobalt alloys, and nickel alloys which are acceptable and in many cases, are preferred for the service. When these materials are substituted they must meet the requirements of NACE MR0103. For instance, standard materials for Rockwell Edwards Y-pattern globe valves include an ASTM A-638 Grade 660 stem and a Stellite seat and disk. Orbit Valve Company's standard materials include an ASTM A-638 Grade 660 stem and stem pin.

   It is not possible to itemize all of the acceptable trim materials for the "18-8" classification; however, the following guidelines should be followed:

   CS Valve Body

   When 18-8 SS trim is specified with a CS body, it is usually required for resistance to wet H_2S cracking (see GN-8). In general, any materials that satisfy the requirements of NACE Standard MR0103 are acceptable for this application, except that copper alloys and nickel-copper alloys should be avoided due to their poor H_2S and NH_3 corrosion resistance.

   18-8 SS Valve Body

   When 18-8 SS trim is specified with an 18-8 SS body, it is usually required for resistance to high temperature H_2-H_2S corrosion (see GN-3). In addition, it should also satisfy the requirements of NACE Standard MR0103 for sulfide stress cracking resistance during plant shutdowns. Some guidelines and commonly used materials are summarized below:

   a) Nickel alloys such as Alloy 400 and electroless nickel plating should be avoided due to poor high temperature H_2-H_2S corrosion resistance.

   b) Copper alloys should be avoided due to their poor H_2S and NH_3 corrosion resistance.

   c) Duplex stainless steel is only acceptable with written permission of CLG.

   d) Cobalt alloys with greater than 18% Cr are acceptable. (Stellite 1 and 6 are commonly used materials.)

   e) Austenitic precipitation hardening SS's with greater than 13% Cr are acceptable, but should conform to NACE MR0103 hardness limits. [ASTM A638 Grade 660 (HRC 35 max.) is a commonly used material.]

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f) Martensitic and semi austenitic precipitation hardening SS like 17-4 PH, 17-7 PH, and AM-350 are not recommended for high temperature service. These materials suffer a loss in ductility and impact toughness from "tempered martensite embrittlement" after long-term exposure in the 290°C (550°F) to 480°C (900°F) temperature range.

8. 12 Cr should not be utilized when 18-8 SS is specified. NACE MR0103 will allow its use with hardness below HRC 22, but CLG experience has found it to be an unacceptable material for these services.

GN-14. DESIGN LIFE

The corrosion allowances[2] for materials have been selected based on the following estimated life expectancies:

Equipment	Design Life[3]
Thick-Walled Reactors and Vessels	30 yrs but not less than 3 mm (1/8 in.) CA. For clad or weld overlays see GN-15.
Columns, Vessels, and Exchanger Shells of Carbon and Alloy Steel Construction (Including Non-Removable Internals)	20 yrs but not less than 3 mm (1/8 in.) CA unless clad or weld overlay, see GN-15.
Furnace Tubes	100,000 hours (11.4 yrs), per API 530. Corrosion allowance for 11.4 yrs, but not less than 3 mm (1/8 in.) CA for carbon steel through C-1/2 Mo, 2.5 mm (0.10 in.) CA for low alloy steels, through 5 Cr-1/2 Mo, and not less than 1.3 mm (0.05 in.) CA for 7 Cr-1/2 Mo through SS.

[2] All corrosion allowances are total, i.e., the corrosion allowance specified takes into account the expected corrosion whether it occurs from one side or both sides.
[3] Design life is an estimate, not a guarantee.

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CATALYTIC DEWAXING UNIT
MANALI, CHENNAI, INDIA

Equipment	Design Life[3]
Process Piping, Removable Vessel Internals, and Minor Equipment	12 yrs but not less than 1.5 mm (1/16 in.) CA for carbon steel. CA for 15 yrs design life for alloy and stainless steels, but not less than 1/16 in. (1.5 mm) CA for alloy steels and not less than 1 mm (0.04 in.) CA for SS.
Heat Exchanger Bundles[4] of SS Construction	12 yrs design life, except high pressure feed/effluent exchangers, 20 years. Not less than 14 ga. thickness.
Heat Exchanger Bundles[4] of CS and Alloy Steel Construction	12 yrs but not less than 14 ga. for 19 mm (3/4 in.) CS tubes, not less than 12 ga. for 25 mm (1 in.) CS tubes, and not less than 14 ga. for alloy steel tubes.

GN-15. CLADDING AND WELD OVERLAY

Cladding and weld overlay thickness requirements are based on the specified design life from GN-14 using the anticipated corrosion rate for the cladding or weld overlay material. Corrosion allowance is not typically required for the backing material, and the cladding or weld overlay thickness should not be used when calculating equipment design stresses. CLG's practice is to specify a minimum cladding/weld overlay thickness of 3 mm (1/8 in.). In the case of weld overlay, the deposited weld metal must meet the required chemistry limits of the specified overlay material for a minimum of this depth.

Linings that are not fully bonded to the backing material are unacceptable, except in special situations where fully bonded cladding is impractical (such as wear plates). All joints in clad plate should be weld overlaid with suitable filler metal, considering weld dilution, such that the final weld metal deposit has similar alloy content to the cladding. For joints in Type 316L, 317L, 321 or 347 SS cladding, CLG's practice is to weld one pass with Type 309L (309MoL for 317L) SS filler metal. The final pass is then made with consumables that will produce a minimum of 3 mm (1/8 in.) of material meeting the chemistry requirements of the cladding or weld overlay.

[4] Exchanger tubes are specified by gauge. The selected gauge is based on the wall required for pressure and a service-specific corrosion allowance. If the heat exchanger supplier proposes a design revision, they shall ensure the tube gauge is sufficient for mechanical design plus a corrosion allowance of at least 0.02" (20 mils) or the tube corrosion allowance given on the datasheet, whichever is greater. The revised design shall be provided to CLG for review.

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Clad Tubesheets

When solid alloy tubesheets are specified, clad or weld overlaid tubesheets may be substituted if they meet the following requirements:

- The tubesheet base metal must be the higher alloy of the shell and channel base metal.
- Weld overlay shall have sufficient thickness to provide the required chemical composition for a depth of at least 3 mm (1/8 in.).
- For the side(s) requiring cladding or weld overlay, the minimum clad/weld overlay thickness measured at the bottom of the pass partition groove shall be as follows:
    - The minimum cladding and/or weld overlay thickness on the tube side face of a tubesheet shall be 10 mm (3/8 in.) when tubes are expanded only, and 5 mm (3/16 in.) when tubes are welded to the tubesheet.
    - The minimum cladding and/or weld overlay thickness on the shell side face of a tubesheet shall be 13 mm (1/2 in.).
- For the bare side of the tubesheet, the corrosion allowance shall be at least half of that of the corresponding side of the heat exchanger (shell or channel). The minimum corrosion allowance for carbon steel and alloy steel is 3 mm (1/8 in.). For stainless and high alloys, the minimum is 1 mm (0.04 in.).

When solid 321 stainless steel tubesheets are specified, and weld overlaid tubesheets are substituted, the weld overlay shall be 347SS rather than 321. When weld overlaid tubesheets are to be substituted for solid 825 tubesheets, use 625 weld overlay, rather than 825.

Austenitic Stainless Steel Clad Plate

Austenitic stainless steel clad plate shall be examined to verify that it is not sensitized. A minimum of four locations shall be examined in accordance with ASTM A262 Practice A by in-situ metallography. These locations shall represent different clad plates unless the maximum number of clad plates in the vessel is less than four. In this case at least one test shall be performed on each plate. Clad plates that fail to pass these Practice A criteria shall be rejected: The “step” or “dual” microstructures are acceptable; the presence of the “ditch” microstructure indicates sensitization and intergranular corrosion, and is therefore cause for rejection.

Suppliers of Clad Plate

If clad plate is selected, the vessel fabricator should obtain written approval of the clad plate supplier from the user. The supplier shall demonstrate that the cladding is fully bonded to the backing material by ultrasonic examination in accordance with CLG Specification PVM-SU-1322.

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GN-16. FABRICATION OF CHROME-MOLY STEELS

For fabrication of heavy wall reactors and pressure vessels, refer to the vessel specifications.

For other Cr-Mo steel equipment and piping, CLG specifies that a minimum preheat ranging from 150-232°C (300-450°F), depending on the alloy, must be maintained during welding. We also specify that these materials must be fully PWHT'd following fabrication. In general, PWHT of 1-1/4 Cr-1/2 Mo and 2-1/4 Cr-1 Mo vessel and piping welds should be performed at ~660-720°C (1225-1325°F), 1 hour per 25 mm (1 inch) of thickness, 2 hours minimum. For 9 Cr-1 Mo the PWHT temperature is ~720-760°C (1325-1400 °F) for 2 hours minimum. Note that these are general requirements. **See the appropriate CLG specification for specific preheat and PWHT requirements for Cr-Mo steels.**

For 2% Cr and higher alloy steels, CLG further specifies that the preheat temperature must be maintained continuously until all welding is completed, and until all welds are PWHT'd. (This is to prevent room temperature cracking of non-tempered material from hydrogen that is absorbed into the material during welding.)

CLG recommends careful selection of filler metals for Cr-Mo materials to achieve reliable welds. We specify that the weld metal chemistry must be within the limits specified for the wrought material (except that it may be lower in carbon content). **CLG requires verification of the chemistry of production welds by weld metal sampling.** We further specify that the strength of the weld deposit must be in reasonably good balance with the strength of the base metal. Many of these welds can develop considerably higher strength than the base metal.

Completed welds (after PWHT) in Cr-Mo materials must not exceed a macro-hardness of 215 Brinell. This hardness limit may be relaxed to **225 Brinell** by CLG for welds on higher strength ASME Code Class 2 materials to allow for future heat treatments without tempering the material below the minimum required tensile strength. Note for 9 Cr-1 Mo piping, the weld hardness shall not exceed **225 Brinell**.

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GN-17. POSTWELD HEAT TREATMENT GUIDELINES

CLG's practice for PWHT of piping and equipment are as follows:

1. PWHT is shown on the MSD (materials selection diagram) only where required for mitigation of environmentally-related stress cracking, namely wet H_2S cracking, amine stress corrosion cracking, or caustic stress corrosion cracking. PWHT that is required by applicable codes is not shown on the MSD diagram. Equipment may require PWHT for the additional reasons given below.

2. All welds on carbon steel piping above 19 mm (¾ in.) thick shall be post-weld heat treated. Note that Chevron has maintained the more stringent requirements of ASME B31.3 2012 (has NOT adopted the 2014 version requirements). This is to minimize the possibility of brittle fracture in thick sections. This is especially important in high pressure hydrogen service, as hydrogen can affect crack growth and brittle fracture.

3. All Cr-Mo welds shallbe PWHT'd as specified in GN-16.

4. All 18-8 SS cold bends shall be thermally stress relieved as specified in GN-7.

5. All U-tube exchanger bundles and cold-formed pressure heads and transition sections constructed from carbon steel,[5] low alloy steels,[5] austenitic stainless steel,[6] Alloy 400 and copper alloys, shall be given a stress relieving heat treatment regardless of process fluid or operating temperature. Tubes made of titanium, high nickel alloys, ferritic and duplex stainless steel shall be stress relieved only when specified by the Client. The exception is Alloy 825 U-tube exchanger bundles, which shall be stress relieved at 940°C ± 14°C (1725°F ± 25°F)

6. All welds of CS vessels and major equipment for MEA, DEA, MDEA, and DIPA service shallbe PWHT'd and cold formed parts should be heat treated under the same PWHT conditions, regardless of operating temperature. CLG does not recommend PWHT for longer times at lower PWHT temperatures as permitted by some codes. All CS piping welds in amine service should also be PWHT'd, regardless of operating temperature.

---

[5] Rapid local electric resistance heat treatment of U-bends is permitted provided carbon steel is heated to 650-704°C (1200-1300°F). Cr-Mo steels shall be heated to a maximum of 100°F above the minimum tempering temperature specified in SA-213 for the specific grade of material. The minimum heating temperature for Cr-Mo material shall the minimum tempering temperature specified in SA-213 for the specific grade of material.

[6] Rapid local heat treatment of austenitic stainless steel U-bends is prohibited except for Types 304L, 316L, 321, and 347 SS. Local electric resistance heat treatment of Types 304L, 316L, 321 and 347 SS shall be at 1038°C ± 15°C (1900°F ± 25°F) followed by forced air cooling.

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GN-18. TEMPER EMBRITTLEMENT

Long-term exposures in the 345-595°C (650-1100°F) temperature range can result in temper embrittlement of 2-1/4 Cr-1 Mo, 2-1/4 Cr-1 Mo-1/4 V, and 3 Cr-1 Mo-1/4 V steels. Temper embrittlement increases the temperature at which the steel becomes susceptible to brittle fracture (in comparison to unexposed material). In addition, residual hydrogen in the walls after service increases the temperature where brittle fracture becomes a concern.

CLG requires the vessel fabricator to set a minimum pressurizing temperature (MPT) for each vessel to take into account the code minimum design metal temperature (MDMT) plus the service effects mentioned above. CLG also has limits on the coldest MDMT and MPT values that we deem acceptable, as defined in PVM-SU-5004. When determining the MPT, the fabricator estimates the effect of temper embrittlement based on the results of the CLG required step-cool screening tests of plate, forgings, and weld metal. The effect of hydrogen in the steel is usually estimated from the work of Sakai[7] or Pillot et al.[8]

At temperatures below the MDMT for new vessels or below the MPT for vessels that have been in service, the pressure must not exceed 22.5% of the design pressure for 2-1/4 Cr-1 Mo and 2-1/4 Cr-1 Mo-1/4 V equipment built per Div. 2 (ASME Code, Sect. VIII, Div. 2), or 30% for equipment built per Div. 1 (ASME Code, Sect. VIII, Div. 1).

GN-19. HYDROGEN OUTGASSING

When a reactor is in service a small amount of hydrogen continually diffuses from the inside of the reactor through the wall to the atmosphere. The equilibrium concentration that the hydrogen in the wall reaches depends on the temperature and the hydrogen partial pressure. The higher the temperature and the hydrogen partial pressure, the higher the equilibrium concentration of hydrogen.

During hydroprocessing reactor cool-down, the solubility of hydrogen in the reactor steel decreases. In addition, the hydrogen diffusion rate in the reactor steel is a function of temperature. As the temperature drops, the rate of hydrogen diffusion decreases. Therefore, a rapid cool-down can result in a **significant** concentration of hydrogen remaining entrapped in the reactor wall.

It is desirable to minimize the amount of hydrogen remaining in the reactor wall subsequent to the unit shutdown. The entrapped hydrogen can cause existing fabrication defects to grow. It can also cause cracking and separation (disbonding) of stainless steel overlay from the reactor base material.

[7] T. Sakai et al., PVP Vol. 1, High Pressure Technology, p. 85.
[8] S. Pillot et al., Corrosion 2008, Paper 08559.

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The industry experience indicates that the potential for hydrogen-induced damage increases as the tensile strength of the reactor material increases. For this reason, modern reactor specifications, including the specifications for your reactors, have limitations on the maximum tensile strength.

It is not practical to eliminate the residual hydrogen in the reactor wall after shutdown. However, there are two common methods of reducing the concentration of the entrapped hydrogen.

One method is to reduce or eliminate hydrogen from the reactor feed while maintaining a constant and a relatively high temperature. This method is commonly known as isothermal outgassing. The hold time for isothermal outgassing varies depending on the desired reduction in hydrogen concentration. Isothermal outgassing is time consuming and can significantly add to the reactor shutdown cycle.

A less rigorous method, predominately used in the industry, is to control the cooling rate during shutdown. This allows a significant amount of hydrogen to diffuse out of the reactor wall while the reactor is gradually being cooled. For this reason, CLG has specified a strict cooling rate requirement that should be observed during all normal shutdown operations.

GN-20. PIPING MATERIAL BREAKS

Piping material breaks are shown schematically for main piping lines on the Materials of Construction diagrams. For critical material breaks the maximum process temperature which the lower alloy material should experience is indicated as “NOT TO EXCEED (temp limit) °F.”

GN-21. DISSIMILAR WELDS

Many combinations of dissimilar metals can be joined with high reliability. A materials engineer should be consulted with regard the suitability for service. The suitability of a dissimilar weld depends on many variables such as the compatibility of the filler metal with the base metals, the mechanical properties, thermal fatigue considerations, and corrosion resistance. A complete discussion of the effect of all mechanical and environmental variables on the suitability of dissimilar welds is beyond the scope of this document. However, it is important that users be aware that, whenever possible, dissimilar welds between stainless steel and carbon or low alloy steels should be avoided in wet H_2S and other services where hydrogen embrittlement is possible. This is because of a hard zone that occurs at the fusion line of austenitic weld metal and the carbon or low alloy steel base metal, and potential cracking that can result in this area.

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This hard zone cannot be completely removed with PWHT. For further information on dissimilar welds see Barry Messer et al.[9]

GN-22. PROTECTION OF STAINLESS STEEL FROM THE ATMOSPHERE

Although stainless steels have excellent corrosion resistance to many environments, rapid attack can occur when chlorides are present. For this reason, CLG has specified limits for chlorides in the feed and the test water. However, attack also can occur from moisture and chlorides in the atmosphere. There are numerous reports in the literature of both internal and external pitting and stress corrosion cracking occurring in marine and other chloride containing atmospheres.

Some of the typical measures that have been taken to protect against atmospheric chloride attack are:

1. Externally coat stainless steel pipe which operates in the range of 60-175ºC (140-350ºF)

2. Maintain a dry air or dry nitrogen pressure in reactors and other stainless equipment until just prior to operation.

3. Whenever possible, store stainless steel pipe inside in a dry environment.

4. For outside storage of stainless steel (and other) pipe, keep the material covered, off the ground and place it so any moisture that collects both inside and outside will drain. Note that experience has shown that end plugs breathe and therefore do not protect pipe from internal corrosion.

5. Periodically inspect stored material to insure pockets of moisture, etc., do not exist.

6. Keep the temperature in fired heaters above the dew point during shutdowns.

CLG recommends that their licensees discuss protection of stainless steel equipment with their general contractor and that a mutually agreed program be established for protection of stainless steel equipment during construction and startup.

GN-23. FIRED HEATER STEAM TUBES

Fired heaters often have steam tubes in the convection section. The selection of materials for these tubes normally is outside the scope of CLG's design package. However, CLG has experienced a number of failures of these steam coils in their own units. Therefore, we offer the following minimum guidelines that CLG has developed.

[9] Barry Messer et al., "Selection of Dissimilar Metal Welds in Severe Environments for Today's Petrochemical Plants," Corrosion 2007, Paper 07568.

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Material	Maximum Design Metal Temperature, °C (°F)
Carbon Steel	482 (900)
C-1/2 Mo	510 (950)
1-1/4 Cr-1/2 Mo	593 (1100)
2-1/4 Cr-1 Mo	621 (1150)
18 Cr-8 Ni*	816 (1500)
Alloy 800	816 (1500)

* There have been numerous cases of stress corrosion cracking of 18 Cr-8 Ni steam tubes reported due to solids carry over in the steam. Therefore, 18 Cr-8 Ni steam tubes should be avoided if possible.

GN-24. THERMAL FATIGUE

Thermal fatigue has occurred at hydrogen injection points and other mix points throughout the industry, including at CLG licensee plants. Mix points can be areas of high delta temperature and resultant thermal stresses, potentially leading to thermal fatigue.

For quench hydrogen and other process mix points where the temperature difference (delta T) between the two streams is 139°C (250°F) or greater, CLG recommends using a thermal sleeve design to reduce thermal stresses and prevent thermal fatigue. If applicable to the project, CLG provides a set of thermal sleeve drawings for such designs. The thermal sleeve drawings are based on typical 321 SS materials for inter-reactor quench locations. The detailed design contractor (DDC) will need to confirm the materials and design for the specific project.
For other mix points, such as the wash water injection ahead of the REAC, CLG does not provide a specific design, but rather a concept drawing of the water injection nozzle site (RA-961161). However, if the delta T is above 139°C (250°F) in this location, the DDC should consider the potential for thermal fatigue in this location, and design to avoid thermal stresses. CLG also provides a concept drawing for intermittent water wash locations, and, again, if the delta T in these locations exceeds 139°C (250°F), the DDC should consider thermal fatigue in the design.

Thermal fatigue is dependent on the frequency and extent of thermal cycling. The overall temperature difference and flow rate variation between the streams at the mix point affects the likelihood and severity of thermal fatigue. Although there is some debate in the industry regarding the most appropriate threshold, CLG recommends considering thermal fatigue in areas where the delta T is greater than about 139°C (250°F).
As discussed in the Process Manual, all mix points should be subject to periodic inspection to monitor for thermal fatigue. See also API RP 574 and NACE SP0114 for further industry guidance on mix points.

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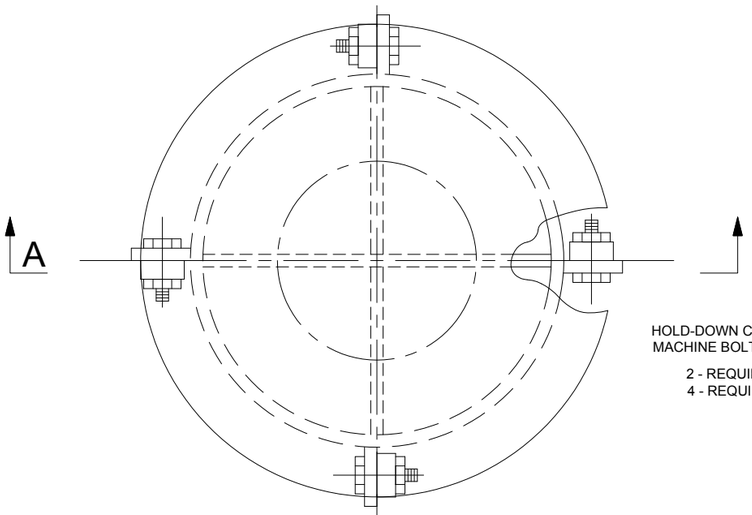
NOTES

1. FABRICATE STRAINER CYLINDER OF 1/4" STEEL PLATE OR SEAMLESS PIPE, MINIMUM THICKNESS 1/4". ENTIRE SURFACE OF CYLINDER SHALL BE PERFORATED WITH 3/8" DIAMETER HOLES AT 1/2" CENTERS ON TRIANGULAR PITCH. OMIT HOLES IN TOP COVER. USE 1/4" STEEL PLATE FOR CROSS MEMBERS AND TOP COVER. WELD CYLINDER AND TOP COVER TO CROSS MEMBERS.
2. MATERIALS: SEE APPLICABLE VESSEL DRAWING FOR MATERIAL REQUIREMENTS.
3. TWO CROSS MEMBERS AS SHOWN FOR ALL MARK NO'S, BUT FOR MARK NO'S 1 THRU 3 TERMINATE ONE CROSS MEMBER AT BOTTOM OF DIMENSION "B".
4. NOTE THAT FOR NOZZLES 10" AND LARGER, THESE STRAINERS MAY NOT FIT THROUGH A MANWAY AND MUST BE INSTALLED BEFORE CLOSING UP THE VESSEL, OR FABRICATED IN TWO PARTS AND WELDED TOGETHER INSIDE THE VESSEL.

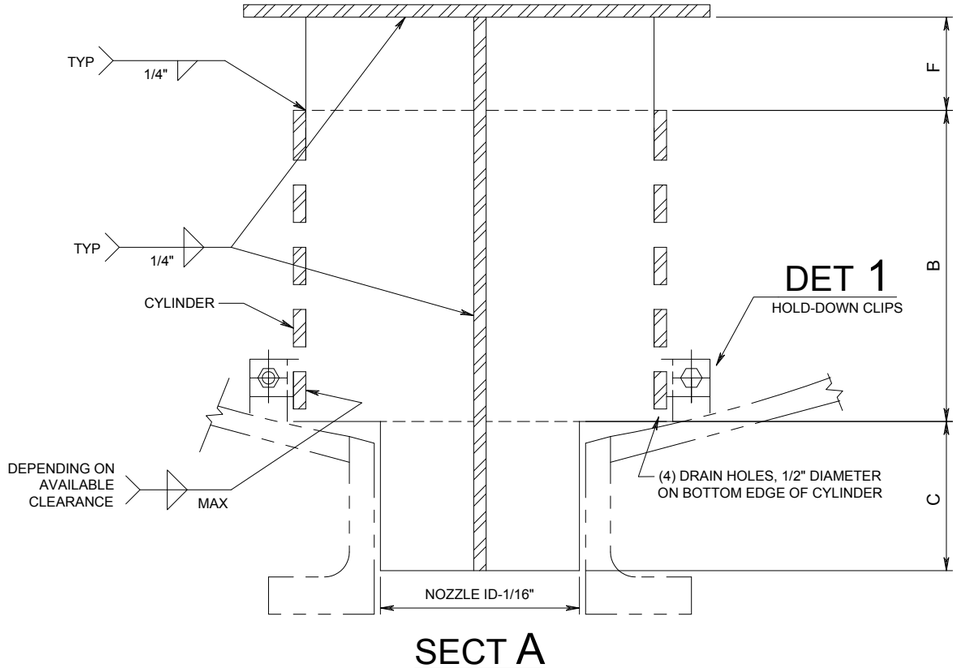


PLAN

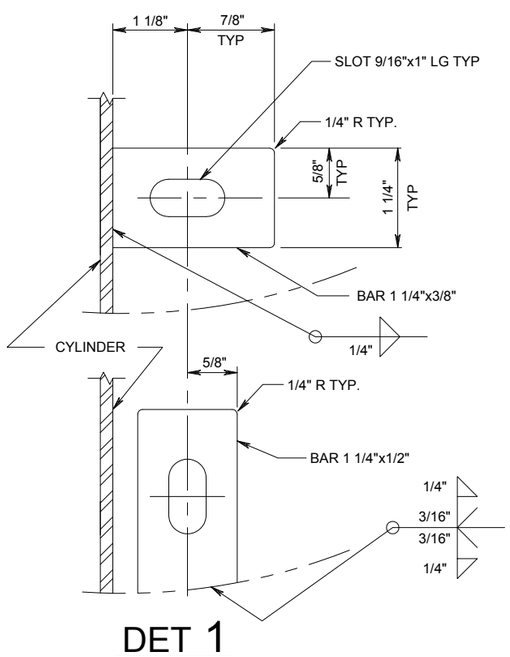
HOLD-DOWN CLIPS WITH 1/2" DIA x 1 1/2" LG CS MACHINE BOLT WITH DOUBLE HEX NUT.

2 - REQUIRED FOR MARK NO. 1 THRU 6
4 - REQUIRED FOR MARK NO. 8 THRU 20

MARK NO	1	1 1/2	2	3	4	6	8	10	12	14	16	18	20
NOZZLE SIZE	1"	1 1/2"	2"	3"	4"	6"	8"	10"	12"	14"	16"	18"	20"
A	3"	3"	4"	5"	6"	8"	12"	14"	16"	18"	20"	24"	27"
B	3"	3"	4"	5"	6"	6"	6"	8"	10"	11"	14"	14"	16"
C	4"	4"	4"	4"	4"	4"	5"	5"	5"	6"	6"	6"	6"
E	4"	4"	5"	6"	7 1/2"	10"	14 1/2"	17 1/2"	20"	23"	26"	30"	33"
F	3/4"	3/4"	1"	1 1/4"	1 1/2"	2"	3"	3 1/2"	4"	4"	5"	6"	7"



SECT A



DET 1

REFERENCES	REVISIONS			
	12 CONVERTED TO CAD CHANGED CHEVRON DECAL	12-31-96	TAV	
	13 GENERAL REVISION	7/1/16	EKQN	JTEF

Chevron

SCALE NONE DATE 7-7-55 APPROVED
DR. JJM CH. LIL DR. APP. ENGR. JM
OPR'G DEPT. ENG. DEPT.

STANDARD DETAIL OF STRAINER
FOR BOTTOM OUTLET NOZZLES

C | C.C. | S.O. | GC-C31554 | 13

STANDARD DRAWING

7/1/16

GC-C31554-13.DWG

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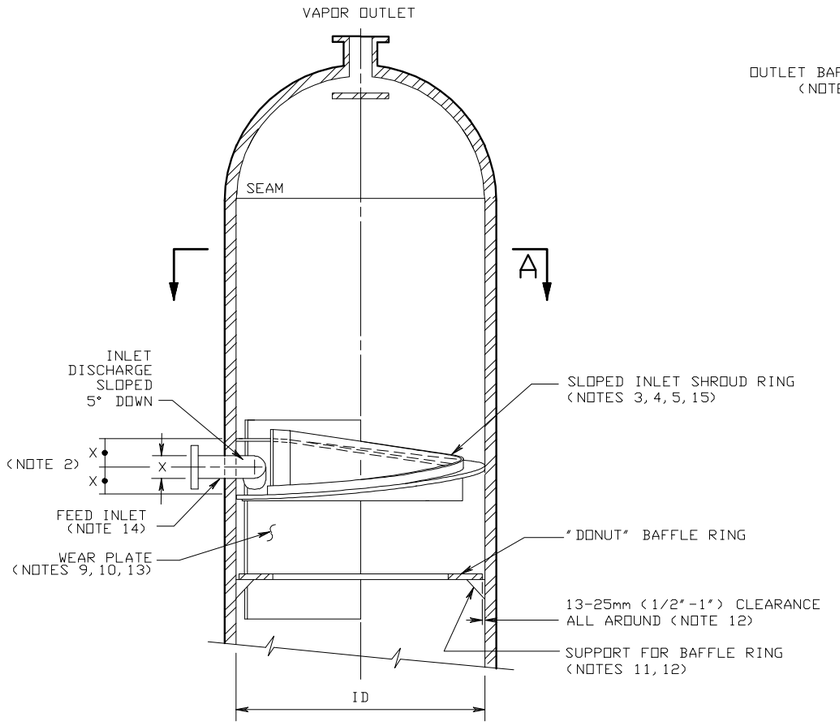
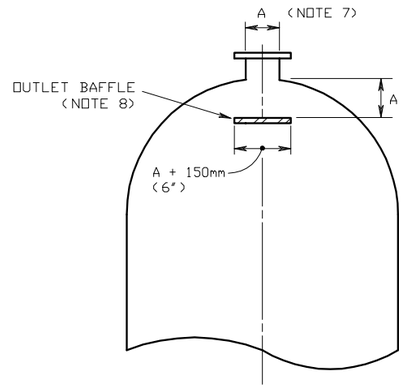
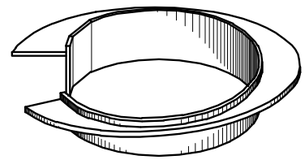
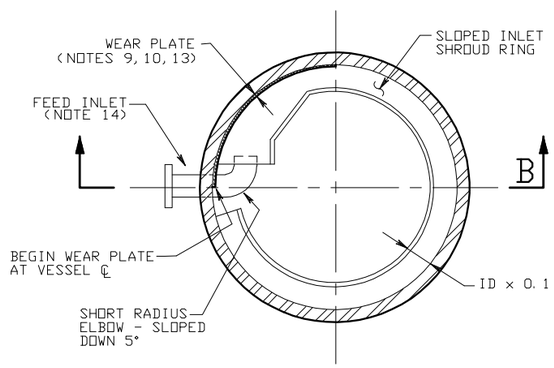




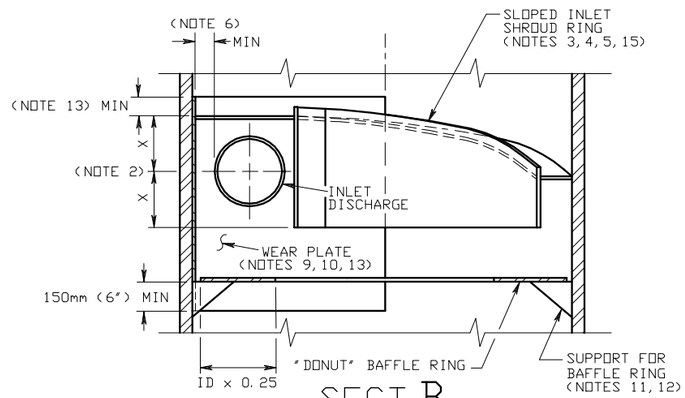
STANDARD VAPOR OUTLET NOZZLE



ISOMETRIC VIEW
SLOPED INLET SHROUD RING



SECT A



SECT B

NOTES

1. EXTERNAL PIPING TO THE FEED INLET SHOULD USE LONG RADIUS ELBOWS.
2. DIMENSION "X" IS DEFINED AS THE INLET PIPE ID. THE TOP OF THE SHROUD RING IS "X" ABOVE THE CENTER LINE OF THE INLET PIPE AND THE BOTTOM OF THE SHROUD RING EXTENDS "X" BELOW THE CENTER LINE OF THE INLET PIPE. THUS, THE TOTAL HEIGHT OF THE SHROUD RING AT THE INLET IS 2X.
3. SUPPORT GUSSETS (IF ANY) FOR INLET SHROUD RING SHOULD BE PLACED ABOVE SHROUD RING TO AVOID INTERFERENCE WITH INLET SWIRL.
4. SLOPED INLET SHROUD RING INCLUDED TO ELIMINATE TURBULENCE WHICH WOULD RESULT FROM SWIRLING LIQUID IMPACTING WITH THE BACK OF THE FEED INLET PIPE.
5. THE BASIC WIDTH OF THE SHROUD RING IS 0.1 X ID. SHOULD THE INLET PIPE BE TOO LARGE TO FIT INTO THE SHROUD RING, THE SHROUD RING ENTRANCE MAY BE ENLARGED THEN TAPER DOWN TO THE DESIRED WIDTH AS SHOWN.
6. IT IS VERY IMPORTANT THAT THE INLET PIPE ELBOW TERMINATE AS CLOSE AS POSSIBLE TO THE SEPARATOR WALL.
7. DIMENSION "A" INDICATES VAPOR OUTLET NOZZLE INSIDE DIAMETER. ALL OTHER DIMENSIONS GIVEN IN TERMS OF "A".
8. OUTLET BAFFLE REQUIRED FOR STANDARD VAPOR OUTLET. BAFFLE IS CIRCULAR PLATE OF "A" + 150mm (6") DIAMETER LOCATED "A" BELOW VAPOR OUTLET. OUTLET BAFFLE IS NOT REQUIRED FOR VESSELS WITH DEMISTER PAD.
9. UNLESS OTHERWISE SPECIFIED, THE WEAR PLATE SHALL BE TYPE 321 OR 347 STEEL. TYPE 304L MAY ALSO BE USED IF WEAR PLATE IS TO BE INSTALLED AFTER POSTWELD HEAT TREATMENT OF SEPARATOR.
10. WEAR PLATE TO COVER MINIMUM 90° OF VESSEL CIRCUMFERENCE.
11. "DONUT" BAFFLE RING SUPPORT WITHIN SHELL AREA COVERED BY WEAR PLATE SHALL BE OF SAME MATERIAL AS WEAR PLATE.
12. ENSURE THAT SUPPORTS FOR BAFFLE RING DO NOT SIGNIFICANTLY BLOCK CLEARANCE.
13. WEAR PLATE EXTENDS JUST ENOUGH ABOVE SHROUD RING TO AVOID WELD INTERFERENCE.
14. ON THREE-PHASE SEPARATORS, THE FEED NOZZLE AND OTHER INLET PIPING MUST BE CLAD WITH 347 STAINLESS STEEL WHERE THE FLUID VELOCITY EXCEEDS 6 m/s (20 ft/s). ALL STAINLESS STEEL PIPING OF TYPE 304L, 316L, 321 OR 347 MAY BE USED INSTEAD OF CLADDING.
15. INNER SECTION OF SLOPED INLET SHROUD RING EXTENDS APPROXIMATELY 50mm (2") ABOVE THE SHROUD RING RAMP.

REFERENCE DRAWINGS

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4	REVISED NOTE 9	8-17-99	DJT	LIL

Chevron Lummus Global

SCALE NONE DATE 3-10-94
DR. JST CH. DGL/DNB DR. APP. ENGR. JLR

APPROVED DGL/DNB

INLET AND VAPOR OUTLET DETAILS
FOR VERTICAL SEPARATOR WITH SIDE ENTRY

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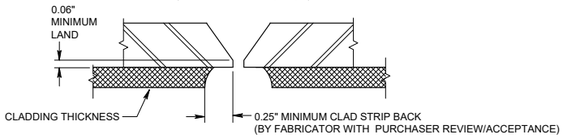
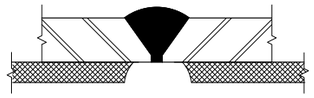
WELDING DETAILS

PROCEDURE A
(SEE NOTE D4)

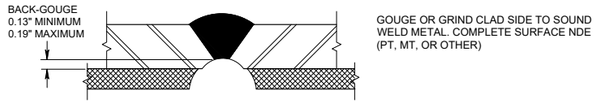
A1. WELD PREPARATION
(DETAILS BY FABRICATOR)



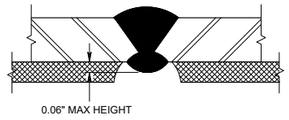
A2. ROOT AND BASE METAL WELDS



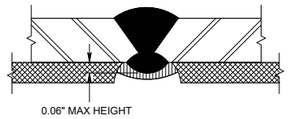
A3. BASE METAL BACK-GOUGE



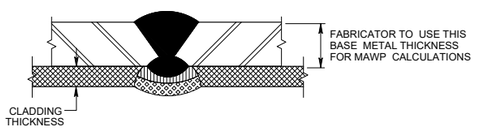
A4. BASE METAL BACK-WELD



A5. 1st PASS - DILUTED OVERLAY WELD

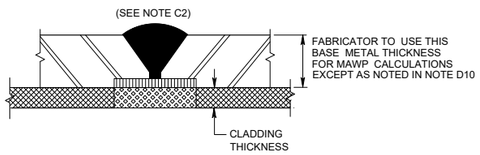


A6a. 2nd PASS MIN. OVERLAY WELD



FINISH WELDING CLAD SIDE USING APPROPRIATE ALLOY WELD PROCEDURE. AT LEAST TWO PASSES ARE REQUIRED TO MINIMIZE DILUTION OF ALLOY AT THE SURFACE EXPOSED TD THE PROCESS.

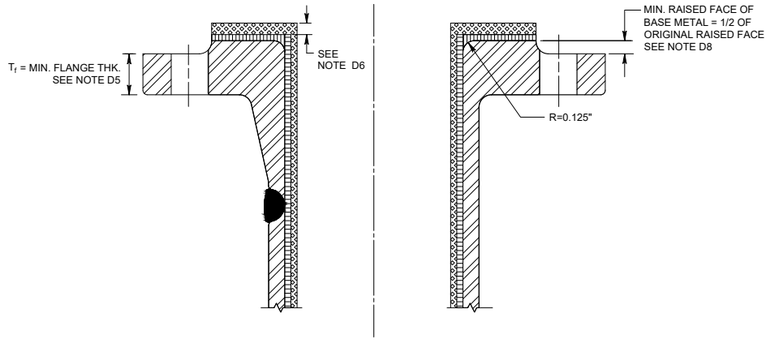
A6b. ALTERNATIVE FINAL WELD RESTORATION DETAIL*



* IF GRINDING FLUSH IS REQUIRED ON THE ID AND UNDILUTED CLADDING CHEMISTRY IS REQUIRED FOR SPECIFIED CLADDING THICKNESS.

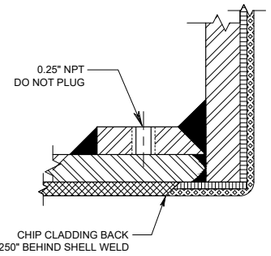
NOZZLE DETAILS

B1. NOZZLE FLANGE DETAILS

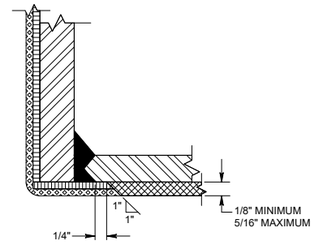


B2. NOZZLE CONNECTION TO SHELL DETAILS
(SEE NOTE G5)

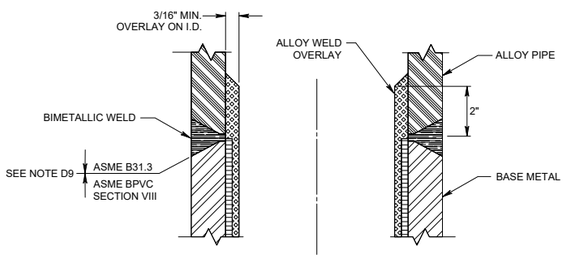
B2a. NOZZLE WITH REINFORCEMENT



B2b. SELF-REINFORCED NOZZLE



B3. NOZZLE ATTACHMENT TO ALLOY PIPING DETAILS
(SEE NOTE D7)



GENERAL NOTES

G1. THIS DRAWING HAS BEEN PREPARED TO DEFINE CERTAIN REQUIREMENTS FOR NOZZLES AND MANIFOLDS AND MANHOLES ON VESSELS PER PVM-SU-1322 WHICH EXCEEDS THE REQUIREMENTS OF THE LATEST EDITION OF THE ASME BOILER AND PRESSURE VESSEL CODE, SECTION VIII.

G2. USE OF SLIP-ON OR SOCKET WELDED FLANGES IS PROHIBITED.

G3. USE OF LINERS IN NOZZLES IS PROHIBITED.

G4. USE OF TONGUE AND GROOVE FLANGES IS PROHIBITED.

G5. REFERENCE DRAWING [PVM] GF-C14311 FOR SCHEDULE OF NOZZLE DIMENSIONS AND NON-CLAD WELDING DETAILS.

DESIGN NOTES

D1. THESE DETAILS ARE DESIGNED FOR CODE PRESSURE VESSELS OF CARBON AND LOW ALLOY STEEL WITH ALLOY CLADDING. ALL ADDITIONAL REQUIREMENTS OF PVM-SU-1322 SHALL APPLY.

D2. MINIMUM DIAMETER OF CLAD VESSELS IS 24 INCHES. AS FULL ACCESS SHALL BE AVAILABLE FOR WELDING, AT LEAST ONE MANHOLE IS REQUIRED.

D3. FINISH ON GASKET FACE OF ALL RAISED FACE FLANGES SHALL BE COMMERCIALLY SMOOTH WITH 125 - 250 RMS ROUGHNESS PRODUCED BY A ROUND-NOSE TOOL IN A CONCENTRIC OR PHONOGRAPHIC PATTERN TO BE JUDGED BY VISUAL COMPARISON WITH COMMERCIAL ROUGHNESS STANDARDS. FOR RING JOINT, FINISH PER ASME B16.20.

D4. PROCEDURE "A" MAY BE MODIFIED FOR PLATES WITH $t \geq 0.75$ INCH TO INCLUDE BEVELING ON BOTH SIDES, BUT OVERLAY STEPS ARE REQUIRED.

D5. FLANGE MINIMUM THICKNESS, T_f, PER ASME B16.5, B16.47, OR SECTION VIII. DO NOT MACHINE PAST T_f, AS SHOWN.

D6. MINIMUM THICKNESS OF ALLOY DEPOSIT = CLADDING/ LINER THICKNESS + 0.06 INCH (FOR FUTURE MACHINING).

D7. DETAIL B3 IS AN OPTION BEST EMPLOYED IN SITUATIONS WHERE A FLANGE CANNOT BE IMMEDIATELY ATTACHED TO A NOZZLE STUB, SUCH AS INSIDE OF A VESSEL SKIRT. EXTENT OF INTERMEDIATE ALLOY MAY NEED TO BE ADJUSTED DEPENDING ON SPECIFIC ALLOY COMBINATIONS USED.

D8. RAISED FACE MAY BE ENTIRELY COMPOSED OF OVERLAY MATERIAL.

D9. CODE BOUNDARY SCOPE BREAK SHALL BE SHOWN CLEARLY ON DRAWINGS AND NOTED ON MANUFACTURER'S DATA REPORT. WELD DEPOSIT IS IN SCOPE OF ASME B31.3

D10. WHEN DETAIL A6b IS USED AND THE CLAD RESTORATION ALLOY AND INTERMEDIATE ALLOY ARE IN A SERVICE WHERE THERE IS A VULNERABILITY TO CRACKING, ONLY THE BASE METAL THICKNESS IS USED FOR CALCULATIONS. THE INTERMEDIATE ALLOY THICKNESS IS EXCLUDED.

CONSTRUCTION NOTES

C1. WELDING DETAILS ARE NOT INTENDED TO PROVIDE A COMPLETE WELDING PROCEDURE. ADDITIONAL DETAILS OF PROCEDURE AND EVIDENCE OF QUALIFICATION OF THE PROCEDURE IN ACCORDANCE WITH SECTION IX OF THE ASME BOILER AND PRESSURE VESSEL CODE SHALL BE FURNISHED BY THE VESSEL FABRICATOR.

C2. WHEN DETAIL A6B IS USED, THE INTERMEDIATE ALLOY WELD METAL MUST HAVE A MINIMUM TENSILE STRENGTH EQUAL TO OR GREATER THAN THAT OF THE BASE METAL.

C3. DEBURR ALL SHARP EDGES AFTER FINAL MACHINING.

LEGEND

- BASE METAL
- BIMETALLIC WELD
- SOLID ALLOY
- BASE METAL WELD
- ALLOY CLAD
- INTERMEDIATE ALLOY WELD
- ALLOY WELD

REFERENCES

PRESSURE VESSEL NOZZLES	[PVM] GF-C14311	RULES FOR CONSTRUCTION OF PRESSURE VESSELS	ASME BPVC SECTION VIII
METALLIC GASKETS FOR PIPE FLANGES	ASME B16.20		
PRESSURE VESSELS WITH HIGH-ALLOY CLADDING OR OVERLAY	PVM-SU-1322	WELDING, BRAZING, AND FUZING PROCEDURES; WELDERS; BRAZERS; AND WELDING, BRAZING AND FUSING OPERATORS	ASME BPVC SECTION IX
PIPE FLANGES AND FLANGED FITTINGS NPS 1/2 THROUGH NPS 24 METRIC/INCH STANDARD	ASME B16.5		
LARGE DIAMETER STEEL FLANGES NPS 26 THROUGH NPS 60 METRIC/INCH STANDARD	ASME B16.47	PROCESS PIPING	ASME B31.3

REVISIONS

15	GENERAL REVISION	09-11-11	BJR	RSM	BOSD	C. Padour
16	CHANGES TO "PROCEDURE A" WELDING DETAILS. MINOR LEGEND AND DESIGN NOTES CHANGES	01-02-13	AQRC	WDMA		R. Darby
17	MAJOR REVISION	22 DEC 2016				J. Taagepera
18	GENERAL REVISION	1 OCT 2019				J. Taagepera

SCALE NONE DATE 10-13-52 APPROVED
DR. DEBERRY CH. E.T.CHAN DR. APP. ENGR. CAR

STANDARD CONNECTIONS AND WELDING DETAILS FOR CLAD STEEL VESSELS

C C.C. S.O. PVM-GD-C99663 18

PVM-GD-C99663-18

1 OCT 2019

STANDARD DRAWING

प्रेशर वेसल्स के सामान्य विनिर्देश

GENERAL SPECIFICATION

FOR

PRESSURE VESSELS

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
7	12.03.2024	REVISED AND REISSUED AS STD. SPEC.	PS	TKh	KA/NK	MN
6	28.08.2018	REVISED AND REISSUED AS STD. SPEC.	SK	TK	KJH	RKT
5	30.06.2010	REVISED AND REISSUED AS STD. SPEC.	KA	RKT	AKM/DM	N.DUARI
4	20.11.2009	REVISED & REISSUED AS STD. SPEC.	VB	RKG	AKM	N.DUARI
3	15.05.2000	REVISED AND REISSUED AS STD. SPECN.	RKT	AKM	CRMN	MI
2	15.12.1994	REVISED AND REISSUED AS STD. SPECN.	DD	AKM	VC	AS

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Abbreviations:

AI	:	Authorized Inspector
ASME	:	American Society of Mechanical Engineers
DNV	:	Det Norske Veritas
IBR	:	Indian Boiler Regulations
ISO	:	International Organisation for Standardisation
NB	:	Nominal Bore
PESO	:	Petroleum & Explosives Safety Organisation
PWHT	:	Post Weld Heat Treatment

Static Equipment Standards Committee

Convenor: Mr. Nalin Kumar

Members:
Mr. K. Anjaneyulu (Co-Convener)
Mr. Tarun Kumar (Emp. No. A328)
Mr. Tarun Khurana (Coordinator)
Mr. P V S Satyanarayana
Mr. Anish Trehan
Mr. P Barik
Mr. Saikat Chakraborty
Mr. Piyush Suryavanshi
Mr. Mittal Kumar Patel
Mr. Srikanth Karanam
Mr. Aasheesh Handa (Projects)
Mr. Prabhakar Choudhary (SMMS)
Mr. Avdhesh Agarwal (SCM-Inspection)

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1.0 SCOPE

1.1 This specification covers the general requirements for design, fabrication, workmanship, erection, inspection, testing and supply of unfired pressure vessels and is intended to supplement the minimum requirements of the applicable codes.

1.2 Supplementary specification indicating special or additional requirements shall form addenda to this specification and shall govern. Supplementary specification, when made addenda to this specification shall be referred to in the Material Requisition and / or Purchase Order. Following are the supplementary specifications:

a) 6-12-0002 Supplementary specification for carbon steel vessels.
b) 6-12-0003 Supplementary specification for low alloy steel vessels.
c) 6-12-0006 Supplementary specification for austenitic stainless steel vessels.
d) 6-12-0007 Supplementary specification for stainless steel clad vessels.
e) 6-12-0008 Supplementary specification for 3½ %Ni steel pressure vessels.

1.3 In case of conflict, the order of precedence shall be as follows:

a) Statutory Requirement
b) Engineering drawing
c) Specifications and standards
d) Codes

In general, the most stringent requirement out of the above shall govern. However, in such a case, vendor shall promptly refer the conflicts to client / EIL in writing to obtain client / EIL instruction before proceeding with manufacture. Decision of client / EIL shall be binding without any time and cost implication.

2.0 REFERENCES

2.1 Design Codes

The following codes in the edition and addenda referred to in the datasheet shall form the basis for design, fabrication, inspection, testing and acceptance of equipment:

(a) ASME Boiler and Pressure Vessel Code Section VIII Div. 1

(b) ASME Boiler and Pressure Vessel Code Section IX and Sec V.

(c) Indian Boiler Regulations- IBR (whenever applicable).

(d) The Static and Mobile Pressure Vessels (Unfired) Rules 2016 (whenever applicable).

2.2 Material Codes

Material to be used shall conform to:

(a) ASME Boiler and Pressure Vessel Code Sec. II.

(b) Indian Standard specification.

Other specifications of equivalent grade can be used only after written approval from EIL.

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2.3 Regulations

National laws and regulations together with any local by-laws for the country or state wherever the vessels are to be used must be complied with by the fabricator.

3.0 DESIGN

3.1 Design Pressure

3.1.1 Design pressure specified shall be at the top of vertical vessel or at the highest point of horizontal vessel.

3.1.2 The design pressure at any lower point shall be determined by adding the maximum operating liquid head and any pressure gradient within the vessel.

3.2 Corrosion Allowance

3.2.1 Corrosion allowance shall be added to both sides of tray support rings and other fixed internal non-pressure parts.

3.2.2 Removable internal parts (with the exception of trays) which are bolted or clamped in place, shall be provided with extra thickness equal to half the specified corrosion allowance on each surface exposed to vessel contents.

3.2.3 Full corrosion allowance shall be added to the throat thickness necessary for strength or sealing in case of fillet and seal welds on internal attachments.

3.3 During fabrication or shop/site hydro-test conditions in horizontal position, the supports for shell shall be so provided that combined stresses in any shell component (based on corroded thickness for site test) do not exceed the following:

(a) Tensile :	90% of ambient yield
(b) Compressive:	Code allowable stress

3.4 All conical reducers shall be designed as tori conical type with knuckle radius at each end as minimum 10% of the adjoining diameter, unless otherwise specified in requisition. 100% radiography shall be carried out for all weld seams of tori cone.

3.5 Statutory Approvals

3.5.1 For vessels coming under the purview of Indian Boiler Regulations, it shall be vendor's responsibility to get approval from IBR authorities pertaining to design, drawings, material of construction, fabrication, inspection and testing etc.

3.5.2 For vessels coming under the purview of Static and Mobile Pressure Vessel rules, it shall be vendor's responsibility to get approval from Petroleum & Explosives Safety Organization (PESO) pertaining to design, drawings, material of construction, fabrication, inspection and testing etc.

3.6 Internals

3.6.1 For support beams, allowable deflection shall be limited to L/325 where L is the length of the beam.

3.6.2 For plates, allowable deflection shall be limited to T/2 where T is the corroded thickness of plate.

3.6.3 Allowable stress at design temperature shall be as per applicable Code.

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4.0 FABRICATION

4.1 Head/Dished Ends

4.1.1 Dished ends shall preferably be of seamless construction. However, dished ends with one chordal weld seam are acceptable. In such cases, the width of chordal plate shall not be less than one third of the blank diameter and chordal seam shall clear nozzle opening. Intermediate heat treatment, if considered necessary, shall be carried out by the fabricator.

4.1.2 Whenever a dished end is made of more than two plates, it must have a seamless crown plate. Whenever a nozzle or a manhole is positioned at the center of the dished end, the crown plate should be larger than the nozzle /manhole reinforcing pad.

4.1.3 Wherever hot forming and subsequent heat treatment is involved, adopted procedure shall not impair the mechanical properties of the material beyond the limits specified in respective material specification.

4.2 Nozzles, Manholes, Handholes and Attachments

4.2.1 Manhole cover shall be provided with a davit or hinge as per EIL standard and Handhole cover shall be provided with suitable lifting handles.

4.2.2 Inside edges of Manholes/Handholes shall be rounded off smooth with a minimum radius of 3mm.

4.2.3 All nozzles shall be set-in type, unless otherwise specified in requisition. Set on type nozzle for size 50 NB and below can be used with prior approval.

4.2.4 Reinforcing pads whenever required as per drawings or Code shall be of the same material or equivalent as the vessel component to which it is welded. All reinforcing pads shall be provided with two 1/8" (3 mm) NPT tapped holes located 180 apart for air soap solution test with a pressure of 1.25 kg/cm2 (g). This test shall also be required to be carried out for slip on flanges. Higher test pressures are not recommended because of accompanying risks and also because the soap bubbles have a chance to blow off. Tell-tale holes in the reinforcing pads shall be plugged with hard grease unless otherwise indicated after the hydro test of the vessel.

4.2.5 Wrapper plates, reinforcing pads, saddle plates or stiffeners of higher thickness than specified can be used provided there is no change in basic dimensions and with the approval from Inspection Agency. No separate deviation permit is required for the same.

4.3 Flanges

4.3.1 When tongue and groove or male and female faced flanges are used, groove or female face shall be in nozzle flange. However, when the nozzle is located in the bottom, the groove or female face shall be in the cover.

4.3.2 Dimensions of flanges shall be as per ASME B16.5 for sizes up to 600 mm NB (24" NB) and as per ASME B 16.47 series 'B' for sizes greater than 600 mm NB (24" NB).

4.4 Fasteners and Gaskets

4.4.1 All bolts/studs shall have ISO threading unless otherwise specified. Studs shall extend beyond nuts at least by 2 threads & studs shall be threaded to full length. Bolts/studs (to be tightened by hydraulic bolt tensioner) shall be longer than normal length by minimum 1 nut diameter.

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Hydraulic bolt tensioner shall be used for following bolt sizes and conditions and shall be supplied by vendor unless otherwise specified in the Requisition:

Nominal Bolt Size	Conditions
All Sizes	When specified by Process Licensor/Project specifications
50 mm and Over	All flange joints
38 mm and to 50 mm	Class 600 and above
	Hydrogen service (Partial Pressure of Hydrogen > 7 Kg/cm2g)
	Design Temperature above 370oC
	Inlet, outlet and quench nozzle flanges of reactors/separators (#)
25 mm to 38 mm	Design Temperature above 370oC
	Inlet, outlet and quench nozzle flanges of reactors/separators (#)

Applicable only for Hydrocracker/DHDT/DHDS/VGO-HDT Reactors and Hot & Cold Separators.

4.4.2 All internal bolts shall be provided with double nuts.

4.4.3 Threads on external bolting shall be lubricated with graphite grease for working temperature up to 200°C and with Molybdenum Di-sulphide for higher temperature.

4.4.4 In addition to stamping of the specifications & manufacturer's symbol as specified in ASME material specifications, size shall be clearly punch marked on one of the ends of the stud. Similarly, the nuts shall have the size punch marked on one of the faces. In case of tapped hole the size shall be punch marked near the hole without disturbing the gasket seating area. Further for all alloy/SS metallurgy bolts & nuts shall also be identified by distinct color marking at the stud end/bolt side face.

4.4.5 Gaskets shall conform to ASME B16.20 and ASME B 16.21.

4.5 Internals and Externals

4.5.1 All removable internals shall pass through the manhole.

4.5.2 Internal baffles, tray support beams or other internals spanning a chord or diameter of the vessel shall be provided with means to allow differential thermal expansion between the part and the vessel shell.

4.5.3 Internal pressure piping shall be seamless and of same specification as the external connected piping. However internal non-pressure piping can be either seamless or welded type.

4.5.4 Internal flanges for pressure piping shall be forged flanges. However internal flanges for non-pressure piping can be fabricated from plate.

4.5.5 Tray/seal pan support rings, downcomer bolting bars and all internals, beam supports welded to the vessel shall be supplied and welded by vessel fabricator in accordance with the details furnished by EIL.

4.5.6 The internal baskets, mesh screens, support grid, distributor/mixing trays, distributor pipes, outlet collector etc. of equipment shall be fitted at shop unless otherwise specified in requisition. Equipment shall be transported along with internals fitted. Temporary wedges (if required) shall be provided for supporting the internals which shall be removed by mechanical contractor at site under vendor's supervision unless otherwise specified. Additionally, manway panels shall be provided for mesh screens, support grids, distributor/mixing trays etc. to ensure accessibility for future inspection, loading/unloading of catalyst/inert balls etc.

In case the internals are to be supplied loose as per requisition, the internal baskets, mesh screens, support grid, distributor/mixing trays, distributor pipes, outlet collector etc. of equipment shall be trial fitted,

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then removed and crated for dispatch with vessel. These items should be tagged for field identification and installation.

4.5.7 All externals shall be supplied and welded by vessel fabricator in accordance with the details furnished by EIL.

4.5.8 Internal perforated plate inside the vessel shall be removable type unless specified otherwise and should be able to pass through nearest manhole.

4.5.9 Unless specified otherwise, Full support rings shall be provided below catalyst bed to support same.

4.6 Name Plate

Each vessel shall be supplied with EIL name plate and manufacturer name plate as per EIL Standards.

4.7 Welding

4.7.1 Flame cut edges shall be ground by the fabricator as required to remove slag, detrimental discoloration and non-uniformity of edges.

4.7.2 All pressure bearing butt welds shall be full penetration, double welded joints. When second side welding is not possible due to inaccessibility, single welded butt joints with root run by Tungsten Inert Gas Process (TIG) can be used to ensure full penetration. Backing strip can be used only after obtaining prior approval from EIL. For sub-zero temperatures, backing strips shall not be used.

4.7.3 Nozzles and Manways and their reinforcement pads shall be attached to vessel with full penetration welds.

4.7.4 Seams in supporting skirt shall be made with full penetration butt welds. Connections between skirt and vessel head shall be made with a smooth flat faced weld unless otherwise indicated in the engineering data sheet. Width of the weld at skirt end shall be equal to the skirt thickness and its height shall at least be twice its width.

4.7.5 All main weld seams shall be clear of nozzles, reinforcement pads, internals, tray support rings, cleats and stiffening rings by 50 mm minimum (weld edge to weld edge). In case the same is unavoidable following requirements shall apply:

a) Nozzles without reinforcing pad

i) Any weld seam having distance (weld edge to weld edge) to nozzles within 50 mm (but not fouling with weld seam) shall be fully radiographed and dye penetrant examined to a length equal to 100 mm on each side measured from nearest point to nozzle edge.

ii) Any weld seam fouling with nozzle opening shall be fully radiographed and dye penetrant examined to a length equal to 3 times of outside diameter of nozzle i.e. 1.5 times of outside diameter of nozzle on each side after installation of nozzles. Nozzle to vessel fillet weld shall be provided with smooth concave radius.

b) Nozzles with reinforcing pad

i) Any weld seam having distance (weld edge to weld edge) to reinforcing pad within 50 mm (but not fouling with weld seam) shall be fully radiographed and dye penetrant examined to a length equal to 100 mm on each side measured from nearest point to reinforcing pad edge.

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ii) Any weld seam not fouling with nozzle opening but coming under reinforcement pad shall be ground flush, fully radiographed and dye penetrant examined to a length equal to portion of weld seam below reinforcement pad + 100 mm on each side.

iii) Any weld seam fouling with nozzle opening shall be ground flush, fully radiographed and dye penetrant examined to a length equal to higher of 3 times of outside diameter of nozzle i.e. 1.5 times of outside diameter of nozzle on each side or length equal to portion of weld seam below reinforcement pad + 100 mm on each side.

In case other attachments like internals, tray support rings, cleats etc. is fouling with weld seam, the weld seam portion coming under the attachment plus 100 mm length on each side shall be ground flush, fully radiographed and dye penetrant examined before welding of any such attachment.

4.7.6 Vendor shall submit welding procedure specification and qualification record to purchaser's inspector or authorized representative for approval as per ASME Boiler and Pressure Vessels Code Section IX.

Previously qualified welding Procedure Qualification Record (PQR) under EIL, CEIL, Lloyds, DNV, TUV and Bureau Veritas shall also be acceptable.

4.7.7 Welding consumables shall be as per ASME Boiler and Pressure Vessel Code Sec. II Part C and shall be indicated in fabrication drawings.

4.7.8 Welding shall not commence unless the concerned procedures are approved.

4.7.9 Only welders who are qualified in the accepted procedure shall be employed for welding.

4.7.10 All internal/external attachments (nozzles, cleats etc.) with fillet welds to the vessel pressure components in case of Hydrogen service, cyclic service and vessels with design temperature (-)29oC and lower, vessels with design temperature 370oC and higher for carbon steel and 425oC and higher for low alloy steel shall be ground smooth and generous concave contour shall be provided.

4.7.11 Minimum shell course width shall be 1 meter. Maximum no. of longitudinal seams shall be as follows:

(a)	vessels up to 2 meter diameter	:	1 seam
(b)	vessels from 2 meter up to 4 meter diameter	:	2 seam
(c)	vessels from 4 meter up to 6 meter diameter	:	3 seam
(d)	vessels beyond 6 meter diameter	:	seam nos. can be proportionately decided

4.8 Post Weld Heat Treatment

4.8.1 Vessels shall be post weld heat treated when specified on the engineering drawings. In no case shall the post weld heat treatment performed be less than that specified in the code.

4.8.2 Vessels shall be post weld heat treated as a complete unit including skirt/support, wherever practicable.

4.8.3 All flange faces shall be suitably protected against oxidation during post weld heat treatment.

4.9 Tolerances

4.9.1 Tolerances shall be as per drawing/standard/code.

4.9.2 For dimensions not provided with tolerances, fabricator shall maintain dimensions as per good engineering practice.

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5.0 INSPECTION & TESTING

5.1 All vessels shall be offered for inspection to purchaser or his authorized inspector.

5.2 Inspectors shall have free access to all workshops of contractors or sub-contractors.

5.3 Inspection shall be carried out both during fabrication and before delivery and also for sub-ordered materials, if any. In addition to final inspection and certification by Inspector, Inspector's written approval shall be obtained by the manufacturer at all stages of fabrication including, but not limited to the following:

a) Raw material identification

b) Edge preparation for welding, including visual check for laminations.

c) Alignment of longitudinal seams.

d) Rolling tolerance on individual sections

e) Alignment of sections

f) Root pass clearing before welding.

g) Nozzle setting.

h) Dimensional check.

i) Radiographic / Ultrasonic Examination.

j) Dye penetrant examination/Magnetic particle examination.

k) Stress relieving.

l) Calibration records of measuring instruments.

m) Pressure Test (Hydrostatic/Pneumatic test).

n) Any other special test such as for leak, corrosion, hardness etc.

o) Surface preparation, primer and painting.

5.4 Radiography

5.4.1 The extent of radiography shall be as specified on the engineering drawings. In no case shall the radiographic examination be less than that specified in the code. However, spot radiography is the minimum requirement for all vessels.

5.4.2 When spot radiography is specified, the following requirements shall supplement the requirements specified in ASME Section VIII Division I:

Each category A or B pressure containing weld shall be spot radiographed in accordance with ASME Section-VIII Division-I, paragraph UW-52 as a minimum requirement. Each Spot radiograph shall be a minimum of six inches (150mm) in length and minimum one spot shall be selected in each circumferential & longitudinal seam. Additionally, at least one T-joint in each circumferential seam shall also be selected. Welds from each welding procedure, welded/welding operator shall be examined. DIA shall be consulted in marking the areas to be radiographed.

5.4.3 All nozzles fabricated from plate, irrespective of thickness of plate, shall be 100% radiographed. When Full radiography is specified due to Service requirement, 100% radiography shall be carried out of all butt welds including nozzle flange to nozzle neck, pipe to pipe and pipe to fitting.

5.4.4 Weld seams of formed ends shall be 100% radiographed after forming and heat treatment, if any.

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5.4.5 The specified radiography of welds may be performed before or after post weld heat treatment (PWHT). If performed before PWHT, an additional radiography or alternatively, ultrasonic examination shall be performed after PWHT.

5.4.6 The technique employed and the weld quality achieved shall meet the requirement of the code.

5.4.7 If vendor intends to carry out radiography by Computed Radiography technique, then demo should be carried out under AI to prove vendor's capability to produce acceptable images meeting the Code requirements.

5.4.8 In case Supplier want to perform UT in lieu of RT based on Code para UW 51 (a) (4), the same is acceptable for Carbon Steel and Low Alloy Steel materials subject to the following:

a) The thickness of weld is greater than 13 mm.

b) The joint is either Category A or Category B joint except shell to hemispherical dished end joint. (Category of joint as defined by Code).

c) Automated recordable UT machine incorporating TOFD and pulse echo probes or TOFD and Phased Array mounted on the same chassis that automatically traverses along the joints to be inspected and displaying both results simultaneously on a single screen may be considered in lieu of RT.

d) Calibration block of similar material & thickness shall be used. Calibration block shall have suitable notches to simulate longitudinal as well as transverse cracks on outside and inside surface. Setup should be capable of detecting defects on outside as well as inside while scanning from one surface only.

e) The system that is proposed to be deployed in lieu of radiography shall be submitted along with the track record prior to use for EIL review. The Supplier shall demonstrate successfully the capability of AUT machine and the defects evaluation in the same screen during the site visit of EIL representative. Based on which, the approval of the parties and their AUT system shall be given.

f) Qualified NDT persons shall be deployed to perform UT.

g) Acceptance criteria shall be as per Code para UW 51.

5.5 All nozzle to shell welds (Root and Final run) shall be examined by magnetic particle/ Dye- penetrant examination.

5.6 Hydrostatic Test

5.6.1 All necessary precautions shall be taken to guard against the risk of brittle fracture during hydrostatic test in the shop and at site. The temperature of testing medium shall be as per Code. Prior to hydrostatic test, all weld spatter, weld stubs, scale, dirt etc. shall be removed from vessel.

5.6.2 Hydrostatic test shall be conducted at pressures mentioned in engineering drawings after complete fabrication and post weld heat treatment, as applicable. After hydrostatic testing, Equipment shall be completely dried by passing hot air for sufficient time until no further increase in relative humidity of outgoing air is observed. Alternatively vacuum drying of the equipment is also acceptable.

5.6.3 Clean potable water shall be used for hydrotest. Sea water shall not be used.

5.6.4 During hydrotest, care shall be taken to avoid local stresses in shell from exceeding 90 % of the yield strength of the material at the temporary saddle supports.

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5.6.5 Vessels shall not be painted or coated either internally or externally prior to the hydrostatic pressure tested.

5.7 For all weld overlays used in hydrogen or H2S service with design temperature greater than 350oC, Hydrogen disbonding test shall be carried out. The test condition shall be representative of the actual design conditions and the procedure shall be submitted to Inspection Agency for approval. Rate of cooling shall be 100oC/hr (min.) unless specified otherwise in datasheets. Holding time shall be 48 hours unless specified otherwise in datasheets.

5.8 Vendor shall ensure that all alloy steel and stainless steel material are properly identified and finally check tested by a PMI analyzer before dispatch of equipment as per EIL Specification 6-81-0001.

6.0 SUPPLY

6.1 Surface Cleaning and Painting

6.1.1 Surface cleaning and painting shall be as per standard specification for shop and field painting (6-44-0004) or as per job specification as applicable. Paint system shall be selected as per the environments specified on engineering drawing.

6.1.2 All completed equipment shall be cleaned internally and externally to remove scale, dirt, sand, water and foreign matter.

6.1.3 All flanged faces and other machined surfaces shall be greased or protected with rust preventive coating.

6.1.4 Except for machined surfaces, all exterior surfaces of vessels and columns including skirts and integral supports shall be painted to prevent rust, corrosion or damage during transit and storage before erection and final painting.

6.2 Marking

In addition to General purchase conditions, the following requirements shall also be complied with:

6.2.1 All loose components such as studs, nuts, washers, gaskets etc. shall be packed in crates and shall be marked for the project, consignee, consigner, job number, item number, order number, gross and net weight, dimensions etc.

6.2.2 Additional indications such as North/East/South/West along with center of gravity shall be clearly marked with white paint.

6.2.3 Vessels which have been post weld heat treated or have an applied lining, e.g. lead, glass, rubber etc., shall have a suitable warning printed on the visible portion on the outside of vessel.

6.2.4 Specific marking with white paint for slinging shall be provided for all heavy lifts weighing 5 tons and above.

6.2.5 A copy of packing list shall accompany the material enclosed in a water tight envelope fastened inside a shell connection with an identifying arrow sign "Documents" applied with indelible paint.

6.2.6 If it is necessary to separate the unit into different parts for transportation all components and subassemblies shall be carefully identified and match marked to prevent any error in assembly.

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6.3 Packing and Shipment

In addition to General Purchase Conditions, the following requirement shall also be complied with:

6.3.1 Packing

a) Vessels, unless provided with their own steel saddles for entire protection, shall be provided with suitable wooden/steel saddles with steel ties and tension rods. The minimum height of the saddle shall correspond to the maximum projected length of the connected attachments, plus an additional clearance of 45mm. Saddles spacing shall depend on the length of the equipment.

b) Equipment shall be provided with suitable type and adequate number of supports to prevent any deformation during transportation and handling. These supports shall not be removed until the equipment is placed in position at job site.

c) All connections/protrusions shall be suitably protected. Flanges shall be provided with bolts on metal covers (minimum 5 mm thk) using at least four bolts. (Wiring on covers is not acceptable). For ocean shipment, flanged openings shall be additionally covered with heavy plastic bags taped to nozzle. All tell-tale holes shall be plugged with hard grease before dispatch. Tapped orifices shall have threaded plugs.

d) Fragile or machined components shall be especially protected against nature during handling and transit.

e) Pre-fabricated sections shall be protected by temporary stiffeners at each non-supported end. Edges of plates for section to be welded shall also be protected.

6.3.2 Shipment

All dispatches of equipment shall be done in accordance with the relevant terms of the Purchase Order.

7.0 GUARANTEE

Unless otherwise specified in General Purchase conditions regarding guarantee, the following shall govern:

7.1 Manufacturer shall guarantee that all materials used in the equipment are new and have been submitted to regular acceptance procedure and are free from any defect regarding quality, form and appearance.

7.2 Vessel(s) shall be guaranteed for design, raw materials and workmanship for a duration as defined in General Purchase Conditions. When design has been carried out by EIL, the word design shall be excluded from the guarantee clause. The manufacturer shall be completely responsible for any design work carried out by him. EIL's approval of his design will not relieve him of his responsibility to ensure satisfactory performance of such item.

7.3 Approval of work by EIL or release of vessels for shipment shall in no way release or relieve the manufacturer of any responsibility for carrying out all provisions of this specification.

8.0 SITE FABRICATION AND ERECTION

8.1 Where size or shape of vessel makes it impossible to ship it in one piece, the fabricator shall ship number of shop fabricated sections as defined in requisition. Assembly and testing shall be completed by vessel fabricator at site in horizontal position (for erection by others) in strict accordance with the provisions of applicable order/specification.

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8.2 Where size or shape of vessel makes it impossible to erect in single piece, the fabricator shall ship number of shop fabricated sections as defined in requisition. Erection, assembly and testing shall be completed by vessel fabricator at site in vertical position in strict accordance with the provisions of the applicable purchase order/specifications.

8.3 In either of the above two cases, the following additional requirements shall apply:

8.3.1 All pieces shall be shop fit up into sections and each section fit to the adjacent one by the fabricator and all pieces match marked thereafter.

8.3.2 Suitable erection lugs /tailing lugs and locating pins shall be provided by the fabricator to ensure proper fit up & handling of the equipment.

8.3.3 All radiographic requirements for welds completed in the shop shall be made by the fabricator before the part of section leaves the shop.

8.3.4 For equipment requiring PWHT, complete or local PWHT may be carried out at site.

8.4 All lifting lugs/trunnions and tailing lugs shall be designed with an impact factor of minimum 1.5 unless otherwise specified.

9.0 DATA FOLDER

Manufacturer shall complete requisite copies of data folder as required in purchase order. This folder shall contain the following information duly certified by Inspector:

a) Manufacturer's Code Certificate

b) Fabrication drawing of vessel showing 'As Built' dimensions and 'As Built' erection weight.

c) Material Test Certificates with their cast /heat and test numbers

d) Welding procedure qualification reports

e) Welder qualification reports

f) Radiographic results

g) Ultrasonic, Magnetic particle, Dye-penetrant test results (if applicable)

h) Hardness, corrosion and leak test records (if applicable)

i) Record charts showing complete heat treatment cycle (if applicable)

j) Production test coupon results (if applicable)

k) Charpy V notch test results (if applicable)

l) Record chart of pressure test (Hydrostatic and/or pneumatic)

m) Rubbing of name plate

n) Any other documentation as required in the purchase requisition/purchase order.

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कार्बन स्टील वैसल्स के पूरक विनिर्देश

SUPPLEMENTARY SPECIFICATION FOR CARBON STEEL VESSELS

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
9	23.11.2023	REVISED AND REISSUED AS STD. SPEC.	PSV	TK	KA/NK	MN
8	26.06.2018	REAFFIRMED AND REISSUED AS STD. SPEC.	SK	TK	KJH	RKT
7	30.06.2010	REVISED AND REISSUED AS STD. SPEC.	KA	RKT	AKM/DM	N.DUARI
6	25.09.09	REVISED AND REISSUED AS STD. SPEC.	VB	RKG	AKM	N.DUARI
5	12.04.04	REVISED AND REISSUED AS STD. SPEC.	DNN	AKM	SSA	SKG

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Abbreviations:

ASME	:	American Society of Mechanical Engineers
BHN	:	Brinell Hardness Number
HAZ	:	Heat Affected Zone
LWN	:	Long Weld Neck
NB	:	Nominal Bore
PWHT	:	Post Weld Heat Treatment
UTS	:	Ultimate Tensile Strength

Static Equipment Standards Committee

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Mr. Avdhesh Agarwal (SCM- Inspection)

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4.0 FABRICATION 4

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1.0 SCOPE

This supplementary specification indicating additional requirements for fusion welded carbon steel vessels shall form addenda to General Specification for Pressure Vessels (6-12-0001) in its latest revision.

2.0 DESIGN

Reinforcement for nozzle openings shall be self reinforced type for equipment where nominal thickness of shell/head exceeds 50 mm. Nozzles of size up to 50 NB LWN can be used in place of self reinforced nozzles.

3.0 ELECTRODES

3.1 Electrodes conforming to following specification shall be used. In case of any deviation, prior approval from EIL shall be obtained.

S. No.	Material to be welded		Electrode Specification
1.	General Structure Welding	:	ASME IIC SFA 5.1 E-6013
2.	Pressure Parts		
a.	Steels with UTS upto 70,000 psi	:	ASME IIC SFA 5.1 E7016/ E7018
b.	Steels with UTS upto 80,000 psi	:	ASME IIC SFA 5.5 E8016-X/ E8018-XX depending upon the alloy element.
c.	Low temperature service		
	Below 0°C and upto (-) 46°C Steels to SA-516 / 516M, SA-537 / 537M, SA-333 / 333M Gr. 6 and SA-350 / 350M LF2	:	ASME IIC SFA 5.1 E 7016-1/ E7018-1

3.2 All electrodes and fluxes shall be properly baked/dried as per manufacturer's recommendation before use.

3.2.1 All bare electrodes and fluxes shall be selected as per ASME IIC .All weld metal shall have equal or better mechanical properties than the parent metal.

3.2.2 Only dry flux shall be used.

4.0 FABRICATION

4.1 Plate Forming

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4.1.1 Forming of shell plates and heads shall be carried out by machine, either hot or cold, in such a way so as to preserve the specified material properties and to produce a regular finish.

4.1.2 Magnetic particle/Dye penetrant examination shall be carried out on the outside and inside surfaces including edges of torispherical or elliptical heads in knuckle zone, after forming, for detection of cracks.

4.2 Edge Preparation

4.2.1 The preparation of edges to be welded shall be done by machining, chipping, grinding, cold shearing, Oxy- acetylene flame cutting or a combination of these.

4.2.2 Chipping shall be followed by grinding to a smooth and regular finish.

4.2.3 Oxy-acetylene flame cutting done in any circumstances, shall be followed by machining or grinding to eliminate any discolouration of material affected.

4.2.4 All welding edges shall be checked by Magnetic particle/Dye penetrant examination for detection of cracks, laminations or segregations.

4.2.5 No welding shall be carried out when ambient temperature is less than 10° Celsius unless preheating is carried out.

4.3 Heat Treatment

4.3.1 Heat treatment of formed parts shall be carried out as per following:

a. Cold formed dished ends or knuckles shall be stress relieved, unless otherwise specified in requisition.

b. Hot formed dished ends or similar parts, which have not been uniformly heated in the normalising range in the final stages of manufacture shall be normalised.

c. When the completed vessel involves post weld heat treatment, heat treatment recommended in (a) above shall not be applicable.

4.3.2 Vessels in Hydrogen/Amine/ Sour (Wet H_2S)/Cyclic/Caustic/HIC service shall be PWHT.

4.4 Production Weld Tests

Production Weld Tests shall be applicable for vessels over 50 mm nominal thickness. The following requirements shall apply:

a. Two production test plate coupons representative of one longitudinal and another circumferential seam shall be provided for each procedure, position and thickness in each vessel shell.

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b. One production test plate representative of the weld seams shall be provided for each procedure, position and thickness for welded dished ends/cones.

c. The production test plate shall be from material of the same heat and thickness as of shell/head/cone as applicable. During and after welding, the test plates shall be subjected to same heat treatment as and together with the course they represent. Extra coupons shall be preserved to take care of eventuality of retests.

d. The tests mentioned below shall be carried out as per methods of testing in governing codes:

i. One transverse tension test

ii. Two side bend tests with weld located in the centre of bend.

iii. Hardness test on production test coupon weld & HAZ (Hardness limitations shall comply with requirement of EIL Spec. 6-15-0091)

iv. Micro & macro examination of welds

v. Charpy V notch tests on weld and HAZ.

a. For low temperature service (0°C & colder) impact test temperature shall be lowest of minimum ambient temperature, design temperature and minimum design metal temperature.

b. For vessel over 50 mm nominal thickness used for warmer service, the test temperature shall be lower of 0°C or lowest ambient temperature of the site.

c. The acceptance criteria for energy absorption shall be as per Table A 2.15 of SA-20 / 20M. In case the acceptance criteria is not available in SA-20, then applicable design code shall be referred unless otherwise specified in requisition.

4.5 Hardness Survey

Hardness Limitation for Vessels, where PWHT is required shall comply with requirement of EIL Spec. 6-15-0091.

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निम्न मिश्र स्टील वैसल्स के पूरक विनिर्देश

SUPPLEMENTARY SPECIFICATION FOR LOW ALLOY STEEL VESSELS

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
6	02.04.2024	ISSUED FOR WIDER CIRCULATION	SC	TKh	KA/NK	MN
5	28.08.2018	REVISED AND REISSUED AS STD. SPEC.	SK	TK	KJH	RKT
4	30.06.2010	REVISED AND REISSUED AS STD. SPEC.	KA	RKT	AKM/DM	N. DUARI
3	29.06.09	REAFFIRMED & REISSUED AS STD. SPECN.	VB	RKG	AKM	N. DUARI
2	23.03.04	REVISED & REISSUED AS STD. SPECN.	SCDD	DNN	SSA	SKG

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Abbreviations:

ASME	:	American Society of Mechanical Engineers
ASTM	:	American Society for Testing and Materials
BHN	:	Brinell Hardness Number
HAZ	:	Heat Affected Zone
HIC	:	Hydrogen Induced Cracking
LPG	:	Liquified Petroleum Gas
LWN	:	Long Weld Neck
NB	:	Nominal Bore
PWHT	:	Post Weld Heat Treatment

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1.0 SCOPE

This supplementary specification indicating additional requirements for fusion welded low alloy steel vessels (C-½Mo, 1Cr-½Mo, 1¼ Cr-½Mo, and 2¼ Cr-1Mo) shall form addenda to general specification for pressure vessels (6-12-0001) in its latest revision.

2.0 DESIGN

2.1 Reinforcement for nozzle openings shall be self-reinforced type for equipment where nominal thickness of shell/head exceeds 50 mm. Nozzles of size up to 50 NB LWN can be used in place of self-reinforced nozzles.

2.2 All flanges shall be of weld neck type irrespective of design condition, size and service.

3.0 MATERIALS

All external and internal attachments such as tray support rings, bolting bars, internal gussets, pipe/ladder/platform support clips, saddles, skirts, leg supports, lifting lugs etc. welded to vessel directly shall be of same material as that of shell.

4.0 WELDING AND WELDING CONSUMABLES

4.1 Welding of various materials shall be carried out using one or more of the following processes.

- Shielded Metal Arc Welding
- Gas Tungsten Arc Welding
- Submerged Arc Welding

Use of any other process shall be subject to EIL approval. All pressure bearing butt welds shall be full penetration double welded butt joints. Single welded full penetration joints may be provided when second side welding is not possible. Gas Tungsten Arc Welding shall be used for the root run of all the single welded butt joints. Backing strip joints may also be used with prior approval of EIL. Backing strip, if used, shall be removed.

4.2 Proper technique and welding procedure shall be adopted to prevent the risk of cracking of welds. All fitment and alignment clips (Bridge or C-clamps) shall be of same material as that of shell. Tack welding shall be avoided as far as possible.

4.3 Electrodes and fluxes shall be properly baked and dried as per manufacturer's recommendation before use.

4.4 The adopted welding procedure shall be such, that least amount of hydrogen is produced in the weld.

4.5 No welding shall be carried out when the ambient temperature is less than 15 C. Any welding once started should be completed in one operation as far as possible.

4.6 Welding Consumables

The manufacturer shall provide evidence that the deposited weld metal is suitable in all respects for the intended service and conform to the classification by conducting tests and/or obtaining test certificates of the batch supplied. The tests shall at least consist of chemical analysis and all weld tensile test as a minimum. The welding consumables proposed shall conform to one of the classifications listed in ASME SEC.II Part C unless specifically approved otherwise.

The welding consumables shall be selected such that for welds of similar metallurgy, the chemistry of weld shall match the chemical composition of the base material and the physical strength of weld shall

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be better or equal to the base material. Also, the corrosion properties of the filler material shall be in line with the base metal.

4.7 Hardness on base metal, welds and HAZ for all applicable materials, shall be taken and complied as per EIL spec. 6-15-0091.

5.0 FABRICATION

5.1 Dish heads and knuckles shall preferably be hot formed to prevent possible cracks during cold-forming, with intermediate checking of hardness. However, if vendor has past experience of cold forming, same is acceptable.

5.2 Magnetic particle examination shall be carried out on the formed knuckle surface (both inside and outside) of dished ends and toricones for detection of cracks.

5.3 The preparation of the edges shall be made by oxyacetylene flame cutting or by machining. Cold shearing is permitted for cutting to size up to a thickness of 10mm. Oxy acetylene flame cutting done in any circumstances shall be followed by machining or grinding to eliminate any discoloration of material affected.

5.4 Non-Destructive Test

5.4.1 All welding edges shall be checked by Magnetic Particle examination for detection of cracks, laminations or segregations.

5.4.2 The ultrasonic examination shall be carried out within 2t of material thickness (t) of all weld edges prior to welding as per ASTM A 578 with acceptance level-B.

5.4.3 External and internal surfaces of all welds shall be checked by Magnetic Particle examination after hydro-test for vessels having thickness above 50mm.

6.0 PRE AND POST WELD TREATMENT

6.1 Fabricator shall propose the preheat required for each type of weld including tack welds in the welding procedure specifications. The preheat requirements shall be established at the time of approval of welding procedure.

6.2 Preheating performed by using resistance or induction heating method shall be preferred. However, preheating by gas burners, using oxy-acetylene or LPG with neutral flame may be acceptable when permitted by the inspecting authority.

6.3 Pre heating shall extend uniformly to at least three times the thickness of the joint but not less than 50mm on both sides of the weld. Temperature indicating crayons shall be used for assessment of correct preheat temperature.

6.4 Post Heating

6.4.1 In case of P4 and P5 materials, if post-weld heat treatment is not carried out immediately after welding, the weld joint and adjacent portion of the joint at least 50mm on either side of weld shall be uniformly heated to 300°C. This temperature shall be maintained for a duration of 1 hr./inch (minimum 20 minutes) and then wrapped in asbestos or insulation before allowing it to cool to room temperature. Post weld heat treatment as applicable shall be carried out later on.

6.4.2 Post weld heating shall also be necessary for highly restrained joints or where heavy welding is carried out in a localized area for example full throat nozzle welding etc.

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6.5 Post Weld Heat Treatment

6.5.1 When PWHT is required as per code or specified in vessel design data, the procedure and scheme for PWHT shall be submitted to inspecting authority for review.

6.5.2 Heat treatment of formed parts shall be carried out as per following:

a) All cold formed dished ends or knuckles shall be stress relieved.

b) When hot forming of any part is carried out, the complete procedure including heat treatment shall be submitted to inspecting authority.

c) When any of the treatments referred to above involves a final treatment which coincides with the final post weld heat treatment of completed vessel, that part of earlier treatment may be omitted.

d) Vessels in Amine, sour, HIC, Hydrogen service, Caustic Service and cyclic services shall be stress relieved.

e) It shall be fabricator's responsibility to ensure that mechanical properties of material are not impaired during any cold / hot forming process and subsequent stress relieving heat treatment. Tests to demonstrate suitability of process shall be carried out by fabricator.

7.0 PRODUCTION WELD TESTS

Production weld tests shall be applicable for vessels over 25mm thickness. The following requirements shall apply:

a) Two production test plate coupons representative of one longitudinal and another circumferential seam shall be provided for each procedure, position and thickness in each vessel shell.

b) One production test plate representative of the weld seams shall be provided for each procedure, position and thickness for welded dished ends/cones.

c) The production test plate shall be from material of the same heat and thickness as of shell/head. During and after welding the test plates shall be subjected to same heat treatment as and together with the course they represent, extra coupons shall be preserved to take care of eventually of retests. The production test plate shall incorporate additional PWHT cycle in case the equipment closing seam is to be post weld heat treated.

d) The tests mentioned below shall be carried out as per methods of testing in governing codes:

i) One transverse tension test.

ii) Two side bend test with weld located in the center of bend.

iii) Hardness test on production test coupon weld & HAZ (Hardness limitations shall comply with requirement of EIL Spec. 6-15-0091).

iv) Micro and macro examination of welds. Microstructure shall be free from untampered martensite.

v) Charpy V notch tests on weld and HAZ.

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क्लैड वैसल्स के पूरक विनिर्देश

SUPPLEMENTARY SPECIFICATION FOR CLAD VESSELS

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
6	27.03.23	REVISED AND REISSUED AS STD. SPEC	NSK 27/03/23	TK 27/03/23	NK 29/3/23	SM 31/3/23
5	24.08.18	REAFFIRMED AND REISSUED AS STD. SPEC.	SK	TK	KJH	RKT
4	30.07.10	REVISED AND REISSUED AS STD. SPEC.	KA	RKT	AKM/DM	N.DUARI
3	20.11.09	REVISED AND REISSUED AS STD. SPEC.	VB	RKG	AKM	N.DUARI
2	15.04.04	REVISED AND REISSUED AS STD. SPEC.	DNN	AKM	SSA	SKG

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Abbreviations:

ASME	:	American Society of Mechanical Engineers
ASTM	:	American Society for Testing & Materials
AWS	:	American Welding Specification
GMAW:	:	Gas Metal Arc Welding
GTAW	:	Gas Tungsten Arc Welding
IGC	:	Intergranular Corrosion
NB	:	Nominal Bore
ppm	:	Parts Per Million
SAW	:	Submerged Arc Welding

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1.0 SCOPE

1.1 This supplementary specification covers the additional requirements to be followed in the manufacture of cladded steel vessels with carbon steel or low alloy steel base, clad with stainless steel or Monel. This is intended as an addenda to the General Specification for Pressure Vessels (6-12-0001)/ Supplementary Specification for C.S. Vessels (6-12-0002)/ Supplementary Specification for Low Alloy Steel Vessels (6-12-0003).

1.2 Any ambiguity or contradiction between various specifications and drawings shall be informed by fabricator to EIL and shall be resolved taking into account the following documents in the order of preference.

i) EIL Engineering Drawing/datasheets.
ii) This supplementary specification.
iii) Specifications 6-12-0001, 6-12-0002 and 6-12-0003.
iv) Vessel Design Code.

2.0 DESIGN

2.1 The design thickness, in accordance with the specified code, shall not include any material used for the corrosion resistant lining or the cladding material of the integrally clad plate. The minimum thickness of base plate shall be 5 mm.

2.2 Corrosion resistant alloy lining for shells and heads shall be integrally cladded to the base material.

2.3 Nozzles fabricated from integrally clad plates or from weld overlay construction are only permitted. Nozzles with loose liners are not permitted.

2.4 Nozzles of solid stainless steel construction on clad vessels are not permitted unless otherwise specified on the Engineering Drawing.

2.5 Flanges of nozzles and manways in clad portions of vessels shall be weld overlayed with the recommended alloy material and faced. The flange facing shall be made with weld metal overlay which is at least as thick as the vessels lining but not greater than twice the lining thickness. Weld metal overlay shall be made with minimum two passes (layers). The finish machined weld overlay shall exhibit undiluted weld metal composition at a depth of 2/3rd of the cladding thickness from the finished surface.

2.6 Necks of manways and nozzles greater than 150 mm NB shall be made from integrally clad plate. Nozzles 150 mm NB and below shall be of weld overlay construction.

3.0 MATERIALS

3.1 Clad steel plates used for the fabrication of vessels shall conform to SA 263 or SA 264 of ASME Boiler and Pressure Vessels Code Section II Part A and Standard Specification for Clad Plates (6-12-0015).

3.2 In the alloy clad portions of the vessel, all internal parts like support rings, beam seats supports and others shall be made of the alloy corresponding to the cladding in the vessel or as specified in the engineering data sheets.

4.0 FABRICATION

Cold working shall be preferred. If hot working is carried out, it should be followed by

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suitable heat treatment to ensure that the required material properties are achieved in the final product.

5.0 WELDING

5.1 Alloy weld deposit / restoration at weld seam location shall be made of at least two layers.

5.1.1 For weld cladding, appropriate welding consumables that will yield the compatible composition as that of clad material shall be used.

5.1.2 The recommended SMAW welding consumables for welding different clad plates shall be as listed below. However for achieving the undiluted composition at the desired depth of 2/3rd of the cladding thickness from finished surface, intermediate grinding or half bead technique shall be used. Alternatively low dilution processes such as pulsed Gas Metal Arc Welding (P-GMAW) or Electro-slag Welding process with appropriate welding consumables shall be used for achieving the correct dilution and composition requirement.

CLADDING	APPLIED LINING	BARRIER /FIRST LAYER	TOP LAYER
A 263	A 240 TYPE 405,410S	(2)	(8)
A 264	A 240 TYPE 304	(1)	(3)
A 264	A 240 TYPE 304L	(1)	(4)
A 264	A 240 TYPE 316	(2)	(5)
A 264	A 240 TYPE 316L	(2)	(6)
A 264	A 240 TYPE 321 & 347	(2)	(7)
A 265	B-127	(10)	(9)

	SPECIFICATION	ELECTRODE CLASSIFICATION
(1)	AWS A 5.4, ASME SFA 5.4	E 309L
(2)	AWS A 5.4, ASME SFA 5.4	E 309 MoL
(3)	AWS A 5.4, ASME SFA 5.4	E 308
(4)	AWS A 5.4, ASME SFA 5.4	E 308L
(5)	AWS A 5.4, ASME SFA 5.4	E 316
(6)	AWS A 5.4, ASME SFA 5.4	E 316L
(7)	AWS A 5.4, ASME SFA 5.4	E 347
(8)	AWS A 5.4, ASME SFA 5.4	E 410S / E 430/ E 309LMo/ E NiCrFe-2 /ENiCrFe-3
(9)	AWS A 5.11, ASME SFA	E Ni Cu7
(10)	AWS A 5.11, ASME SFA	ENi1

ENiCrFe-2/ENiCrFe-3/ENiCrMo-3 shall be as per AWS 5.11, ASME SFA 5.11.

5.1.3 Stainless steel Filler metals for GTAW, GMAW, SAW Process made in accordance with corresponding AWS/ASME SFA specification with the compositions similar to those listed above shall be used after approval.

5.1.4 For welding of clad piping components such as pipe to pipe/pipe fitting from single side where approach from clad side is not accessible, shall be done with a welding consumable matching the clad material followed by an intermediate layer of pure iron welding filler material and then by a welding consumable matching the base material.

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5.1.5 Consumables selected by fabricator shall be procured from reputed consumable manufacturer with the requisite approvals and shall have written consent of authorised inspector before use.

5.1.6 Weld deposition cladding procedure shall be qualified separately in presence of authorised inspector to yield a weld deposit that exhibits undiluted weld metal composition at a depth of 2/3rd of the cladding thickness from top of the finished overlay surface. Previously qualified & approved procedures, not older than 5 years & meeting the requirements shall be acceptable.

5.2 When integrally bonded clad plate is used, the lining shall be cut back at all the seams and the nozzle to shell/head joints to permit complete welding of the base metal. Weld metal shall be ground flush with the backing material surface prior to clad restoration and fully covered with the applicable weld metal overlay. The weld metal overlay shall be at least as thick as the lining but no greater than twice its thickness or 3mm above clad surface whichever is less, but consistent with the requirement given in 5.1.7 & 5.6.

5.3 The weld joint in the base plate shall be radiographed as specified in the Engg. Data sheet for detection of cracks/flaws, before clad side welding is carried out. Just before clad restoration welding, the prepared surface shall be inspected by 100% dye penetrant testing.

5.4 The weld overlay procedure shall be qualified as per ASME Code Section IX. However, the chemical analysis requirement shall be as per clause 5.6 hereunder.

5.5 The weld metal overlay shall be relatively smooth with no notches and undercuts that would act as stress intensifiers. Flaws on the surface of the base metal that would interfere with bonding of the overlay shall be removed by grinding.

5.6 A minimum of two samples of the weld metal overlay shall be taken from each overlaid shell section and each head to confirm required chemical analysis. Each manual weld overlay, such as those on nozzles and flanges, shall also be sampled. Analysis at a depth of 2/3rd of the cladding thickness from top of the finished overlay thickness shall conform to the chemistry requirements applicable for the corrosion resistant alloy specified for cladding on the engineering drawing.

5.7 Hardness of martensitic stainless steel (410S) and ferritic stainless steel (405) clad welds and heat affected zones shall have maximum hardness of 252 BHN when welded with 410S welding consumables. The welding procedure shall be suitably qualified with preheat, post heat and/or post weld heat treatment to meet this hardness requirement.

5.8 Bend test of clad weld procedure qualification shall be performed. In case of ferritic stainless steel (410S / 405) the failure of bend test shall be ignored.

5.9 When flush finish is required for the clad restoration side, back grinding of the base material to a depth of 2 mm can be permitted provided this reduction of thickness is compensated by the external reinforcement of the base metal weld.

5.10 Iron Content in Monel overlay deposit shall be limited to 2.5 % maximum.

6.0 INSPECTION

6.1 Shell, dished end and toricones formed from a clad plate shall be ultrasonically examined for lack of bond as detailed below:

6.1.1 100% of surface of Knuckle area and Straight face of dished ends and toricones.

6.1.2 Other than 6.1.1, a minimum of 10% of clad surface, not less than one square foot in each 10 square foot or fraction thereof shall be examined for lack of bond after forming.

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6.1.3 Dished ends / toricone made from clad plate shall be ultrasonically examined after final heat treatment for lack of bond. Inspection and repair procedure shall be as per 6.1.5 below.

6.1.4 100% Ultrasonic examination shall be carried out of areas where attachments are to be welded directly to the cladding. The above areas shall include 50mm width of adjacent areas on both sides of attachment.

6.1.5 Unbonded areas that cannot be encompassed by a 1 inch diameter circle shall be repaired by weld deposit overlay in accordance with 5.1.2 of this specification. When repairs in excess of 3 percent of the total examined area are required, the vessel shall be 100 percent ultrasonically examined. Repaired areas and weld deposit overlay at weld seams shall be 100% liquid penetrant examined. Ultrasonic examination shall be in accordance with ASME A578 with acceptance standard Level C and meeting the supplementary requirement S7.

6.2 All weld deposit overlays, whether by manual or automatic procedures, shall be 100% liquid penetrant examined after each layer of deposit including final layer. Weld deposit overlay on machined surfaces shall be 100% liquid penetrant examined after final heat treatment. Weld deposit overlay shall be spot liquid penetrant examined (a minimum of 10% of the surface, including no less than 1 square foot in each 10 square feet or fraction thereof) after final heat treatment and shop hydrostatic testing.

6.3 Cracks, Fissures and circular defects greater than 1/16 inch (1.6mm) diameter in weld deposit overlay shall be completely removed and repaired. Repaired areas shall be subjected to 100% liquid penetrant examination.

6.4 Liquid dye penetrant examination shall be in accordance with methods described in ASTM E 165.

6.5 Unless otherwise specified Intergranular corrosion test (IGC) shall be carried out as per ASTM A 262. Practice-E for all 300 series material and specimen shall be inspected under a magnification of 200X. The bent specimen shall be free of any cracks or grain dropping. The microstructure shall be submitted to the Authorized Inspector for approval. This test shall be applicable for the following cases:

a) Welding consumable qualification testing and welding procedure qualification testing.

b) Hot forming procedure qualification and heat treatment by using production test coupons testing.

7.0 HEAT TREATMENT

7.1 Cladded vessels or parts of vessels shall be post weld heat-treated as per the requirements of mentioned in the respective engineering data sheets.

7.2 When carbon steel vessels clad with unstabilized austenitic stainless steel other than low carbon grade e.g. SS 304 and SS 316 are to be post weld heat treated, they shall be heat-treated at 510 ± 20°C for a period of 10 hours per inch of thickness or 10 hours whichever is longer.

7.3 Heat treatment of ferritic stainless steel welds and heat affected zone shall be carried out in case of hardness exceeds the allowable limit as defined in clause 5.7 above.

7.4 The following precautions should be observed during heat treatment of clad vessels:

i) The clad material should not be exposed to a direct flame if gas or oil is used as a fuel.

ii) Stainless steel clad vessels require a neutral or oxidising atmosphere during heating.

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iii) Maximum sulphur content in the fuel should not exceed 0.5%.

iv) Marking paint and protective oil should be removed completely from the clad side before heating.

8.0 SURFACE CLEANING

8.1 Stainless steel clad surfaces must be cleaned of oxides, scale, welding flux etc. by using stainless steel wire brushes. Paints shall be removed by using suitable solvent (free from chlorides & fluorides). Oil and grease shall be removed using hot clean water and with a suitable detergent.

8.2 After the completion of fabrication and testing, all stainless steel clad surfaces shall be pickled and passivated in accordance with ASTM A-380. The pickling and passivation procedure shall be submitted to EIL for review.

8.3 The stainless steel surfaces shall be free of iron pick up. This shall be checked by Ferroxyl test in accordance with ASTM A-380.

8.4 Procedure for surface cleaning of Monel clad (SB 127) is as follows:

a) In case Surface Oxidation due to welding and iron pick up on the surface is noticed, the following pickling procedure may be adopted for cleaning the internal surfaces of the Monel cladding.

b) Swab with a solution made of
Water : 1 gal.
Nitric acid (42 D Be): 1 gal
Common Salt : □ lb.
Temp.: 70-100 (ambient) □F
Time : 5 Sec.

c) Rinse clean repeatedly with hot water (180 oF)

d) Second swab with a solution with
Water : 1 gal.
Nitric acid (42 D Be): 1 gal
Temp.: 70-100 (ambient) □F
Time : 5 Sec.

e) Neutralisation by swabbing 1-2% Soda ash solution followed by hot water rinse.

f) Cleaning with saw dust or dry cloth cleaning.

The procedure shall be prequalified before being employed on the vessel.

8.5 Pickling/Passivation procedure of SS410S clad surface shall be as follows:

A) Pre-cleanings of the surface contaminated with oily/greasy substances.

1) Solvent cleaning by uncontaminated organic solvents such as aliphatic petroleums, followed by detergent solution cleaning.

2) Alternately high pressure DM water jetting containing detergent at water pressures upto 70 mpa (10,000 psi) can be used.

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B) Pickling/Passivation Procedure

1) The entire surfaces of the parent metal and weldment shall be swabbed with HNO3. 10-15 percentage by volume containing Na2Cr2O7 H2O (care shall be taken to avoid accidental discharge of the effluent containing Na2Cr2O7 H2O into drains. It should be collected and disposed of properly in accordance with the prevailing environmental regulations) of 2-6% by weight at 21-38□C for 30-60 minutes. However, the actual time period shall be decided by a pickling procedure qualification test. This shall be followed by a DM water rinse with the rinse water pH controlled in the range of 6.5 - 7.5

2) Immediately after acid cleaning and water rinsing, the surface shall be swabbed with a caustic solution containing KOH, 10% by weight percent and KMNO. 4% at a temperature of 70-80□C for 5-15 minutes followed by second DM water rinsing and drying by hot air circulations at a temperature of 121-149□C for 24 hours.

3) Inspection after cleaning

Visual inspection shall be conducted to assess freedom from paint, oil, grease welding flux, slag, heat treating and heat forming scale, dirt, trash, metal and abrasive particles and clamps and other gross contamination under adequate light levels of 2690 IX (250 footcandles) supplemented by boroscope, mirror or other aids as necessary to properly examine inaccessible or difficult to see surfaces.

4) Ferroxyl Test Procedure

Ferroxyl test for free iron is highly sensitive test and should be used only when even traces of free iron or iron oxide might be objectionable. The test can be used on stainless steel to detect iron contamination, including iron-tool marks, residual iron salts from pickling solutions, iron dust, iron deposits in welds, embedded iron or iron oxide etc., The test solution is prepared by first adding nitric acid to distilled water and then adding potassium ferricynide, in the following proportions.

Distilled water	94 Weight %	1000 Cu.cm.
Nitric Acid (60-70%)	3 Weight %	20 Cu.Cm.
Potassium Ferricynide	3 Weight %	30 gms.

Apply solution with an aluminium, plastic, glass, or rubber atomizer having no iron or steel parts, or a swab (atomizer spray is preferred).

The appearance of a blue stain (within 15 sec. of application) is evidence of surface iron contamination (Several minutes may be required for detection of oxide steel) the solution should be removed from the surface as quickly as possible after testing using DM water or, if necessary white vinegar or a solution of 5 to 20 weight % acetic acid and scrubbing with a fiber brush. Flash the surface with DM water several times after use of vinegar or acetic acid.

Note : Potassium ferricynide is not a dangerous poison as are the simple cynides, however care should be taken not to heat treat or bring it in contact with concentrated acids. Solution should be handled with gloved hands only. Always use freshly prepared solution only.

9.0 WELD REPAIR

To the extent possible, the repair of clad stainless steel should be avoided. If unavoidable, the number of repairs at the same spot should not exceed two. The repair procedure should be

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specifically qualified with IGC testing under the supervision of EIL inspection. The acceptance criteria for IGC testing shall be the same as specified for original clad overlay welding for 300 series stainless steel materials.

10.0 HYDRO-TESTING

For hydrostatic test, clean fresh water shall be used unless use of a different medium is approved by the purchaser. In case of austenitic stainless steel lined vessels, chloride content in water shall be less than 25 ppm. Vessel shall be dried thoroughly using hot air, immediately after draining, to prevent the possibility of evaporation and concentration of chlorides and fluorides.

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बायलर क्वालिटी कार्बन स्टील प्लेटों का मानक विनिर्देश

STANDARD SPECIFICATION FOR BOILER QUALITY CARBON STEEL PLATES

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
9	08.02.2022	REVISED AND REISSUED AS STD. SPEC.	JK	KA	NK	SM
8	19.01.2017	REVISED & REISSUED AS STD. SPEC.	RNK	SK/KJH	RKT	RN
7	30.06.2010	REVISED & REISSUED AS STD. SPEC.	KA	RKT	AKM/DM	N.DUARI
6	10.09.09	REVISED & REISSUED AS STD. SPEC.	VB	RKG	AKM	N.DUARI
5	16.04.04	REVISED & REISSUED AS STD. SPEC.	DNN	AKM	SSA	SKG

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Abbreviations:

ASME	American Society of Mechanical Engineers
EN	European Norm
HIC	Hydrogen Induced Cracking
IBR	Indian Boiler Regulations
SSC	Sulphide Stress Cracking

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1.0 SCOPE

1.1 This specification covers requirements for carbon steel plates intended primarily for pressure vessels/heat exchangers. The steel plates shall meet the requirements of ASME Boiler and Pressure Vessel Code Section II (latest). This is intended to supplement the minimum applicable requirements of the material specification indicated in the material requisition.

1.2 Following codes, standards etc shall be followed in their latest edition and addenda, errata, amendments unless specified otherwise:

1.2.1 ASME Sec VIII Div 1

1.2.2 ASME Sec II part A

1.2.3 EN 10163: Delivery Requirements for Surface Conditions.

1.2.4 EN 10204: metallic products – Types of Inspection Documents.

2.0 GENERAL

2.1 Plates supplied to this specification shall conform to specification SA-20 of ASME sec II part A with additional requirements mentioned herein.

2.2 The tolerance on thickness of steel plates shall be positive only.

2.3 Final Rolling shall be lengthwise.

2.4 The plates shall be free from injurious defects and shall have workmanlike finish. Reconditioning/repair of plates by welding shall not be permitted. Surface conditions shall meet requirements of EN 10163 Class A Subclass 3.

3.0 SUPPLEMENTARY TECHNICAL REQUIREMENTS

3.1 All plates shall be supplied in normalised condition except when the applicable material specifications require supply of plates in quenched and tempered condition.

3.2 a. One product analysis of each heat shall be carried out and reported. Chemical analysis shall be as per applicable specification.

b. The carbon content for plates shall not exceed 0.23%.

Additionally, one of the following requirements for carbon equivalent based on heat analysis, shall be also satisfied:

$$Ceq = C + \frac{Mn}{6} \leq 0.42 \quad \text{.....} \quad \text{(Eqn. - 1)}$$

$$Ceq = C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Cu+Ni}{15} \leq 0.43 \quad \text{.....} \quad \text{(Eqn. - 2)}$$

Equation-1 shall be used when applicable material specification specifies C and Mn only.

Equation-2 shall be used when applicable material specifies the above elements or restricted chemical requirements are specified or supplementary requirements S19 and S21 of SA-20 are specified in material requisition.

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For HIC services, these requirements shall be as listed separately.

3.3 Ultrasonic Examination of Plates

a. Plates having thickness 16 mm to 50 mm (both inclusive) shall be examined ultrasonically as per SA-435.

b. For thicknesses above 50mm ultrasonic examination shall be carried out as per SA-578 and shall have acceptance standard of level-B.

c. For quenched and tempered steel plates, ultrasonic examination shall be done after the heat treatment of plates.

3.4 Simulated Heat Treatment of Test Coupons

The following heat treatment shall be conducted on the test coupons representative of heat treated plates before the specified mechanical testing like tensile, bend, impact tests, etc. to meet minimum ASME Sec. II Part - A requirements and these details shall also be recorded on the test certificates.

a. All plates supplied in Normalised condition and intended for hot rolling / hot forming:

Heat Treatment Cycle

One normalising* + One stress relieving as per UCS-56 of ASME Sec. VIII Div.1 complying with UCS-85 of ASME Section VIII Div. 1.

Note: Any other special requirement shall be specified in MR.

b. All plates supplied in Quenched & Tempered condition and intended for hot rolling/ hot forming:

Heat Treatment Cycle

One normalising* + quenched & tempering + One stress relieving as per UCS-56 ASME Section VIII Div.1 complying with UCS-85 of ASME Section VIII Div.1.

Note : Tempering temperature shall be at least 20° C above highest stress relieving temperature.

* Normalising cycle shall be as per Material Test Certificate (MTC).

c. All plates supplied in Quenched & Tempered condition and intended for cold forming:

Heat Treatment Cycle

One stress relieving as per UCS-56 ASME Section VIII Div.1 complying with UCS-85 of ASME Section VIII Div.1.

3.5 Impact test requirements shall be ascertained as per ASME Sec VIII Div.1 or Div.2 as applicable. When required, Impact testing shall be performed as per supplementary requirement of S5 of specification SA-20 and acceptance criteria for energy absorption shall be as per table A2.15 of SA-20.

3.6 If specified in the material requisition, plates shall meet the requirements of Indian Boiler Regulations (IBR).

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3.7 Additional Requirements for High Thickness Plates

Plates above 50mm thickness shall meet following additional requirements:

a. Vacuum Degassing treatment as per the supplementary requirement S1 of specification SA-20. If vacuum degassing is not reported in the test certificates, then through thickness tests as per SA 770 shall be conducted and minimum reduction in area of 35% shall be ensured.

b. Charpy V-notch impact test as per the supplementary requirement S5 of specification SA-20.

Material meant to be used for design temperature warmer than 0°C, impact test shall be carried out at 0°C or MDMT whichever is lower and acceptance criteria for energy absorption shall be as per Table A2.15 of SA-20. Incase the acceptance criteria is not available in SA-20, then applicable design code shall be referred unless otherwise specified in material requisition.

c. Simulated heat treatment of test coupons for all plates as per Clause 3.4 mentioned above.

4.0 CERTIFIED DOCUMENTS

The supplier shall furnish certificates/documents (number of copies as specified in requisition) inclusive of all the following tests required as per specification duly certified by the Inspecting Authority before shipment of plates. The actual values obtained shall be recorded in the test certificates/documents. Material certificates shall conform to EN 10204 Type 3.1/3.2 as specified.

a. Chemical Analysis

b. Mechanical Tests

c. Data of heat treatment i.e. initial temperature, heating rate, soaking temperature, cooling rate, etc.

d. Simulated Heat Treatment of Mechanical Test coupons (S3 of SA-20) at indicated Heat Treatment Cycle (if specified in the requisition or whenever applicable)

e. Ultrasonic Examination (S8 or S12 of SA-20)

f. Charpy V-notch impact tests (S5 of SA-20 if specified in the requisition or whenever applicable)

g. Certification as per IBR (if specified in the requisition)

h. SSC and/or HIC tests (if specified in material requisition)

i Additional tests (if specified in requisition).

5.0 PAINTING AND COATING

No painting/coating of any kind is permitted on the steel plates, except stencil marking. However steel plates shall be carefully protected and packed against any damage during transit and shall be of sea worthy condition.

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6.0 INSPECTION AUTHORITY

Material test certificates, duly certified by Mill's Quality Assurance Department are acceptable i.e. 3.1 certification as per EN 10204. However, if third party inspection is required specifically for plates in material requisition, all test certificates and documents shall be duly certified by the third party. i.e. 3.2 certification as per EN 10204.

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भंडारण टैंक एवं वैसल के लिये वेल्डिंग योग्य स्ट्रक्चरल क्वालिटी स्टील प्लेटों का मानक विनिर्देश

STANDARD SPECIFICATION FOR WELDABLE STRUCTURAL QUALITY STEEL PLATES FOR STORAGE TANKS AND VESSELS

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
7	27.03.2023	REVISED AND REISSUED AS STD. SPEC.	NSK	TK	NK	SM
6	22.09.2017	REVISED AND REISSUED AS STD. SPEC.	SK	KJH	RKT	RN
5	30.06.2010	REVISED AND REISSUED AS STD. SPEC.	KA	RKT	AKM/DM	N.DUARI
4	15.09.09	REVISED & REISSUED AS STD. SPEC.	VB	RKG	AKM,	N.DUARI
3	08.04.04	REVISED & REISSUED AS STD. SPEC.	DNN	AKM	SSA	SKG

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Abbreviations:

EN	:	European Norm
IS	:	Indian Standard

Static Equipment Standards Committee

Convenor : Mr. Nalin Kumar

Members :

Mr. K. Anjaneyulu
Mr. Tarun Kumar (Emp. No. A328)
Mr. Tarun Khurana (Coordinator)
Mr. Anish Trehan
Mr. P V Satyanarayana
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1.0 SCOPE

1.1 This specification covers additional and supplementary requirements for weldable structural quality Steel Plates to IS: 2062 (latest), intended primarily for storage tanks and vessels.

1.2 Following Codes, standards etc. shall be followed in their latest edition and addenda, errata, amendments unless specified otherwise:

1.3 IS:2062, IS:1852, IS:10842 (All Parts)

1.4 EN 10163 (All Parts): Delivery for Surface Condition.

1.5 EN 10204: Metallic products – Types of Inspection Documents.

2.0 GENERAL

2.1 The maximum under tolerance permissible on the thickness of plate shall be 0.25 mm. Tolerances on other dimensions of plates shall be as per IS: 1852.

2.2 Direction of final rolling shall be lengthwise.

2.3 Reconditioning/Repair of plates by welding shall not be permitted. Surface finish shall be ground/flush smooth and shall be free from any surface imperfection.

3.0 SUPPLEMENTARY TECHNICAL REQUIREMENTS

3.1 Heat analysis and product analysis shall be carried out for each heat and chemical composition shall meet the limits as specified in IS: 2062.

3.2 Y-groove weld crackability test as per IS: 10842 shall be carried out for plates conforming to Gr. E250 C having thickness 12mm and above.

3.3 Charpy impact test shall be carried out on the plates having thickness greater than 12 mm conforming to Grade E250 BR, E250 B0 & E250 C. Test temperature and acceptance criteria for energy absorption shall be as mentioned in Table-2 of the specification IS: 2062.
For Thickness less than 12 mm (for materials E250 Gr.B0 & E250 Gr.C), if required, the minimum impact energy values of reduced sizes shall be as per Fig. 2 of the specification IS: 2062.

4.0 CERTIFIED DOCUMENTS

The supplier shall furnish certificates/documents (number of copies as specified in requisition) inclusive of all the following tests required as per specification, duly certified by the Inspecting Authority before shipment of plates. The actual values obtained during tests shall be recorded in the test certificates/documents. Material certificates shall conform to EN 10204 Type 3.1/3.2 as required.

a. Chemical Analysis
b. Mechanical Tests
c. Data of heat treatment
d. Charpy V-notch impact tests
e. Y- groove crackability test

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5.0 PAINTING AND COATING

No painting/coating of any kind is permitted on the steel plates except stencil marking. However steel plates shall be carefully protected and packed against any damage during transit and shall be of sea worthy conditions.

6.0 INSPECTION AUTHORITY

Material test certificates, duly certified by Mill's Quality Assurance Department are acceptable i.e. 3.1 certification as per EN 10204. However, if third party inspection is required specially for plates in requisition, all test certificates and documents shall be duly certified by the third party. i.e. 3.2 certification as per EN 10204.

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क्लेड प्लेटों के लिए मानक विनिर्देश

STANDARD SPECIFICATION

FOR CLAD PLATES

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
6	23.11.23	REVISED AND REISSUED AS STD. SPEC.	PSV	TK	KA/NK	MN
5	26.06.18	REVISED AND REISSUED AS STD. SPEC.	SK	TK	KJH	RKT
4	30.07.10	REVISED AND REISSUED AS STD. SPEC.	KA	RKT	AKM/DM	N.DUARI
3	30.09.09	REVISED AND ISSUED AS STD. SPEC.	KA	RKG	AKM	N. DUARI
2.	29.03.04	REAFFIRMED & REISSUED AS STD. SPEC.	DNN	AKM	SSA	SKG

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Abbreviations:

ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing & Materials
BHN	Brinell Hardness Number
EN	European Standards
IBR	Indian Boiler Regulations
IGC	Inter Granular Corrosion

Static Equipment Standards Committee

Convenor: Mr. Nalin Kumar

Members: Mr. K. Anjaneyulu
Mr. Tarun Kumar (Emp. No. A328)
Mr. Tarun Khurana (Coordinator)
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1.0 SCOPE

This specification covers clad plates of a carbon steel or low alloy steel base integrally and continuously bonded on one side with a layer of chromium steel, chromium nickel steel, nickel chromium alloys and Monel. The base as well as the alloy cladding metal shall meet the requirements of ASME Boiler and Pressure Vessel Code Section II and EIL specifications 6-12-0011, 6-12-0017 and 6-12-0018. This is intended to supplement the minimum applicable requirements of material specification indicated in the material requisition.

2.0 GENERAL

2.1 Plates supplied to this specification shall conform to the latest issue of applicable specification SA-263, SA-264, SA-265, SA-20 and SA-578 with additional requirements mentioned herein.

2.2 The tolerance on thickness of base metal as well as cladding metal shall be positive only.

2.3 Final rolling shall be lengthwise.

2.4 Outside surface of the base metal shall meet requirements of EN 10163 class- A, sub class 3.

2.5 Unless otherwise specified in the requisition, the surface of alloy cladding shall have No.1 finish for hot rolled and 2B finish for cold rolled, as per SA-480.

2.6 Acid pickling shall be done for chromium nickel stainless steel cladding as per ASTM A-380. Manufacturer's standard practice for pickling shall be followed for chromium steel cladding.

2.7 Repair of cladding surface by welding shall not exceed 3% of surface area of individual plate.

2.8 The integral & continuous cladding shall be produced by roll bonding or explosion bonding process with bond integrity Class 1.

3.0 SUPPLEMENTARY TECHNICAL REQUIREMENTS

3.1 Chemical composition of the clad metal shall meet the requirement of the applicable material specification. But in case of straight chrome (13% Cr) cladding material, carbon content shall be less than 0.06%.

3.2 Unless otherwise specified in the requisition, heat treatment of the clad steel plate shall be as per the applicable clad plate material specification. All stabilized grades of austenitic Stainless Steel (SS 321, SS 347etc.) clad plate shall be in stabilization heat treated condition in addition to solution annealing.

3.3 Alloy cladding representative of each heat shall be subjected to inter-granular corrosion test as per ASTM A-262, Practice E for all 300 series material. IGC test shall be carried out after cladding process.

3.4 During the mill tension test of all clad steel plates, the cladding shall be removed and the tensile properties shall meet the requirements of the base material. Test specimen shall be taken transverse to rolling direction.

3.5 For all clad steel plates, shear test shall be conducted. Strength of bond shall be at least 20,000 psi in shear.

3.6 All clad steel plates shall be ultrasonically examined in accordance with SA-578 with acceptance standard level-C and shall meet the supplementary requirements S1. Integrity of clad plate shall be ensured Quality level: class-1 as per SA 263 or SA 264 or SA 265.

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3.7 The ultrasonic examination shall be conducted after the specified final heat treatment of the clad steel plates.

4.0 CERTIFIED DOCUMENTS

The supplier shall furnish the following certificates/documents inclusive of all tests required as per specification, duly certified by the Inspecting Authority before shipment of plates. The actual values obtained shall be recorded in the test certificates/documents. Purchaser's technical requirements shall be clearly shown in the test certificates. Material certificate shall conform to EN10204 Type 3.1/3.2 as specified.

4.1 Chemical Analysis for both cladding metal and base metal.

4.2 Mechanical test for base metal and shear strength for clad plate.

4.3 Data of heat treatment of clad plate i.e. initial temperature, heating rate, soaking temperature/time, cooling rate etc. For simulation heat treatment of base plate test coupon, chart of heat treatment is to be attached to Mill's Certificate.

4.4 Ultrasonic examination.

4.5 Inter-granular corrosion test.

4.6 Finish for a cladding & base metal.

4.7 Method of cladding.

4.8 Certification as per IBR (if specified in material requisition)

4.9 Date of manufacture.

4.10 Pickling procedure.

5.0 PAINTING AND COATING

Plates are not to be painted/coated but shall be covered with plastic foils or paper or by other means for careful protection and shall be packed against any damage during transit and sea weather conditions.

6.0 INSPECTION AUTHORITY

Material test certificates, duly certified by Mill's Quality Assurance Department are acceptable i.e. 3.1 certification as per EN 10204. However, if third party inspection is specified in material requisition, all test certificates and documents shall be duly certified by the third party i.e. 3.2 certifications as per EN 10204.

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2.25 % क्रोमियम– 1% मालीब्डेनम स्टील प्लेटों का मानक विनिर्देश

STANDARD SPECIFICATION FOR 2.25% Cr – 1% Mo STEEL PLATES

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
5	08.02.22	REVISED AND REISSUED AS STD SPEC.	TK	KA	NK	SM
4	22.12.15	REVISED AND REISSUED AS STD SPEC.	TK	KJH	RKT	SC
3	15.09.09	REVISED AND REISSUED AS STD.SPEC.	VB	RKG	AKM	N.DUARI
2	12.04.04	REVISED AND REISSUED AS STD. SPEC.	DNN	AKM	SSA	SKG
1	05.01.01	REVISED AND REISSUED AS STD. SPEC.	DNN	AKM	CRMN	MI
0	01.03.96	ISSUED AS STANDARD SPECIFICATION	DNN	CRMN	VC	AS

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Abbreviations:

ASME	:	American Society of Mechanical Engineers
ASTM	:	American Society for Testing and Materials
ACC	:	Accelerated Cooling Cycle
BHN	:	Brinell Hardness Number
EN	:	European Standard
IBR	:	Indian Boiler Regulations
ISR	:	Intermediate Stress Relieving
N	:	Normalised
PWHT	:	Post Weld Heat Treatment
T	:	Tempered

SMED-I (Static Equipment) (Group code 12,13,15,26)

Mr. Nalin Kumar (Convener)
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1.0 SCOPE

1.1 This specification covers additional and supplementary requirements of SA-387 / SA-387M Gr.22 Cl.2 (ASME Sec. II Part A) plates intended primarily for pressure vessels/heat exchangers.

1.2 Following codes, standards etc. shall be followed in their latest edition and addenda, errata, amendments unless specified otherwise:

1.2.1 ASME Sec II part A

1.2.2 EN 10163: Delivery Requirements for Surface Conditions.

1.2.3 EN 10204: Metallic products – Types of Inspection Documents

1.2.4 API RP 934

2.0 GENERAL

2.1 Plates supplied to this specification shall conform to specification SA-20 with additional requirements mentioned herein.

2.2 The tolerance on thickness of steel plates shall be positive only.

2.3 Final rolling shall be lengthwise.

2.4 The plates shall be free from injurious defects and shall have workmanlike finish Reconditioning, repair of plates by welding shall not be permitted. Surface conditions shall meet requirements of EN 10163 (Part: 2) Class A Subclass 3.

3.0 SUPPLEMENTARY TECHNICAL REQUIREMENTS

3.1 The steel shall be made by electric furnace or basic oxygen process and shall be vacuum degassed.

3.2 Plates shall be supplied in Normalised and Tempered condition (N+T). Alternatively accelerated cooling from an austenitizing temperature by liquid quenching followed by tempering is also acceptable (N + ACC. + T).

Tempering temperature, in no case, shall be less than PWHT temperature.

3.3 The tensile strength at room temperature shall not exceed 690 N/mm^2 and the yield strength should not exceed 620 N/mm^2

3.4 Certified chemical analysis shall be carried out on heat as well as on product. In addition to chemical analysis required by applicable material specification, analysis of Ni, Cu, As, Sn, and Sb shall be reported. However, Copper and Nickel content shall be limited to 0.20% (max.) and 0.30% (max.) respectively. Material shall have a 'J' factor, as defined as (Si + Mn) x (P + Sn) x 10^4 less than or equal to 100, where the concentration of elements are in percent.

3.5 The material shall have an austenitic grain size 6 (Six) or finer as determined by the method of ASTM E 112.

3.6 Ultrasonic Examination of Plates

Plates shall be ultrasonically examined after specified heat treatment as per following:

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a. Plates having thickness 16 mm to 50 mm (both inclusive) shall be examined ultrasonically as per SA-435.

b. For thicknesses above 50 mm ultrasonic examination shall be carried out as per SA-578 and shall have acceptance standard of level-B.

3.7 Simulated Heat Treatment on Test Coupons

All test specimens representative of heat treated (N+T) / (N+ACC+T) plates shall be subjected to simulated heat treatments. Simulated heat treatments shall include all heat treatments involved during fabrication of the equipment. This shall include PWHT of the equipment plus two additional PWHT cycle (one for shop repair and one for future) including all intermediate heat treatments (ISR). Simulated heat treatment shall be conducted on the test coupons before the specified mechanical tests like tensile, bend, impact tests etc. All tests shall be carried out with minimum PWHT condition and maximum PWHT condition including all ISR. These details shall be recorded on test certificates clearly indicating all heat cycle conditions. Simulated heat treatments shall be decided by fabricator based on fabrication technique employed.

3.7.1 Impact tests (Charpy V-Notch) shall be carried out at minus 29 °C with 55J average value for three specimens and with no single specimen below 48J. The orientation of test specimen shall be transverse to the direction of final rolling.

3.8 a) Impact energy versus temperature (transition) curves shall be developed for each heat of plate.

i) A minimum of eight sets of three impact tests of completely heat treated material with minimum PWHT condition and maximum PWHT condition including all ISR shall be performed for each curve. Sample location shall be as specified in ASME code.

ii) The eight sets of impact test shall be performed at different temperature but shall include minus 29 °C. The generated transition curve shall clearly define the transition zone and the upper shelf. The maximum test temperature shall correspond to the upper shelf energy level.

b) Step cool tests shall be performed on completely heat treated material with minimum PWHT condition and maximum PWHT condition including all ISR from each heat of plate.

i) Samples shall be heated from 316°C to 593°C with maximum rate of 56°C.

ii) Step cooling shall be in accordance with the following temperatures, holding times and cooling rates to the next lower temperature:

Temperature °C	**Holding time Hrs.**	**Cooling rate to the next temperature °C/hr**
593	1	5.6
538	15	5.6
524	24	5.6
496	60	2.8
468	100	27.8
315	-	air cooling

iii) Impact test of each step cooled sample shall be performed and transition curve shall be developed following the procedure outlined in clause 3.8(a) above.

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c) Acceptance criteria for materials shall be in accordance with the following:

$vTr\ 55 + 3.0 \times \Delta\ vTr\ 55_{sc} \leq 10\ ^{\circ}C$

Where $vTr\ 55$ = Charpy V-notch 55J impact energy transition temperature of completely heat treated specimens before step cooling.

$\Delta vTr\ 55_{sc}$ = The shift in Charpy V-notch 55J impact energy transition temperature after step cooling tests.

3.9 Hardness of plates shall not exceed 225 BHN.

3.10 Additional Tests

Following additional tests shall be applicable, if specified in Purchase/tender documents:

a. High temperature tension test for each heat as per S7 of SA-20 / SA- 20M at temperature specified in Material Requisition with maximum PWHT condition including all ISR. Test results shall also be furnished in the test certificates. The acceptance criteria shall be 90 % of the value specified in AME Sec II Part D Table U for the test temperature.

b. Through-thickness tension test for plates 25 mm and above in thickness shall be conducted with frequency and test procedure as per SA-770 / SA-770M with maximum PWHT condition including all ISR and determination of reduction of area as per SA-370. Minimum reduction area shall not be less than 35%. Test results shall also be furnished in the test certificates.

4.0 CERTIFIED DOCUMENTS

The supplier shall furnish certificates/documents (number of copies as specified in the requisition) inclusive of all the following tests required as per specification, duly certified by the Inspecting Authority before shipment of plates. The actual values obtained shall be recorded in the test certificates/documents. Material certificates shall conform to EN 10204 Type 3.1/3.2 as specified.

a. Chemical Analysis

b. Mechanical Tests

c. Additional Tests (if specified in the requisition)

d. Data of heat treatment i.e. initial temperature including heat rate soaking temperature soaking time /cooling rate, etc.

e. Simulated Heat Treatment of Mechanical Test Coupons (S3 of SA-20 / SA-20M)

f. Ultrasonic Examination (S 12 of SA-20 / SA-20M)

g. Certification as per IBR (if specified in the requisition).

h. Charpy V-notch impact tests (S5 of SA-20 / SA-20M)

i. Impact - temperature transition curves and acceptance criteria.

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5.0 PAINTING

No painting of any kind is permitted on the steel plates except stencil marking. However, steel plates shall be carefully protected and packed against any damage during transit and shall be of sea-worthy condition.

6.0 INSPECTION AUTHORITY

Material test certificates, duly certified by Mill's Quality Assurance Department are acceptable i.e. 3.1 certification as per EN 10204. However, if third party inspection is specified in material requisition, all test certificates and documents shall be duly certified by the third party i.e. 3.2 certifications as per EN 10204.

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स्टेनलैस स्टील प्लेटों का मानक विनिर्देश

STANDARD SPECIFICATION FOR STAINLESS STEEL PLATES

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
9	29.05.2023	REVISED AND REISSUED AS STD. SPEC.	NSK	TK	NK	SM
8	22.09.2017	REVISED AND REISSUED AS STD. SPEC.	RNK	SK/KJH	RKT	RN
7	30.06.2010	REVISED AND REISSUED AS STD. SPEC.	KA	RKT	AKM/DM	N.DUARI
6	25.09.09	REVISED AND REISSED AS STD. SPEC	VB	RKG	AKM	N.DUARI
5	16.04.04	REVISED AND REISSED AS STD. SPEC	DNN	AKM	SSA	SKG

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Abbreviations:

ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing & Materials
EN	European Standard
BHN	Brinell Hardness Number
SS	Stainless Steel

Static Equipment Standards Committee

Convenor : Mr. Nalin Kumar
Members :

Mr. K.Anjaneyulu
Mr. Tarun Kumar
Mr. Tarun Khurana (Coordinator)
Mr. Anish Trehan
Mr. P V Satyanarayana
Mr. Saikat Chakraborty
Mr. Piyush Suryavanshi
Mr. Mittal Patel
Mr. Srikanth Karanam
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Mr. Avdhesh Agrawal (SCM-Inspection)

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1.0 SCOPE

1.1 This specification covers requirements for chromium, chromium-nickel and chromium-manganese-nickel stainless and heat resisting steel plates intended primarily for pressure vessels/heat exchangers. The steel plates shall meet the requirements of ASME Boiler and Pressure Vessel Code Section II (latest). This is intended to supplement the minimum applicable requirements of the material specification indicated in the material requisition.

1.2 Following codes, standards etc. shall be followed in their latest edition and addenda, errata, amendments unless specified otherwise:

1.2.1 ASME Sec II Part A.

1.2.2 EN10163 (All parts): Delivery Requirements for Surface Conditions.

1.2.3 EN 10204: Metallic products- Types of Inspection Documents.

2.0 GENERAL

2.1 Plates supplied to this specification shall conform to specification SA-480 with additional requirements mentioned herein.

2.2 Adequate tolerance shall be considered on Plate length and width for shear and plasma cutting. The tolerance on thickness of plates shall be positive only.

2.3 Final Rolling shall be lengthwise.

2.4 Plates shall have no. 1 finish on both sides with reference to SA-480. Cold Rolled plates, if permitted by Requisition, shall have no. 2B finish. Repair/reconditioning of plates shall not be permitted.

3.0 SUPPLEMENTARY TECHNICAL REQUIREMENTS

3.1 All plates shall be supplied in hot-rolled, fully annealed and pickled condition. All stabilized grades of Stainless Steel (SS 321, SS 347etc.) shall be given stabilization heat treatment in addition to solution annealing. The soaking temperatures for stabilization heat treatment shall be 915°C ± 10°C and soaking period shall be minimum of 4 hours (2 hours for thickness $\leq$ 3.5mm).

3.2 Unless specified otherwise in material requisition, plate representative of each heat shall be subjected to intergranular corrosion tests as per ASTM A-262 Practice E for all the 300 series materials. The bend test specimen shall be examined at a magnification of 200 x; and bent specimen shall be free of any cracks or grain dropping.

3.3 For straight chrome (13% Cr) material, maximum carbon content shall not exceed 0.06%. Hardness of UNS no.S41000, S41008 and S40500 shall be less than 241 BHN unless otherwise specified in the ASME BPV Section II, material specification.

3.4 a. Plate having thickness 16 mm to 50 mm (both inclusive) shall be examined ultrasonically as per SA-435.

b. For thickness above 50mm ultrasonic examination shall be carried out as per SA-578 and shall have acceptance standard of level-B.

3.5 All mandatory tests as per material specification shall be carried out. However tension test specimen shall be from finished material and shall be selected in transverse direction.

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4.0 CERTIFIED DOCUMENTS

The supplier shall furnish certificates/documents (number of copies as specified in requisition) inclusive of all the following tests required as per specification duly certified by the Inspecting Authority before shipment of plates. The actual values obtained shall be recorded in the test certificates/documents. Material certificates shall conform to EN 10204 Type3.1/3.2 as required.

a. Chemical Analysis

b. Mechanical Tests

c. Data of heat treatment i.e. initial temperature, heating rate, soaking temperature, cooling rate, etc.

d. Ultrasonic Examination

e. Intergranular corrosion test for 300 series

f. Intergranular corrosion test for series other than 300 (whenever specified in the requisition)

g. Type of finish for plate surfaces

5.0 PAINTING AND COATING

Plates are not to be painted/coated but shall be covered with plastic foils or paper or by other means for careful protection and shall be packed against any damage during transit and sea-weather conditions.

6.0 INSPECTION AUTHORITY

Material test certificates, duly certified by Mill's Quality Assurance Department are acceptable i.e. 3.1 certification as per EN 10204. However, as specified, if third party inspection is required specifically for plates, all test certificates and documents shall be duly certified by the third party. i.e. 3.2 certification as per EN 10204.

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वायरमेश डेमिस्टर्स के लिए सामान्य विनिर्देशन

GENERAL SPECIFICATION FOR WIRE MESH DEMISTERS

Rev. No.	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
5	23.03.2026	REVISED & ISSUED AS STANDARD SPECIFICATION	HD	PG	AK	NK
4	31.12.2019	REVISED & ISSUED AS STANDARD SPECIFICATION	HD	PB	KJH	RKT
3	07.11.2013	REVISED & ISSUED AS STANDARD SPECIFICATION	HD	P.Goel /SKM	AP	SC
2	17.09.2008	REAFFIRMED & REISSUED AS STANDARD SPECIFICATION	VDY	HCN	AKG	VC

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Abbreviations:

ASME	:	American Society of Mechanical Engineers
AISI	:	American Iron and Steel Institute
AIA	:	Authorized Inspection Agency
TIG	:	Tungsten Insert Gas
GTAW	:	Gas Tungsten Arc Welding

Mass Transfer Mechanical & System Standards Committee

Convenor: Mr Arun Kumar

Members:
Mr Inder Kumar
Mr Pankaj Gandhi
Mr Jeevan Agrawal (Coordinator)
Mr. VinodKumar Yadav
Mr. Urmilesh Tiwari
Mr. Aasheesh Handa (Projects)
Mr. Prabhakar Choudhary (SMMS)
Mr. Avdhesh Agrawal (SCM)

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1.0 SCOPE

This specification covers the general requirements for design, fabrication, inspection and supply of knitted wiremesh assemblies (Demisters/ Mist Eliminators) in duly packed condition.

The demister assembly(ies) shall consist of wiremesh pad, top and bottom holding grids, major beams (if required), bolting/clamps/tie rods/J-bolts/spacer pipes and any other components which are essential to complete the demister assembly to be installed on supports welded inside the Columns/ Vessels.

2.0 REFERENCES

Following documents/standards/specifications in their latest edition shall form the basis of design, fabrication, inspection, testing and acceptance of the demister:

i	Demister data sheet	-	As per Material Requisition
ii	6-14-0009	-	Packing, Marking and Shipping Specifications for Trays and Tower Internals
iii	7-12-0036*/Licensor's standards	-	Details of Wire Mesh Demister Supports (as applicable)
iv	7-12-0001	-	Vessel Tolerances
v	ASME Sec. II	-	Material Specifications
vi	ASME Sec. VIII Div.1	-	Boiler and Pressure Vessel Code
vii	ASME Sec. IX	-	Welding Qualifications
viii	AISI	-	Material Specification for Wires, Spacer Rods etc.

(* Standard 7-12-0036 shall be used for reference only. Final design of demister supports shall be as per vendor design standards/calculations and shall be submitted to EIL for review prior to start of fabrication.)

3.0 MATERIAL

All the material used for fabrication & supply of Demister shall be new & of first quality and shall be duly supported with material test certificates.

4.0 DESIGN & FABRICATION

4.1 Basic details such as demister specification (surface area, voidage, and bulk density), material of construction, supporting arrangement, manhole size etc. shall be as per drawings/ specification furnished by EIL/ Licenser.

4.2 In the General Arrangement drawing, Demister Vendor shall indicate the following outside dimensions for square / rectangular demister and diameter for circular demister:

a) The oversized outside dimensions / diameter to take care of the tolerances/column ovality defined in applicable Code / EIL Standard 7-12-0001/ drawings for snug fitment. These dimensions /diameters shall be used for shop inspection of the demister mock up assembly.
b) The final fit-up dimensions / diameter.

4.3 Demister assembly shall be designed for a net upward thrust of 100 kg/m^2 (uniform) due to the vapour loadings. The assembly should also be able to withstand a concentrated live load of 60 kg at mid point of any of the demister sub-assembly/component.

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In addition to the above loading, demister shall be designed to withstand the additional pressure surges, if specified.

4.4 All demister sub-assemblies shall be sized to facilitate installation inside the column/vessel through the manhole. These shall rest on support ring/cleats welded inside vessel/column by others.

4.5 Tentative details of the support rings/cleats, mid span beams (wherever required) grid details & fixing arrangements shall be as given in the referenced documents listed in clause 2.0. Vendor is advised to follow the same for the design/fabrication in their scope of work. Vendor may, however, suggest any improvement in the design. Such suggestions will however be reviewed/ approved by EIL/Licenser/ Client and can be followed by the vendor only on written acceptance by EIL/Licenser/ Client.

4.6 All internal bolting shall be minimum M10 size and shall be compatible with Demister material. All bolting shall be provided with locknuts.

4.7 Welding

All welding shall be done by TIG/ GTAW Process only. Welding wherever required, is to be done by qualified and approved welders using the suitable fillers and fluxes recommended for materials used for the fabrication. Where TIG welding cannot be done at site due to access/Practical Constraints, Metal Arc welding may be carried out after taking approval from Engineer-In-Charge. Resistance welding is permitted for minor details where weld is a non-strength weld. Gas or Carbon arc welding shall not be used.

5.0 SPARES

5.1 Mandatory Spares

Following operational spares shall be supplied as Mandatory Spares:

Fasteners - bolts, nuts, clamps, spacer pipes, J-bolts etc.	-	10% (minimum 4 Nos. of each type/size)

5.2 Erection & commissioning Spares

In addition to Operational spares as indicated above, vendor shall also supply following Constructional spares as Erection and Commissioning spares:

Fasteners - bolts, nuts, clamps, spacer pipes, J-bolts etc.	-	10% (minimum 4 Nos. of each type/size)

6.0 INSPECTION AND TESTING

6.1 The material, fabrication and trial assembly shall be Inspected by the EIL/ Licenser/ Client and/or their AIA.

6.2 Complete demister assemblies (along with grids) and Beams etc. shall be assembled as floor mock-up & shall be offered for final inspection. Assembly shall be checked for oversized dimensions as per the approved drawing.

7.0 PACKING & FORWARDING

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The individual components shall be marked with proper item/part number as given in the drawings to facilitate identification at site. Demister shall be wrapped in polythene and supplied in polythene lined wooden cases as per either EIL std. specification or Vendor's proven standard, for damage free transportation. For all other requirements refer EIL Specification No. 6-14-0009.

8.0 GUARANTEE

Vendor shall guarantee the design, performance (i.e. separation efficiency and pressure drop), material, fabrication and workmanship for the demisters for the period as outlined in General Purchase Conditions.

9.0 DOCUMENTATION

Documents/drawings shall be furnished as per the `Vendor Data Requirement' enclosed with requisition / FEED document. All the drawings prepared by the vendor shall be on AUTOCAD.

10.0 AS BUILT DOCUMENTATION

Nos. and type of documents to be supplied as 'As-built' document shall be per 'Vendor Data Requirement' enclosed with requisition / feed document.

Shop changes made by Vendor after approval of drawings under Code-1 by EIL and deviations granted in deviation permits, if any, shall be marked in hard copy of drawings which shall then be stamped 'As-built' by the vendor. These 'As-built' drawings shall be reviewed and stamped by Authorized Inspector also. Vendor shall prepare scanned image files of all marked-up 'As-built' drawings. Vendor shall also incorporate the above changes in the native soft files of the drawings.

Authorized Inspector shall ensure/certify completeness of Vendor's Final/As-built documents before equipment dispatch.

In addition, vendor shall also incorporate site changes, if any, based on mismatch observed at site and resubmit the 'As-built' documents.

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विक्रेता कार्यशाला में सकारात्मक सामग्री पहचान के लिए मानक विनिर्देश

STANDARD SPECIFICATION FOR POSITIVE MATERIAL IDENTIFICATION (PMI) AT SUPPLIER'S WORKS

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
4	23.12.21	REVISED AND RE-ISSUED	HKM	CS	RK	S Majumdar
3	19.09.16.	REVISED AND RE-ISSUED	TKK	HP	RKS	RN
2	20.10.11	REVISED AND RE-ISSUED	RKS	SCG	AKC	DM
1	15.07.08	REVISED AND RE-ISSUED	NKR	SSL	SKP	VC
0	07.12.00	ISSUED AS STANDARD SPECIFICATION	AKC	AKB	AKB	MI

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Abbreviations:

API	:	American Petroleum Institute
AS	:	Alloy Steel
HIC	:	Hydrogen Induced Cracking
LSTK	:	Lump Sum Turn Key
NACE	:	National Association of Corrosion Engineers
PMI	:	Positive Material Identification
RTJ	:	Ring Type Joint
Sch	:	Schedule
SS	:	Stainless Steel
TPI or TPIA	:	Third Party Inspection Agency

Inspection Standards Committee

Convenor: Mr. Rajeev Kumar

Members: Mr. Rajesh Sinha
Mr. Himangshu Pal
Mr. R. Muthuramalingam (RPO Representative)
Mr. Chandrashekhar
Mr. Avdhesh Agrawal
Mr. P V Satyanarayana (Engg. Representative)
Mr. Mahendra Mittal

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1.0 SCOPE

1.1 This specification applies to the requirements for Positive Material Identification (PMI) to be performed at the Supplier's works on Metallic Materials procured either directly by the Owner/EIL/ LSTK contractor or indirectly through the sub-Suppliers.

1.2 This specification covers the procedures and methodology to be adopted to assure that the chemical composition of the material is consistent with the material specifications as specified in purchase documents using 'Alloy Analyzer' at the time of final inspection before dispatch.

1.3 The scope of this specification shall include but shall not be limited to Positive Material Identification (PMI) to be performed on Materials listed below:

1.3.1 For alloy Steel materials as below:

- Alloy Steel Pipes including Clad Pipes
- Alloy Steel Flanges & Forgings
- Alloy Steel Fittings including Clad Fittings
- Alloy Steel Fasteners
- Alloy Cast & Forged steel valves
- Alloy Steel Instrumentation Items (Control Valves, Safety Valves etc.)
- Longitudinal Pipe & Fittings Welds.
- Gaskets (for Ring Type Joints)

1.3.2 For Carbon Steel materials as below:

- All Carbon Steel Piping items under NACE or HIC or H2 or Wet Hydrogen Sulfide (H_2S), Hydrofluoric acid (HF), Sulfuric acid (H_2SO_4) services etc.
- Carbon Steel flanges and valves (Rating 900# and above)
- LTCS items

Following items shall be excluded from scope of PMI examination.

- Gaskets other than for Ring Type Joints
- Internal Components of Valves

1.4 All grades of material supplies including Stainless Steels shall be liable for PMI test at site. In case of any defective materials being found at site, the Supplier shall be responsible to effect replacement of such defective materials at project site without any delays to the satisfaction of EIL site RCM (Resident Construction Manager).

2.0 REFERENCE DOCUMENTS

2.1 API Recommended Practice 578 (Third Edition, Feb 2018) - Material Verification Program for New and Existing Alloy Piping Systems.

3.0 DEFINITIONS

3.1 **Supplier:** Any Supplier or Manufacturer on whom an order is placed for the supply of referred items. This definition shall also include any sub-Supplier or manufacturer on whom a sub-order is placed by the Supplier.

3.2 **Inspection Lot:** A group of items offered for inspection covered under same size, Heat and Heat treatment lot.

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3.3 **Alloy Material:** Any metallic material (including welding filler materials) that contains alloying elements such as Chromium, Nickel, Molybdenum or Vanadium, which are intentionally added to enhance mechanical or physical properties and/or corrosion resistance.

4.0 PMI EXAMINATION

4.1 The Supplier shall submit a procedure of PMI to comply with the requirements of this Specification. Approval of PMI Procedure shall be obtained from Owner / EIL / TPIA prior to commencing manufacture / inspection of product.

4.2 PMI examination of materials is independent of any certification, markings or colour coding that may exist and is aimed at verifying that the alloy used are as per specified grades.

4.3 The Supplier shall identify all incoming alloy materials and maintain full traceability of all alloy
materials, including all off-cuts. Transfer of identification marks shall be undertaken prior to cutting to ensure maintenance of identification on off-cuts.

4.4 The Supplier shall ensure that all alloy materials are segregated and stored in separately identified locations to prevent the mix up of materials of different alloy specifications or alloy material with carbon steel. Non ferro-magnetic materials shall be segregated at all times from ferro-magnetic materials.

4.5 PMI examination is subject to surveillance inspection by Owner / EIL / TPIA.

5.0 ACCEPTABLE METHODS FOR PMI

5.1 The method used for PMI examination shall provide a quantitative determination of the alloying elements like Chromium, Nickel, Molybdenum or Vanadium in Alloy Steel items.

5.2 Instruments or methods used for PMI examination shall be able to provide quantitative, recordable, elemental composition results for positive identification of alloying elements present.

5.3 The acceptable instruments for alloy analyzer shall be either "Portable X-Ray fluorescence" or "Optical Emission" type each capable of verifying the percentage of alloy elements within specified range.

5.4 Chemical spot testing, magnets, alloy sorters and other methods using eddy current or triboelectric testing methods are not acceptable for PMI examination.

5.5 The PMI instrument used shall have the sensitivity to detect the alloying elements in the specified range.

5.6 All PMI instruments shall have been serviced within a 6 month period of the time of use to verify the suitability of batteries, sources etc., and the data of the last service shall be stated on the PMI Report Form (Sample enclosed).

5.7 Each analyzer must be calibrated according to the manufacturer's specification at the beginning and end of each shift. Instrument must be checked against known standard for each alloy type to be inspected during the shift.

5.8 Certified samples, with full traceability, of a known alloy materials shall be available for use as a random spot check on the instrument calibration.

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5.9 The surfaces to be examined shall be prepared by light grinding or abrasive paper and solvent cleaner. Evidence of Arc burn resulting from examination shall be removed by light grinding or abrasive paper. No permanent marks, which are injurious to the usage of product in service, are acceptable.

5.10 Alloy Steel ring type joint Gaskets shall be inspected by using portable X-Ray fluorescence instrument.

5.11 Testing shall be done as per the procedures outlined by the manufactures of alloy analyzer being used. Modification of these procedures if any must be approved by Owner/EIL.

5.12 The persons performing PMI shall demonstrate their capabilities to the satisfaction of Owner/EIL/TPIA visiting engineer. If the Supplier has qualified operator on their rolls, he may perform the examination. Otherwise PMI examination shall be sub-contracted to an independent testing agency approved by EIL.

5.13 Whenever material is identified as not meeting requirements by the visiting engineer a rejection note shall be issued.

6.0 EXTENT OF PMI EXAMINATION

Following sampling plans shall be applicable for PMI examination of various items.

A. Flanges, Fittings Valves, RTJ Gaskets - 100% by Supplier and 10% @ random witness by EIL/TPIA

B. Pipes - 100% (for pipes procured from traders)
2 random samples drawn from each Size/Heat/Lot (for pipes procured directly from mills)

C. Fasteners -

Lot Size	**Sample Size for** EIL/TPIA	Sample size for Supplier
Upto 100	2% (Min 2)	Min 10%
101 to 500	1% (Min 3)	Min 10%
501 and above	0.5% (Min 5)	Min 10%

Note:

a. For Welded Pipes and Fittings and flanges like Fig 8, Spacer and blinds etc. PMI shall be performed on Base Metal as well as weldments.

7.0 ACCEPTANCE CRITERIA

7.1 Base Metal

PMI test results showing presence of characteristic elements upto 10% less than the minimum specified value in the material specification and upto 10% more than the maximum specified value in the material specification shall be acceptable.

7.2 Deposited Weld Metal

For deposited weld metal using welding consumables matching with base metals, the recorded presence of characteristic elements upto 12.5% less than the minimum specified value in the

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welding consumable specification and upto 12.5% more than the maximum specified value in the welding consumable specification shall be acceptable.

8.0 REJECTION CRITERIA

8.1 If PMI test results fall outside the acceptable range as specified in 7.0 above, the supplier shall obtain a quantitative check analysis performed by a laboratory acceptable to Owner / EIL / TPIA for a complete chemical analysis. Alternatively, the item can be tested with a spark analyser for verification. Results of this analysis shall be submitted to Owner / EIL / TPIA for final decision.

Decision of Owner / EIL / TPIA shall be final in this regard.

8.2 If any sample drawn to PMI test on the basis of percentage selection as per 6.0 above, fails to meet specification requirements, 100% of items of lot shall be tested for PMI by supplier. Any failure thereafter during sample check shall mean rejection of whole lot.

9.0 RECORDING AND DOCUMENTATION

The results of PMI examination shall be recorded in a Report Format as enclosed with this specification.

10.0 MARKING

10.1 All alloy materials tested by PMI shall be identified using either of the following methods by indicating "PMI OK"

a) Bar Code/Hologram Sticker
b) A low stress stamp marking

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	POSITIVE MATERIAL IDENTIFICATION REPORT BULK MATERIALS							Page of
Project:		Client						Job No.
PMI Report No.		Supplier/Sub-Supplier						
Purchase Order No.		Testing Agency						
Purchase Requisition No:		PMI Location						
Bulk Item Type (as per Requisition)								
Material Specification/Grade Number of items in Lot								
Requisition Item No./ Description		Major content, Weight Percent						Remarks Accept/Reject
Element	Cr	Ni	Mo	V	Ti *	Cb/ Nb **	Mn ***	
Specified Range								
Actual observations 1.								
2.								
3.								
4.								
5.								
6.								
7.								
8.								
Instrument Type / ID Last Service Date		Inspection Agency						Witnessed By

* To be reported in case of SS321 Material
** To be reported in case of SS347 Material
*** To be reported in case of LTCS Material

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Signature Not Verified
Digitally signed by ARYA SUBASH
Date: 2026.08.27 16:27:33 IST
Location: Chennai Petroleum Corpn. Ltd-CPCL